Mathematics

Class 5 (English Medium)

Solved Exercises — Sindh Textbook Board

9 lessons — 254 questions

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Exercise 1.1 — Reading and Writing Numbers

Question: A. Write the number 171,283,680 in words.

Answer: Step 1: Split the number into groups: 171 million, 283 thousand, 680.

Step 2: Read each group in words: 171 = one hundred seventy one, 283 = two hundred eighty three, 680 = six hundred eighty.

Final answer: One hundred seventy one million, two hundred eighty three thousand, six hundred eighty.

Question: B. Write the following numbers in words: (1) 45,672 (2) 2,670,273 (3) 34,296,127 (4) 100,000,000

Answer: Step 1: Group each number in threes from the right.

Step 2: (1) 45,672 = forty five thousand six hundred seventy two. (2) 2,670,273 = two million six hundred seventy thousand two hundred seventy three.

Step 3: (3) 34,296,127 = thirty four million two hundred ninety six thousand one hundred twenty seven. (4) 100,000,000 = one hundred million.

Final answer: (1) Forty five thousand six hundred seventy two (2) Two million six hundred seventy thousand two hundred seventy three (3) Thirty four million two hundred ninety six thousand one hundred twenty seven (4) One hundred million.

Question: C. Write the following numbers in words: (1) 66,655,522 (2) 96,340,529 (3) 245,672,316 (4) 100,000,000

Answer: Step 1: Split each number into millions, thousands and ones groups.

Step 2: (1) 66,655,522 = sixty six million six hundred fifty five thousand five hundred twenty two. (2) 96,340,529 = ninety six million three hundred forty thousand five hundred twenty nine.

Step 3: (3) 245,672,316 = two hundred forty five million six hundred seventy two thousand three hundred sixteen. (4) 100,000,000 = one hundred million.

Final answer: (1) Sixty six million six hundred fifty five thousand five hundred twenty two (2) Ninety six million three hundred forty thousand five hundred twenty nine (3) Two hundred forty five million six hundred seventy two thousand three hundred sixteen (4) One hundred million.

Question: D. Write the following words in figures: (1) One million two thousand six hundred (2) Nine million ninety nine thousand seventy seven (3) Fifty eight million eight hundred sixty two thousand forty five (4) One billion (5) Three hundred forty five million six hundred seventy one thousand eight hundred six

Answer: Step 1: Read the words in groups: the millions part, the thousands part and the rest.

Step 2: (1) 1,000,000 + 2,000 + 600 = 1,002,600. (2) 9,000,000 + 99,000 + 77 = 9,099,077.

Step 3: (3) 58,000,000 + 862,000 + 45 = 58,862,045. (4) One billion = 1,000,000,000. (5) 345,000,000 + 671,000 + 806 = 345,671,806.

Final answer: (1) 1,002,600 (2) 9,099,077 (3) 58,862,045 (4) 1,000,000,000 (5) 345,671,806.

Exercise 1.2 — Addition

Question: A. Solve: (1) 713,492 + 268,310 (2) 4,318,114 + 313,934 (3) 8,193,860 + 429,177 (4) 6,216,921 + 834,538 (5) 3,941,234 + 590,543 (6) 5,491,869 + 605,304 (7) 7,390,443 + 1,538,006 (8) 2,555,312 + 1,084,881 (9) 5,319,432 + 4,181,518 (10) 34,653,635 + 2,660,477 (11) 56,018,987 + 23,594,431 (12) 438,394,568 + 351,056,939

Answer: Step 1: Write each sum in columns and add from the right (ones), carrying over to the next column.

Step 2: (1) 713,492 + 268,310 = 981,802. (2) 4,318,114 + 313,934 = 4,632,048. (3) 8,193,860 + 429,177 = 8,623,037.

Step 3: (4) 6,216,921 + 834,538 = 7,051,459. (5) 3,941,234 + 590,543 = 4,531,777. (6) 5,491,869 + 605,304 = 6,097,173.

Step 4: (7) 7,390,443 + 1,538,006 = 8,928,449. (8) 2,555,312 + 1,084,881 = 3,640,193. (9) 5,319,432 + 4,181,518 = 9,500,950.

Step 5: (10) 34,653,635 + 2,660,477 = 37,314,112. (11) 56,018,987 + 23,594,431 = 79,613,418. (12) 438,394,568 + 351,056,939 = 789,451,507.

Final answer: (1) 981,802 (2) 4,632,048 (3) 8,623,037 (4) 7,051,459 (5) 4,531,777 (6) 6,097,173 (7) 8,928,449 (8) 3,640,193 (9) 9,500,950 (10) 37,314,112 (11) 79,613,418 (12) 789,451,507.

Question: B. Add the following: (1) 680,563 and 563,168 (2) 3,541,371 and 232,164 (3) 34,399,188 and 6,412,508 (4) 65,943,022 and 22,913,924 (5) 840,233,419 and 65,113,846 (6) 733,050,195 and 188,439,919

Answer: Step 1: Place the numbers in columns and add from the right.

Step 2: (1) 680,563 + 563,168 = 1,243,731. (2) 3,541,371 + 232,164 = 3,773,535. (3) 34,399,188 + 6,412,508 = 40,811,696.

Step 3: (4) 65,943,022 + 22,913,924 = 88,856,946. (5) 840,233,419 + 65,113,846 = 905,347,265. (6) 733,050,195 + 188,439,919 =

921,490,114.

Final answer: (1) 1,243,731 (2) 3,773,535 (3) 40,811,696 (4) 88,856,946 (5) 905,347,265 (6) 921,490,114.

Question: C. Fill the missing digits in the following: (1) 536,2_4 + 3_3_22 = 889,636 (2) 347,845 + 4_2_1_ = 760,159 (3) 5,601,897 + 2_5_4_1 = 8,155,318 (4) 34_5_37 + 2_420_2 = 6,067,149

Answer: Step 1: Find each missing digit from its own column, remembering any carry.

Step 2: (1) The complete sum is 536,214 + 353,422 = 889,636, so the missing digits are: top tens 1, bottom ten-thousands 5, bottom hundreds 4.

Step 3: (2) Missing digits are 1, 3 and 4: 347,845 + 412,314 = 760,159.

Step 4: (3) Missing digits are 5, 3 and 2: 5,601,897 + 2,553,421 = 8,155,318. (4) Missing digits: top ten-thousands 2, top hundreds 1, bottom hundred-thousands 6, bottom tens 1: 3,425,137 + 2,642,012 = 6,067,149.

Final answer: (1) 1, 5, 4 (2) 1, 3, 4 (3) 5, 3, 2 (4) 2, 1, 6, 1.

Question: D. Faisal's father was admitted to the hospital for 1 day for an operation. His operation charges were Rs. 350,000, the room charges were Rs. 1,200 and his medicine charges were Rs. 125,500. How much was the total bill?

Answer: Step 1: Write the three charges: 350,000 + 1,200 + 125,500.

Step 2: Add the first two: 350,000 + 1,200 = 351,200.

Step 3: Add the medicine charges: 351,200 + 125,500 = 476,700.

Final answer: The total bill was Rs. 476,700.

Exercise 1.3 — Subtraction

Question: A. Solve: (1) 2,863,132 − 164,350 (2) 5,634,153 − 393,844 (3)

4,194,312 − 994,208 (4) 4,395,684 − 2,348,736 (5) 50,862,131 − 944,029 (6) 65,309,949 − 8,214,309

Answer: Step 1: Write the numbers in columns and subtract from the right; where the top digit is smaller, borrow from the column on the left.

Step 2: (1) 2,863,132 − 164,350 = 2,698,782. (2) 5,634,153 − 393,844 = 5,240,309. (3) 4,194,312 − 994,208 = 3,200,104.

Step 3: (4) 4,395,684 − 2,348,736 = 2,046,948. (5) 50,862,131 − 944,029 = 49,918,102. (6) 65,309,949 − 8,214,309 = 57,095,640.

Final answer: (1) 2,698,782 (2) 5,240,309 (3) 3,200,104 (4) 2,046,948 (5) 49,918,102 (6) 57,095,640.

Question: B. Subtract: (1) 214,379 from 600,500 (2) 856,394 from 3,767,595 (3) 4,930,109 from 5,851,036 (4) 5,396,138 from 43,547,967 (5) 35,180,962 from 89,086,371 (6) 134,258,369 from 656,148,154

Answer: Step 1: Subtract the smaller number from the bigger number, borrowing in columns where needed.

Step 2: (1) 600,500 − 214,379 = 386,121. (2) 3,767,595 − 856,394 = 2,911,201. (3) 5,851,036 − 4,930,109 = 920,927.

Step 3: (4) 43,547,967 − 5,396,138 = 38,151,829. (5) 89,086,371 − 35,180,962 = 53,905,409. (6) 656,148,154 − 134,258,369 = 521,889,785.

Final answer: (1) 386,121 (2) 2,911,201 (3) 920,927 (4) 38,151,829 (5) 53,905,409 (6) 521,889,785.

Question: C. Fill the missing digits in the following: (1) 873,246 − 43_,_2_ = 442,123 (2) 694,7_2 − 3_3_24 = 381,638 (3) 5_34_98 − 32_274_ = 2,721,554 (4) 681,4_39 − __0,252_ = 3,412,215

Answer: Step 1: To find a missing digit, add the answer to the lower number or subtract from the top number, column by column.

Step 2: (1) 873,246 − 442,123 = 431,123, so the missing digits in the

lower number are thousands 1, hundreds 1 and ones 3.

Step 3: (2) The complete subtraction is 694,762 − 313,124 = 381,638; missing digits: top tens 6, bottom ten-thousands 1 and hundreds 1.

Step 4: (3) The complete subtraction is 5,934,298 − 3,212,744 = 2,721,554; missing digits: top hundred-thousands 9, top hundreds 2, bottom ten-thousands 1, bottom ones 4. (4) The complete subtraction is 6,814,739 − 3,402,524 = 3,412,215; missing digits: top hundreds 7, bottom millions 3, bottom hundred-thousands 4, bottom ones 4.

Final answer: (1) 1, 1, 3 (2) 6, 1, 1 (3) 9, 2, 1, 4 (4) 7, 3, 4, 4.

Question: D. Pakistan imported 107,857 kg dry grapes in 2014, and 154,792 kg dry grapes in 2015. How many more dry grapes were imported in 2015 than in 2014?

Answer: Step 1: Subtract the smaller quantity from the larger: 154,792 − 107,857.

Step 2: Subtract in columns: 154,792 − 107,857 = 46,935.

Final answer: 46,935 kg more dry grapes were imported in 2015.

Question: Activities 1 and 2 (book page 8): Fill the missing digits in the following: (1) 585,463 − 3_1_12 = 234,251 (2) 74_86_ − 3_1_12 = 442,453

Answer: Step 1: Find the missing digits column by column.

Step 2: (1) The complete subtraction is 585,463 − 351,212 = 234,251, so the missing digits are ten-thousands 5 and hundreds 2.

Step 3: (2) The complete subtraction is 743,865 − 301,412 = 442,453, so the missing digits are: top thousands 3 and ones 5; bottom ten-thousands 0 and hundreds 4.

Final answer: (1) 5, 2 (2) 3, 5, 0, 4.

Exercise 1.4 — Multiplication

Question: A. Solve: (1) 4,136 × 10 (2) 34,569 × 10 (3) 21,034 × 10 (4) 15,347 × 100 (5) 27,796 × 100 (6) 155,430 × 100 (7) 41,357 × 1,000 (8) 386,975 × 1,000

Answer: Step 1: For × 10 put one 0 at the end of the number, for × 100 put two 0s and for × 1,000 put three 0s.

Step 2: (1) 4,136 × 10 = 41,360. (2) 34,569 × 10 = 345,690. (3) 21,034 × 10 = 210,340. (4) 15,347 × 100 = 1,534,700.

Step 3: (5) 27,796 × 100 = 2,779,600. (6) 155,430 × 100 = 15,543,000. (7) 41,357 × 1,000 = 41,357,000. (8) 386,975 × 1,000 = 386,975,000.

Final answer: (1) 41,360 (2) 345,690 (3) 210,340 (4) 1,534,700 (5) 2,779,600 (6) 15,543,000 (7) 41,357,000 (8) 386,975,000.

Question: B. Solve: (1) 1,942 × 50 (2) 63,578 × 80 (3) 25,608 × 70 (4) 326,985 × 90 (5) 8,540 × 300 (6) 280,915 × 600

Answer: Step 1: First multiply by the digit without the 0, then put the same number of 0s at the end.

Step 2: (1) 1,942 × 5 = 9,710, so 1,942 × 50 = 97,100. (2) 63,578 × 8 = 508,624, so 63,578 × 80 = 5,086,240. (3) 25,608 × 7 = 179,256, so 25,608 × 70 = 1,792,560.

Step 3: (4) 326,985 × 9 = 2,942,865, so 326,985 × 90 = 29,428,650. (5) 8,540 × 3 = 25,620, so 8,540 × 300 = 2,562,000. (6) 280,915 × 6 = 1,685,490, so 280,915 × 600 = 168,549,000.

Final answer: (1) 97,100 (2) 5,086,240 (3) 1,792,560 (4) 29,428,650 (5) 2,562,000 (6) 168,549,000.

Question: C. Multiply: (1) 25,839 × 33 (2) 243,419 × 86 (3) 65,204 × 75 (4) 467,808 × 92 (5) 76,391 × 22 (6) 298,543 × 44 (7) 349,776 × 53 (8) 531,062 × 68 (9) 12,873 × 425 (10) 859,046 × 710 (11) 357,904 × 486

(12) 809,507 × 572 (13) 598,722 × 235 (14) 914,076 × 572 (15) 743,158 × 377 (16) 865,432 × 444

Answer: Step 1: In each product, multiply by the ones, then by the tens, then by the hundreds, and add the partial products. Example: (1) 25,839 × 3 = 77,517 and 25,839 × 30 = 775,170; sum = 852,687.

Step 2: (2) 243,419 × 86 = 20,934,034. (3) 65,204 × 75 = 4,890,300. (4) 467,808 × 92 = 43,038,336. (5) 76,391 × 22 = 1,680,602.

Step 3: (6) 298,543 × 44 = 13,135,892. (7) 349,776 × 53 = 18,538,128. (8) 531,062 × 68 = 36,112,216. (9) 12,873 × 425 = 5,471,025.

Step 4: (10) 859,046 × 710 = 609,922,660. (11) 357,904 × 486 = 173,941,344. (12) 809,507 × 572 = 463,038,004. (13) 598,722 × 235 = 140,699,670.

Step 5: (14) 914,076 × 572 = 522,851,472. (15) 743,158 × 377 = 280,170,566. (16) 865,432 × 444 = 384,251,808.

Final answer: (1) 852,687 (2) 20,934,034 (3) 4,890,300 (4) 43,038,336 (5) 1,680,602 (6) 13,135,892 (7) 18,538,128 (8) 36,112,216 (9) 5,471,025 (10) 609,922,660 (11) 173,941,344 (12) 463,038,004 (13) 140,699,670 (14) 522,851,472 (15) 280,170,566 (16) 384,251,808.

Question: D. The school library ordered 45 books. The cost of each book was Rs. 325. What is the cost of 45 library books?

Answer: Step 1: Total cost = cost of one book × number of books = 325 × 45.

Step 2: 325 × 40 = 13,000 and 325 × 5 = 1,625.

Step 3: Add both: 13,000 + 1,625 = 14,625.

Final answer: The total cost is Rs. 14,625.

Exercise 1.5 — Division

Question: A. Divide: (1) 295,845 by 33 (2) 569,551 by 89 (3) 639,133 by 97 (4) 576,480 by 60 (5) 269,760 by 480 (6) 135,095 by 205 (7) 444,771 by 321 (8) 466,896 by 822

Answer: Step 1: Do each division by the long division method and write the quotient.

Step 2: (1) 295,845 ÷ 33 = 8,965, remainder 0. (2) 569,551 ÷ 89: 89 × 6,399 = 569,511, so quotient 6,399, remainder 40. (3) 639,133 ÷ 97 = 6,589, remainder 0. (4) 576,480 ÷ 60 = 9,608, remainder 0.

Step 3: (5) 269,760 ÷ 480 = 562, remainder 0. (6) 135,095 ÷ 205 = 659, remainder 0. (7) 444,771 ÷ 321: 321 × 1,385 = 444,585, so quotient 1,385, remainder 186. (8) 466,896 ÷ 822 = 568, remainder 0.

Final answer: (1) 8,965 (2) 6,399, remainder 40 (3) 6,589 (4) 9,608 (5) 562 (6) 659 (7) 1,385, remainder 186 (8) 568.

Question: B. Find the remainder and quotient of the following: (1) 5,678 ÷ 10 (2) 396,785 ÷ 10 (3) 473,405 ÷ 100 (4) 843,216 ÷ 100 (5) 5,230,106 ÷ 100 (6) 8,256,879 ÷ 1,000 (7) 6,456,782 ÷ 1,000 (8) 9,650,000 ÷ 1,000

Answer: Step 1: When dividing by 10, 100 or 1,000, the last digits on the right form the remainder and the rest of the number is the quotient.

Step 2: (1) Quotient 567, remainder 8. (2) Quotient 39,678, remainder 5. (3) Quotient 4,734, remainder 5. (4) Quotient 8,432, remainder 16.

Step 3: (5) Quotient 52,301, remainder 6. (6) Quotient 8,256, remainder 879. (7) Quotient 6,456, remainder 782. (8) Quotient 9,650, remainder 0.

Final answer: (1) 567, r 8 (2) 39,678, r 5 (3) 4,734, r 5 (4) 8,432, r 16 (5) 52,301, r 6 (6) 8,256, r 879 (7) 6,456, r 782 (8) 9,650, r 0.

Question: C. Mr. Anwar cooks 61,500 naans in 500 days. How many naans must he cook every day if he cooks equal number of naans daily?

Answer: Step 1: Naans per day = total naans ÷ days = 61,500 ÷ 500.

Step 2: Cancel two 0s from both numbers: 615 ÷ 5 = 123.

Final answer: He must cook 123 naans every day.

Exercise 1.6 — Real Life Problems (Mixed Operations)

Question: (1) Fouzia had 145,320 rupees. Her father gave her 54,304 rupees more. How many rupees did Fouzia have altogether? (2) Sobia gave a 5,000 rupee note to the shopkeeper for purchasing an electronic doll worth Rs. 2,300. Find the remaining amount received by Sobia. (3) A school collected funds for flood relief. 225 students of a school contributed equally Rs. 23,650. Find the contribution by each student. (4) A factory produced 235,806 floor marble tiles in a day. How many marbles will be produced in 32 days?

Answer: Step 1: (1) Add: 145,320 + 54,304 = 199,624 rupees.

Step 2: (2) Subtract: 5,000 − 2,300 = 2,700 rupees received back.

Step 3: (3) Divide: 23,650 ÷ 225; 225 × 105 = 23,625, remainder 25, so each student contributed Rs. 105 and Rs. 25 were left over (about Rs. 105.11 per student).

Step 4: (4) Multiply: 235,806 × 32; 235,806 × 30 = 7,074,180 and 235,806 × 2 = 471,612; sum = 7,545,792 tiles.

Final answer: (1) 199,624 rupees (2) Rs. 2,700 (3) Rs. 105 per student, Rs. 25 remaining (4) 7,545,792 tiles.

Question: (5) Government spent Rs. 5,380,100 on gas supply in one town. What is the total amount spent on gas supply of 25 such towns? (6) A container holds 9,475 litres of milk. How many litres of milk will 354 such containers hold? (7) Find the number of rolls required to pack 531,675 metre cloths. One roll contains 45 metres. (8) A construction company

wanted to purchase a land for housing scheme of Rs. 52,890,500. It had Rs. 50,456,128 and borrowed the rest from the Bank. How much money did the company borrow from the Bank?

Answer: Step 1: (5) Multiply: 5,380,100 × 25; 5,380,100 × 100 = 538,010,000, one fourth of it = 134,502,500 rupees.

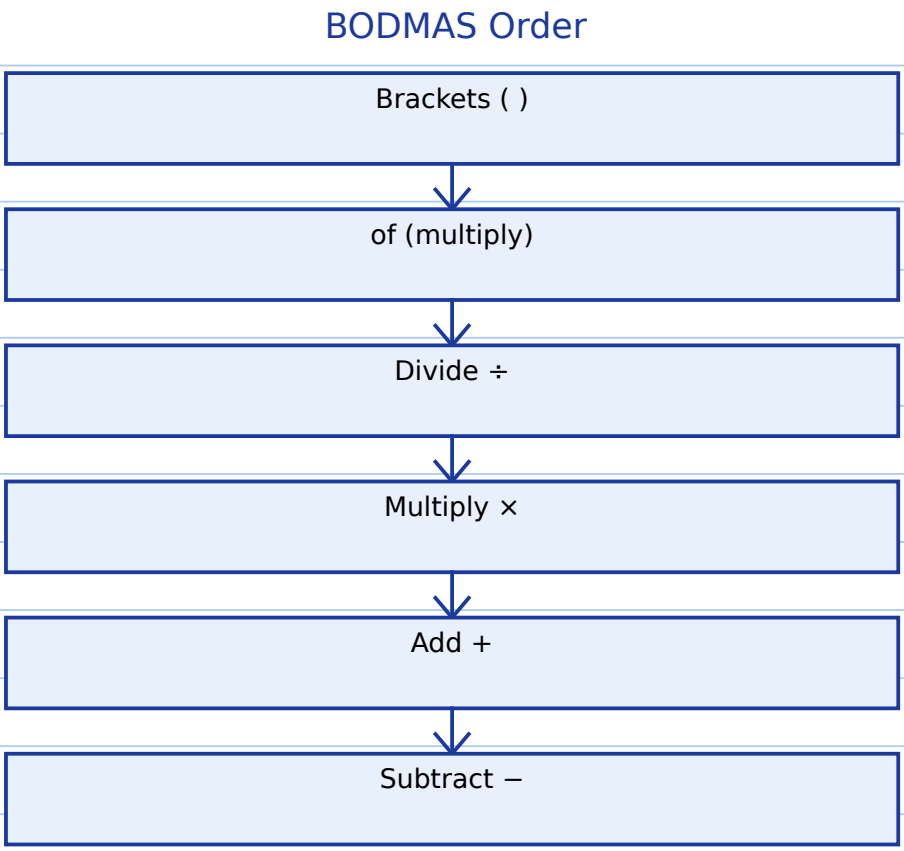
Step 2: (6) Multiply: 9,475 × 354; 9,475 × 300 = 2,842,500; 9,475 × 54 = 511,650; sum = 3,354,150 litres.

Step 3: (7) Divide: 531,675 ÷ 45 = 11,815 rolls.

Step 4: (8) Subtract: 52,890,500 − 50,456,128 = 2,434,372 rupees borrowed.

Final answer: (5) Rs. 134,502,500 (6) 3,354,150 litres (7) 11,815 rolls (8) Rs. 2,434,372.

Exercise 1.7 — BODMAS Rule



Question: A. Salman solved the following question using BODMAS: (213 − 123) + 60 × 5 − (64 ÷ 8) = 382. Write down the steps that Salman followed.

Answer: Step 1: Salman first solved the bracket: 213 − 123 = 90.

Step 2: Then he divided: 64 ÷ 8 = 8.

Step 3: Then he multiplied: 60 × 5 = 300.

Step 4: Then he added: 90 + 300 = 390.

Step 5: At last he subtracted: 390 − 8 = 382.

Final answer: Steps: 1. Solve the brackets 2. Divide 3. Multiply 4. Add 5. Subtract — answer 382.

Question: B. Solve: (1) (9 − 8) × 18 (2) (5 × 9) ÷ 15 (3) 16 ÷ 2 + 5 × 4 − 2 (4) 32 × 21 − 42 ÷ 7 (5) 50 × 5 + (15 + 23) (6) 7 + (15 ÷ 3 + 5) × 4 − 20 (7) 4 of 195 ÷ 13 − 54 (8) (3 × 18) ÷ 3 of 2 + 105 (9) (28 ÷ 4 + 5) × 4

− 11 of 3 (10) 5 + (42 + 7 of 2 − 2) × 8 (11) 100 of (3 × 150 ÷ 10) (12) 60 + (72 ÷ 7 of 3 + 5) × 2

Answer: Step 1: In each question, do brackets first, then "of", then division and multiplication, and at last addition and subtraction from left to right.

Step 2: (1) Bracket: 9 − 8 = 1; 1 × 18 = 18. (2) Bracket: 5 × 9 = 45; 45 ÷ 15 = 3. (3) Divide and multiply: 16 ÷ 2 = 8, 5 × 4 = 20; 8 + 20 − 2 = 26. (4) Multiply and divide: 32 × 21 = 672, 42 ÷ 7 = 6; 672 − 6 = 666.

Step 3: (5) Bracket: 15 + 23 = 38; 50 × 5 = 250; 250 + 38 = 288. (6) Bracket: 15 ÷ 3 + 5 = 10; 7 + 10 × 4 − 20 = 7 + 40 − 20 = 27. (7) of: 4 of 195 = 780; 780 ÷ 13 = 60; 60 − 54 = 6. (8) Bracket: 3 × 18 = 54; of: 3 of 2 = 6; 54 ÷ 6 = 9; 9 + 105 = 114.

Step 4: (9) Bracket: 28 ÷ 4 + 5 = 12; 12 × 4 = 48; of: 11 of 3 = 33; 48 − 33 = 15. (10) In the bracket, "of" first: 7 of 2 = 14; 42 + 14 − 2 = 54; 54 × 8 = 432; 5 + 432 = 437.

Step 5: (11) Bracket: 3 × 150 ÷ 10 = 45; of: 100 of 45 = 4,500. (12) In the bracket, "of" first: 7 of 3 = 21; 72 ÷ 21 does not divide exactly, so as printed the question has no whole-number answer (72 ÷ 21 = 3, remainder 9); the book's answer key prints 1.

Final answer: (1) 18 (2) 3 (3) 26 (4) 666 (5) 288 (6) 27 (7) 6 (8) 114 (9) 15 (10) 437 (11) 4,500 (12) Not exactly divisible; 72 ÷ 21 = 3 remainder 9.

Question: C. Add brackets to these mathematical statements so that the statements become correct: (1) 24 ÷ 4 × 3 = 18 (2) 6 × 2 + 8 ÷ 4 = 15

Answer: Step 1: (1) Without brackets: 24 ÷ 4 × 3 = 6 × 3 = 18 — the statement is already correct from left to right; write it as (24 ÷ 4) × 3 = 18.

Step 2: (2) 6 × 2 + 8 ÷ 4 = 12 + 2 = 14, which is not correct. Put the brackets like this: 6 × (2 + 8) ÷ 4.

Step 3: Check: bracket 2 + 8 = 10; 6 × 10 = 60; 60 ÷ 4 = 15 ✓.

Final answer: (1) (24 ÷ 4) × 3 = 18 (2) 6 × (2 + 8) ÷ 4 = 15.

Exercise 1.8 — Distributive Laws

Question: A. Fill in the blanks: (1) 15 × (5 + 3) = (15 × ____) + (15 × ____) (2) (25 − 18) × 32 = (25 × ____) − (18 × ____) (3) (____ + ____) × 10 = (30 × 10) + (40 × 10) (4) (3 ____ 5) × 14 = (3 ____ 14) + (5 × 14) (5) ____ × (26 + 74) = (5 × 26) ____ (5 × 74) (6) 9 × (13 − ____) = (9 × ____) − (9 × 5) (7) 10 × (20 − ____) = (10 × ____) − (____ × 30) (8) (____ + ____) × 4 = (12 × ____) + (8 × ____)

Answer: Step 1: Distributive law: a × (b + c) = (a × b) + (a × c) and a × (b − c) = (a × b) − (a × c). Fill the blanks so both sides become equal.

Step 2: (1) 5 and 3. (2) 32 and 32. (3) 30 and 40. (4) The + sign in both blanks. (5) First blank 5, second blank the + sign.

Step 3: (6) 5 and 13. (7) As printed, the last part is (____ × 30); in the correct distributive form the blanks are 3, 20 and 10 and the last part should be (10 × 3): 10 × (20 − 3) = (10 × 20) − (10 × 3) = 200 − 30 = 170. (8) 12, 8, 4 and 4.

Final answer: (1) 5, 3 (2) 32, 32 (3) 30, 40 (4) +, + (5) 5, + (6) 5, 13 (7) 3, 20, 10 (the printed 30 appears to be a misprint for 3) (8) 12, 8, 4, 4.

Question: B. Verify distributive laws: (1) 5 × (3 + 2) = (5 × 3) + (5 × 2) (2) 4 × (9 − 5) = (4 × 9) − (4 × 5) (3) (18 + 2) × 10 = (18 × 10) + (2 × 10) (4) 10 × (12 − 3) = (10 × 12) − (10 × 3)

Answer: Step 1: Solve both sides of each statement separately.

Step 2: (1) Left side: 5 × 5 = 25; right side: 15 + 10 = 25. Both equal ✓. (2) Left: 4 × 4 = 16; right: 36 − 20 = 16 ✓.

Step 3: (3) Left: 20 × 10 = 200; right: 180 + 20 = 200 ✓. (4) Left: 10 × 9

= 90; right: 120 − 30 = 90 ✓.

Final answer: In all four, both sides are equal, so the distributive law is verified.

Review Exercise 1

Question: 1. Write the following numbers in words: (i) 246,416,279 (ii) 905,407,608 2. Write the following numbers in numerals: (i) Seventy five million, twenty six thousand four hundred twenty (ii) Four hundred five million, seven hundred forty five thousand eight hundred six

Answer: Step 1: Read the numbers in groups of millions, thousands and the rest.

Step 2: 1(i) 246,416,279 = two hundred forty six million, four hundred sixteen thousand, two hundred seventy nine. (ii) 905,407,608 = nine hundred five million, four hundred seven thousand, six hundred eight.

Step 3: 2(i) 75,000,000 + 26,000 + 420 = 75,026,420. (ii) 405,000,000 + 745,000 + 806 = 405,745,806.

Final answer: 1(i) Two hundred forty six million four hundred sixteen thousand two hundred seventy nine (ii) Nine hundred five million four hundred seven thousand six hundred eight 2(i) 75,026,420 (ii) 405,745,806.

Question: 3. Add the following: (i) 205,617,291 and 5,412,306 (ii) 4,000,405 and 20,136,999 (iii) 214,308,196 and 523,410,018 4. Subtract the following: (i) 412,326,917 from 624,005,123 (ii) 96,105,892 from 712,342,445 (iii) 234,596,501 from 641,884,962

Answer: Step 1: Add and subtract in columns.

Step 2: 3(i) 205,617,291 + 5,412,306 = 211,029,597. (ii) 4,000,405 + 20,136,999 = 24,137,404. (iii) 214,308,196 + 523,410,018 =

737,718,214.

Step 3: 4(i) 624,005,123 − 412,326,917 = 211,678,206. (ii) 712,342,445 − 96,105,892 = 616,236,553. (iii) 641,884,962 − 234,596,501 = 407,288,461.

Final answer: 3(i) 211,029,597 (ii) 24,137,404 (iii) 737,718,214 4(i) 211,678,206 (ii) 616,236,553 (iii) 407,288,461.

Question: 5. Solve: (i) 1,243 × 10 (ii) 962,345 × 45 (iii) 56,729 × 40 (iv) 612,378 × 962 (v) 405,617 × 1,000 6. Divide: (i) 753,400 by 30 (ii) 269,817 by 356 7. Solve: (i) 180 ÷ 10 × (50 of 2 ÷ 4) (ii) 48 − (5 of 2 + 12)

Answer: Step 1: Multiply and divide step by step; in BODMAS do brackets and "of" first.

Step 2: 5(i) 1,243 × 10 = 12,430. (ii) 962,345 × 45: 962,345 × 40 = 38,493,800, 962,345 × 5 = 4,811,725, sum = 43,305,525. (iii) 56,729 × 40 = 2,269,160. (iv) 612,378 × 962 = 589,107,636. (v) 405,617 × 1,000 = 405,617,000.

Step 3: 6(i) 753,400 ÷ 30 = 25,113, remainder 10. (ii) 269,817 ÷ 356 = 757, remainder 325 (356 × 757 = 269,492).

Step 4: 7(i) In the bracket, "of" first: 50 of 2 = 100; 100 ÷ 4 = 25; 180 ÷ 10 = 18; 18 × 25 = 450. (ii) Bracket: 5 of 2 = 10; 10 + 12 = 22; 48 − 22 = 26.

Final answer: 5(i) 12,430 (ii) 43,305,525 (iii) 2,269,160 (iv) 589,107,636 (v) 405,617,000 6(i) 25,113 r 10 (ii) 757 r 325 7(i) 450 (ii) 26.

Question: 8. Verify: (i) 2 × (4 + 7) = (2 × 4) + (2 × 7) (ii) 5 × (96 − 34) = (5 × 96) − (5 × 34) 9. A water tanker holds 24,541 litres of water. How much will be in 35 such tankers? 10. A box contains 30 chalks. How many such boxes are required to pack 24,660 chalks?

Answer: Step 1: 8(i) Left side: 2 × 11 = 22; right side: 8 + 14 = 22 ✓ equal. (ii) Left: 5 × 62 = 310; right: 480 − 170 = 310 ✓ equal.

Step 2: 9. Multiply: 24,541 × 35; 24,541 × 30 = 736,230; 24,541 × 5 = 122,705; sum = 858,935 litres.

Step 3: 10. Divide: 24,660 ÷ 30 = 822 boxes.

Final answer: 8(i) 22 = 22 ✓ (ii) 310 = 310 ✓ 9. 858,935 litres 10. 822 boxes.

یونٹ2: HCF LCM

Exercise 2.1 — HCF (by Prime Factorization and Division Method)

Question: A. Find HCF of the following by Prime Factorization method: (1) 16 and 28 (2) 27 and 36 (3) 24 and 56 (4) 28 and 42 (5) 44 and 66 (6) 52 and 78 (7) 20, 60 and 80 (8) 32, 48 and 96 (9) 35, 49 and 63 (10) 26, 39 and 65 (11) 45, 75 and 90 (12) 21, 35 and 63

Answer: Step 1: Write each number as prime factors and multiply the factors common to all. Example: (1) 16 = 2×2×2×2, 28 = 2×2×7; common factors 2×2 = 4.

Step 2: (2) 27 = 3×3×3, 36 = 2×2×3×3; common 3×3 = 9. (3) 24 = 2×2×2×3, 56 = 2×2×2×7; common 2×2×2 = 8. (4) 28 = 2×2×7, 42 = 2×3×7; common 2×7 = 14.

Step 3: (5) 44 = 2×2×11, 66 = 2×3×11; common 2×11 = 22. (6) 52 = 2×2×13, 78 = 2×3×13; common 2×13 = 26. (7) 20 = 2×2×5, 60 = 2×2×3×5, 80 = 2×2×2×2×5; common 2×2×5 = 20. (8) Common factors of 32, 48, 96 are 2×2×2×2 = 16.

Step 4: (9) 35 = 5×7, 49 = 7×7, 63 = 3×3×7; common 7. (10) 26 = 2×13, 39 = 3×13, 65 = 5×13; common 13. (11) Common factors of 45, 75, 90 are 3×5 = 15. (12) 21 = 3×7, 35 = 5×7, 63 = 3×3×7; common 7.

Final answer: (1) 4 (2) 9 (3) 8 (4) 14 (5) 22 (6) 26 (7) 20 (8) 16 (9) 7 (10) 13 (11) 15 (12) 7.

Question: B. Find HCF of the following by using division method: (1) 28 and 70 (2) 66 and 88 (3) 57 and 95 (4) 51 and 85 (5) 48 and 80 (6) 54 and 90 (7) 40, 60 and 80 (8) 36, 60 and 96 (9) 32, 48 and 80 (10) 60, 75 and 90

(11) 42, 70 and 84 (12) 63, 72 and 81

Answer: Step 1: Divide the greater number by the smaller; then keep dividing the divisor by the remainder until the remainder is 0. The last divisor is the HCF. Example: (1) 70 ÷ 28 = 2, remainder 14; 28 ÷ 14 = 2, remainder 0; HCF = 14.

Step 2: (2) 88 ÷ 66 = 1 r 22; 66 ÷ 22 = 3 r 0; HCF = 22. (3) 95 ÷ 57 = 1 r 38; 57 ÷ 38 = 1 r 19; 38 ÷ 19 = 2 r 0; HCF = 19. (4) 85 ÷ 51 = 1 r 34; 51 ÷ 34 = 1 r 17; 34 ÷ 17 = 2; HCF = 17.

Step 3: (5) 80 ÷ 48 = 1 r 32; 48 ÷ 32 = 1 r 16; 32 ÷ 16 = 2; HCF = 16. (6) 90 ÷ 54 = 1 r 36; 54 ÷ 36 = 1 r 18; 36 ÷ 18 = 2; HCF = 18. (7) HCF(40,60) = 20; HCF(20,80) = 20. (8) HCF(36,60) = 12; HCF(12,96) = 12.

Step 4: (9) HCF(32,48) = 16; HCF(16,80) = 16. (10) HCF(60,75) = 15; HCF(15,90) = 15. (11) HCF(42,70) = 14; HCF(14,84) = 14. (12) HCF(63,72) = 9; HCF(9,81) = 9.

Final answer: (1) 14 (2) 22 (3) 19 (4) 17 (5) 16 (6) 18 (7) 20 (8) 12 (9) 16 (10) 15 (11) 14 (12) 9.

Exercise 2.2 — LCM (by Prime Factorization and Division Method)

Question: A. Find LCM of the following by Prime factorization method: (1) 36 and 54 (2) 33 and 55 (3) 52 and 78 (4) 16, 24 and 40 (5) 27, 48 and 72 (6) 50, 80 and 90 (7) 56, 84 and 98 (8) 44, 66 and 99 (9) 25, 50 and 75 (10) 15, 25, 30 and 45 (11) 10, 20, 32 and 40 (12) 12, 24, 48 and 54

Answer: Step 1: Write the prime factors of each number; take the highest power of every prime factor and multiply them all. Example: (1) 36 = $2^2 \times 3^2$, 54 = 2×3^3; LCM = $2^2 \times 3^3$ = 4 × 27 = 108.

Step 2: (2) 33 = 3×11, 55 = 5×11; LCM = 3×5×11 = 165. (3) 52 = $2^2 \times 13$, 78 = 2×3×13; LCM = $2^2 \times 3 \times 13$ = 156. (4) 16 = 2^4, 24 = $2^3 \times 3$, 40 = $2^3 \times 5$;

LCM = $2^4 \times 3 \times 5 = 240$.

Step 3: (5) LCM = $2^4 \times 3^3 = 432$. (6) LCM = $2^4 \times 3^2 \times 5^2 = 3,600$. (7) LCM = $2^3 \times 3 \times 7^2 = 1,176$. (8) LCM = $2^2 \times 3^2 \times 11 = 396$.

Step 4: (9) LCM = $2 \times 3 \times 5^2 = 150$. (10) LCM = $2 \times 3^2 \times 5^2 = 450$. (11) LCM = $2^5 \times 5 = 160$. (12) LCM = $2^4 \times 3^3 = 432$.

Final answer: (1) 108 (2) 165 (3) 156 (4) 240 (5) 432 (6) 3,600 (7) 1,176 (8) 396 (9) 150 (10) 450 (11) 160 (12) 432.

Question: B. Find LCM of the following by division method: (1) 16 and 24 (2) 20 and 25 (3) 36 and 48 (4) 27, 36 and 45 (5) 28, 35 and 63 (6) 48, 64 and 96 (7) 54, 72 and 90 (8) 55, 88 and 99 (9) 60, 70 and 80 (10) 8, 12, 32 and 48 (11) 18, 27, 36 and 45 (12) 20, 40, 60 and 80

Answer: Step 1: Write all the numbers in one row and keep dividing by prime numbers until all become 1; the product of all the divisors is the LCM. Example: (1) Divide 16, 24 by 2 four times and by 3 once: LCM = $2 \times 2 \times 2 \times 2 \times 3 = 48$.

Step 2: (2) LCM = $2^2 \times 5^2 = 100$. (3) LCM = $2^4 \times 3^2 = 144$. (4) LCM = $2^2 \times 3^3 \times 5 = 540$.

Step 3: (5) LCM = $2^2 \times 3^2 \times 5 \times 7 = 1,260$. (6) LCM = $2^6 \times 3 = 192$. (7) LCM = $2^3 \times 3^3 \times 5 = 1,080$. (8) LCM = $2^3 \times 3^2 \times 5 \times 11 = 3,960$.

Step 4: (9) LCM = $2^4 \times 3 \times 5 \times 7 = 1,680$. (10) LCM = $2^5 \times 3 = 96$. (11) LCM = $2^2 \times 3^3 \times 5 = 540$. (12) LCM = $2^4 \times 3 \times 5 = 240$.

Final answer: (1) 48 (2) 100 (3) 144 (4) 540 (5) 1,260 (6) 192 (7) 1,080 (8) 3,960 (9) 1,680 (10) 96 (11) 540 (12) 240.

Exercise 2.3 — Real Life Problems of HCF and LCM

Question: (1) The lowest number, which is divisible by 15, 25, 40 and 75. (2) Find the greatest number that will divide 42, 66 and 78. (3) Salam

exercises after every 10 days and Nadeem every 6 days. Salam and Nadeem both exercised today. After how many days will they be unite and exercise together again? (4) Hooria is thinking of a number that is divisible by both 15 and 21. What is the smallest possible number that Hooria could be thinking of? (5) Four vans leave Sukkur at the same time. The first van is seen after 3 hours, the second after 4 hours, the third after 5 hours and the fourth after 6 hours. After how many hours from the starting time, will all of them come together?

Answer: Step 1: Where the "smallest common/divisible" number is needed, find the LCM; where the "greatest divisor" is needed, find the HCF.

Step 2: (1) LCM(15, 25, 40, 75): prime factors $2^3 \times 3 \times 5^2 = 600$. (2) HCF(42, 66, 78): $42 = 2 \times 3 \times 7$, $66 = 2 \times 3 \times 11$, $78 = 2 \times 3 \times 13$; common factors $2 \times 3 = 6$.

Step 3: (3) LCM(10, 6) = 30 days. (4) LCM(15, 21) = $3 \times 5 \times 7 = 105$. (5) LCM(3, 4, 5, 6) = 60 hours.

Final answer: (1) 600 (2) 6 (3) After 30 days (4) 105 (5) After 60 hours.

Question: (6) Boxes that are 12 cm tall are being stacked next to boxes that are 18 cm tall. What is the shortest height at which the two stacks will be of the same height? (7) Samina has two pieces of cloth. One piece is 64 cm wide and the other piece is 80 cm wide. She wants to cut both pieces into strips of equal width that are as wide as possible. How wide should she cut the strips? (8) Three milk containers hold 30 litres, 40 litres and 50 litres respectively. Find the capacity of a measuring utensil which can hold the milk in each container an exact number of times. (9) Find the greatest length of a measure of stick which can be used to measure exactly 30 cm, 60 cm or 90 cm. (10) Pencils are in packages of 10. Erasers are in packages of 12. A shopkeeper wants to purchase the smallest number of

pencils and erasers so that he will have exactly 1 eraser per pencil. How many packages of pencils and erasers should shopkeeper buy?

Answer: Step 1: Find the LCM in (6) and (10), and the HCF in (7), (8) and (9).

Step 2: (6) LCM(12, 18) = 36 cm. (7) HCF(64, 80): 64 = 2^6, 80 = $2^4 \times 5$; common 2^4 = 16 cm.

Step 3: (8) HCF(30, 40, 50) = 10 litres. (9) HCF(30, 60, 90) = 30 cm.

Step 4: (10) LCM(10, 12) = 60; packages of pencils = 60 ÷ 10 = 6; packages of erasers = 60 ÷ 12 = 5.

Final answer: (6) 36 cm (7) 16 cm (8) 10 litres (9) 30 cm (10) 6 packages of pencils and 5 packages of erasers.

Review Exercise 2

Question: 1. Tick (✓) the correct answer: (i) HCF of 2, 4, 10 is: (a) 2 (b) 4 (c) 10 (d) 1 (ii) LCM of 3, 6, 9 is: (a) 3 (b) 6 (c) 9 (d) 18 (iii) HCF of 3, 5, 7 is: (a) 3 (b) 1 (c) 7 (d) 5 (iv) LCM of 2, 3 and 5 is: (a) 6 (b) 10 (c) 30 (d) 15 (v) To find HCF by division method we divide greater number by ____: (a) same number (b) twice the number (c) smaller number (d) greater number

Answer: Step 1: Solve each question and choose the correct option.

Step 2: (i) HCF(2, 4, 10) = 2, so (a) 2. (ii) LCM(3, 6, 9) = 18, so (d) 18. (iii) 3, 5, 7 have no common factor except 1, so (b) 1.

Step 3: (iv) LCM(2, 3, 5) = 30, so (c) 30. (v) We divide the greater number by the smaller number, so (c) smaller number.

Final answer: (i) (a) 2 (ii) (d) 18 (iii) (b) 1 (iv) (c) 30 (v) (c) smaller number.

Question: 2. Find the least number which is exactly divisible by 6, 8 and 12. 3. Find the greatest number which divides 12 and 18 exactly. 4. What is the least number of apples which can be distributed equally among 10,

15 or 20 children? 5. Two pieces of cloth 15 metres and 20 metres long respectively are to be cut into small pieces of equal lengths. What will be the greatest length of each piece? 6. What will be the capacity of the smallest drum which can be filled completely using each of the utensils that can fill exactly 15l, 25l, 40l or 75l?

Answer: Step 1: Find the LCM in 2, 4 and 6, and the HCF in 3 and 5.

Step 2: 2. LCM(6, 8, 12) = 24. 3. HCF(12, 18) = 6. 4. LCM(10, 15, 20) = 60 apples.

Step 3: 5. HCF(15, 20) = 5 metres. 6. LCM(15, 25, 40, 75) = 600 litres.

Final answer: 2. 24 3. 6 4. 60 apples 5. 5 metres 6. 600 litres.

یونٹ3 پاگا :

Exercise 3.1 — Addition and Subtraction of Fractions

Question: A. Add the following: (1) 1/3 + 1/2 (2) 3/4 + 1/8 (3) 2/5 + 1/3 (4) 3/8 + 1/3 (5) 2/9 + 3/4 (6) 2/5 + 3/7

Answer: Step 1: Find the LCM of the denominators, make both fractions have the same denominator, then add the numerators.

Step 2: (1) LCM 6: 2/6 + 3/6 = 5/6. (2) LCM 8: 6/8 + 1/8 = 7/8. (3) LCM 15: 6/15 + 5/15 = 11/15.

Step 3: (4) LCM 24: 9/24 + 8/24 = 17/24. (5) LCM 36: 8/36 + 27/36 = 35/36. (6) LCM 35: 14/35 + 15/35 = 29/35.

Final answer: (1) 5/6 (2) 7/8 (3) 11/15 (4) 17/24 (5) 35/36 (6) 29/35.

Question: B. Solve: (1) 1/3 + 1/5 + 1/9 (2) 1/6 + 1/15 + 1/18 (3) 1/8 + 1/12 + 1/16 (4) 1 1/10 + 1 1/5 + 2/20 (5) 2 1/3 + 1 1/15 + 1 1/20 (6) 1 1/24 + 1/32 + 1 1/4

Answer: Step 1: Find the LCM of the three denominators; first change mixed fractions into improper fractions.

Step 2: (1) LCM 45: 15/45 + 9/45 + 5/45 = 29/45. (2) LCM 90: 15/90 + 6/90 + 5/90 = 26/90 = 13/45. (3) LCM 48: 6/48 + 4/48 + 3/48 = 13/48.

Step 3: (4) 11/10 + 6/5 + 1/10; LCM 10: 11/10 + 12/10 + 1/10 = 24/10 = 2 2/5. (5) 7/3 + 16/15 + 21/20; LCM 60: 140/60 + 64/60 + 63/60 = 267/60 = 4 9/20. (6) 25/24 + 1/32 + 5/4; LCM 96: 100/96 + 3/96 + 120/96 = 223/96 = 2 31/96.

Final answer: (1) 29/45 (2) 13/45 (3) 13/48 (4) 2 2/5 (5) 4 9/20 (6) 2 31/96.

Question: C. Subtract the following: (1) 3/4 − 3/8 (2) 2/3 − 1/4 (3) 3/4 −

1/2 (4) 5/6 − 1/2 (5) 4 2/5 − 1 1/4 (6) 3 3/5 − 2 9/10

Answer: Step 1: Make the denominators the same, then subtract the numerators.

Step 2: (1) LCM 8: 6/8 − 3/8 = 3/8. (2) LCM 12: 8/12 − 3/12 = 5/12. (3) LCM 4: 3/4 − 2/4 = 1/4.

Step 3: (4) LCM 6: 5/6 − 3/6 = 2/6 = 1/3. (5) 22/5 − 5/4; LCM 20: 88/20 − 25/20 = 63/20 = 3 3/20. (6) 18/5 − 29/10; LCM 10: 36/10 − 29/10 = 7/10.

Final answer: (1) 3/8 (2) 5/12 (3) 1/4 (4) 1/3 (5) 3 3/20 (6) 7/10.

Question: D. Solve: (1) 7/8 − 1/3 − 1/4 (2) 3/4 − 1/6 − 1/3 (3) 6/7 − 1/14 − 1/2 (4) 2 5/6 − 2 1/3 − 1/4 (5) 2 11/12 − 2 1/6 − 1 1/4 (6) 2 9/10 − 3 1/2 − 2 1/5 (7) 4 8/9 − 3 1/6 − 4 1/3 (8) 3 1/12 − 3 1/4 − 2 1/6 (9) 2 3/15 − 1 2/5 − 1 3/10

Answer: Step 1: Change mixed fractions into improper fractions, take the LCM and subtract from left to right. Where the first fraction is smaller, the answer is negative.

Step 2: (1) LCM 24: 21/24 − 8/24 − 6/24 = 7/24. (2) LCM 12: 9/12 − 2/12 − 4/12 = 3/12 = 1/4. (3) LCM 14: 12/14 − 1/14 − 7/14 = 4/14 = 2/7. (4) 17/6 − 7/3 − 1/4; LCM 12: 34/12 − 28/12 − 3/12 = 3/12 = 1/4.

Step 3: (5) 35/12 − 13/6 − 5/4; LCM 12: 35/12 − 26/12 − 15/12 = −6/12 = −1/2. (6) 29/10 − 7/2 − 11/5; LCM 10: 29/10 − 35/10 − 22/10 = −28/10 = −2 4/5.

Step 4: (7) 44/9 − 19/6 − 13/3; LCM 18: 88/18 − 57/18 − 78/18 = −47/18 = −2 11/18. (8) 37/12 − 13/4 − 13/6; LCM 12: 37/12 − 39/12 − 26/12 = −28/12 = −2 1/3. (9) 33/15 − 7/5 − 13/10; LCM 30: 66/30 − 42/30 − 39/30 = −15/30 = −1/2.

Final answer: (1) 7/24 (2) 1/4 (3) 2/7 (4) 1/4 (5) −1/2 (6) −2 4/5 (7) −2 11/18 (8) −2 1/3 (9) −1/2.

Question: E. Add: (1) 2/5 and 3/10 (2) 1/3 and 1/6 (3) 1/7, 1/14 and 1/21 F. Subtract: (1) 2/3 from 3/4 (2) 1/3 from 4/5 (3) 3/4 from 4/5

Answer: Step 1: "And" means add and "from" means subtract. Take the LCM and solve.

Step 2: E(1) LCM 10: 4/10 + 3/10 = 7/10. (2) LCM 6: 2/6 + 1/6 = 3/6 = 1/2. (3) LCM 42: 6/42 + 3/42 + 2/42 = 11/42.

Step 3: F(1) LCM 12: 9/12 − 8/12 = 1/12. (2) LCM 15: 12/15 − 5/15 = 7/15. (3) LCM 20: 16/20 − 15/20 = 1/20.

Final answer: E(1) 7/10 (2) 1/2 (3) 11/42 F(1) 1/12 (2) 7/15 (3) 1/20.

Question: Activity (book pages 38-39): Multiply with the help of diagrams: (1) 3 parts of 1/8 (2) 3 parts of 1/4 (3) 4 parts of 3/4 — write each as a multiplication of fractions and solve.

Answer: Step 1: "Parts of" means multiply: the fraction is taken that many times.

Step 2: (1) 1/8 × 3 = 3/8. (2) 1/4 × 3 = 3/4. (3) 3/4 × 4 = 12/4 = 3.

Final answer: (1) 3/8 (2) 3/4 (3) 3.

Exercise 3.2 — Multiplication of Fractions

Question: A. Multiply the fraction by the given whole number. Also demonstrate with the help of diagram: (1) 3/4 × 4 (2) 1/3 × 3 (3) 3/5 × 5 (4) 1/2 × 2 (5) 1/4 × 8 (6) 1/3 × 6 (7) 3/2 × 4 (8) 2/3 × 6 (9) 3/5 × 11

Answer: Step 1: Multiply the whole number by the numerator only; the denominator stays the same; then simplify. In the diagram, that many equal parts are shaded.

Step 2: (1) 3×4/4 = 12/4 = 3. (2) 1×3/3 = 3/3 = 1. (3) 3×5/5 = 15/5 = 3. (4) 1×2/2 = 2/2 = 1.

Step 3: (5) 1×8/4 = 8/4 = 2. (6) 1×6/3 = 6/3 = 2. (7) 3×4/2 = 12/2 = 6.

(8) 2×6/3 = 12/3 = 4. (9) 3×11/5 = 33/5 = 6 3/5.

Final answer: (1) 3 (2) 1 (3) 3 (4) 1 (5) 2 (6) 2 (7) 6 (8) 4 (9) 6 3/5.

Question: B. Multiply the given fraction by another fraction: (1) 1/2 by 1/5 (2) 1/4 by 3/5 (3) 1/3 by 2/3 (4) 1 1/4 by 3/4 (5) 2 1/3 by 4/5 (6) 3 1/2 by 1 1/2 C. Simplify: (1) 3/4 × 8/9 (2) 2/5 × 25/4 (3) 7/3 × 9/14 (4) 1 5/18 × 2 1/10 × 8/7 (5) 1 6/7 × 14/15 × 3 1/8 (6) 2 1/3 × 1 2/3 × 20/7

Answer: Step 1: Multiply numerators with numerators and denominators with denominators; first change mixed fractions into improper fractions and cancel where possible.

Step 2: B(1) 1×1/2×5 = 1/10. (2) 1×3/4×5 = 3/20. (3) 1×2/3×3 = 2/9. (4) 5/4 × 3/4 = 15/16. (5) 7/3 × 4/5 = 28/15 = 1 13/15. (6) 7/2 × 3/2 = 21/4 = 5 1/4.

Step 3: C(1) 3/4 × 8/9: cancel 8 with 4 and 3 with 9: = 2/3. (2) 2/5 × 25/4 = 50/20 = 2 1/2. (3) 7/3 × 9/14 = 63/42 = 1 1/2.

Step 4: C(4) 23/18 × 21/10 × 8/7 = 3,864/1,260 = 3 1/15. (5) 13/7 × 14/15 × 25/8 = 4,550/840 = 5 5/12. (6) 7/3 × 5/3 × 20/7 = 700/63 = 11 1/9.

Final answer: B(1) 1/10 (2) 3/20 (3) 2/9 (4) 15/16 (5) 1 13/15 (6) 5 1/4 C(1) 2/3 (2) 2 1/2 (3) 1 1/2 (4) 3 1/15 (5) 5 5/12 (6) 11 1/9.

Exercise 3.3 — Multiplication with Brackets

Question: Solve: (1) 5/6 × 2 1/4 × 16/5 (2) 7/9 × (2 1/4 × 8/7) (3) (4/5 × 10/3) × 9/8 (4) (3/4 × 16/5) × 1 2/3 (5) (1 3/5 × 10 1/2) × 5/21 (6) (1 1/6 × 5/6) × 5 1/7 (7) 1 5/16 × (12 1/2 × 1 11/21) (8) 2 5/6 × (1 3/17 × 2 1/10) (9) (3 9/10 × 20/36) × 1 11/13 (10) (2 4/5 × 1 5/7) × 2 1/12 (11) 4 2/7 × (2 5/8 × 1 1/9) (12) 8 1/3 × (2 1/10 × 1/7)

Answer: Step 1: Solve the brackets first, change mixed fractions into

improper fractions, cancel and multiply.

Step 2: (1) 5/6 × 9/4 × 16/5: after cancelling = 6. (2) Bracket: 9/4 × 8/7 = 18/7; 7/9 × 18/7 = 2. (3) Bracket: 4/5 × 10/3 = 8/3; 8/3 × 9/8 = 3. (4) Bracket: 3/4 × 16/5 = 12/5; 12/5 × 5/3 = 4.

Step 3: (5) Bracket: 8/5 × 21/2 = 84/5; 84/5 × 5/21 = 4. (6) Bracket: 7/6 × 5/6 = 35/36; 35/36 × 36/7 = 5. (7) Bracket: 25/2 × 32/21 = 400/21; 21/16 × 400/21 = 25. (8) Bracket: 20/17 × 21/10 = 42/17; 17/6 × 42/17 = 7.

Step 4: (9) Bracket: 39/10 × 20/36 = 13/6; 13/6 × 24/13 = 4. (10) Bracket: 14/5 × 12/7 = 24/5; 24/5 × 25/12 = 10. (11) Bracket: 21/8 × 10/9 = 35/12; 30/7 × 35/12 = 25/2 = 12 1/2. (12) Bracket: 21/10 × 1/7 = 3/10; 25/3 × 3/10 = 5/2 = 2 1/2.

Final answer: (1) 6 (2) 2 (3) 3 (4) 4 (5) 4 (6) 5 (7) 25 (8) 7 (9) 4 (10) 10 (11) 12 1/2 (12) 2 1/2.

Exercise 3.4 — Verify the Distributive Laws

Question: A. Verify the distributive law of multiplication over addition in the following: (1) 2/5 × (3/7 + 4/5) = (2/5 × 3/7) + (2/5 × 4/5) (2) 7/9 × (1/4 + 1/3) = (7/9 × 1/4) + (7/9 × 1/3) (3) 2/7 × (3/8 + 1/5) = (2/7 + 3/8) + (2/7 × 1/5) [as printed in the book] (4) (1/9 + 4/9) × 3/4 = (1/9 × 3/4) + (4/9 × 3/4)

Answer: Step 1: Solve the left hand side (LHS) and the right hand side (RHS) of each question separately.

Step 2: (1) LHS: bracket 3/7 + 4/5 = 43/35; 2/5 × 43/35 = 86/175. RHS: 6/35 + 8/25 = 30/175 + 56/175 = 86/175. Both equal ✓. (2) LHS: 1/4 + 1/3 = 7/12; 7/9 × 7/12 = 49/108. RHS: 7/36 + 7/27 = 21/108 + 28/108 = 49/108 ✓.

Step 3: (3) In the book, the first bracket of the RHS is printed as (2/7 +

3/8), which is a printing mistake; the correct form is (2/7 × 3/8). LHS: 3/8 + 1/5 = 23/40; 2/7 × 23/40 = 23/140. Correct RHS: 6/56 + 2/35 = 3/28 + 2/35 = 15/140 + 8/140 = 23/140 ✓. (4) LHS: 5/9 × 3/4 = 15/36 = 5/12. RHS: 3/36 + 12/36 = 15/36 = 5/12 ✓.

Final answer: In all of them LHS = RHS, so the law is verified; in (3) the book's printing mistake is corrected and shown.

Question: B. Verify the distributive law of multiplication over subtraction in the following: (1) 1/2 × (3/4 − 2/3) = (1/2 × 3/4) − (1/2 × 2/3) (2) 1/5 × (1/2 − 1/3) = (1/5 × 1/2) − (1/5 × 1/3) (3) (4/5 − 3/4) × 2/3 = (4/5 × 2/3) − (3/4 × 2/3) (4) 5/6 × (4/7 − 1/2) = (5/6 × 4/7) − (5/6 × 1/2)

Answer: Step 1: Solve both sides separately.

Step 2: (1) LHS: 3/4 − 2/3 = 1/12; 1/2 × 1/12 = 1/24. RHS: 3/8 − 1/3 = 9/24 − 8/24 = 1/24 ✓. (2) LHS: 1/2 − 1/3 = 1/6; 1/5 × 1/6 = 1/30. RHS: 1/10 − 1/15 = 3/30 − 2/30 = 1/30 ✓.

Step 3: (3) LHS: 4/5 − 3/4 = 1/20; 1/20 × 2/3 = 1/30. RHS: 8/15 − 1/2 = 16/30 − 15/30 = 1/30 ✓. (4) LHS: 4/7 − 1/2 = 1/14; 5/6 × 1/14 = 5/84. RHS: 10/21 − 5/12 = 40/84 − 35/84 = 5/84 ✓.

Final answer: In all of them LHS = RHS, so the law is verified.

Exercise 3.5 — Word Problems on Multiplication

Question: (1) The height of a door is 2 2/3 metres. Out of which 1/8 part was trimmed off. How much part was trimmed off? (2) On Monday one-tenth of the total number of students were on leave. If the total number of students enrolled is 550, how many students were present?

Answer: Step 1: (1) The trimmed part = 1/8 of 2 2/3. 2 2/3 = 8/3.

Step 2: (1) 8/3 × 1/8 = 1/3 metre.

Step 3: (2) Absent = 1/10 × 550 = 55. Present = 550 − 55 = 495.

Final answer: (1) 1/3 metre (2) Absent 55, present 495 students.

Question: (3) A surgeon advised an operating room technician to arrange 12 drips of dextrose solution. The hospital has only 2/3 of the required dextrose. How many drips are available in the hospital? (4) A lady used 1/5 of 50 kg of wheat for making bread. If she makes bread twice a day, then how much wheat will be used?

Answer: Step 1: (3) Available drips = 2/3 of 12.

Step 2: (3) 12 × 2/3 = 24/3 = 8 drips.

Step 3: (4) Once = 1/5 × 50 = 10 kg. Twice a day = 10 × 2 = 20 kg.

Final answer: (3) 8 drips (4) 10 kg once, 20 kg twice daily.

Question: (5) Find the price of 13 dozen of eggs; if the price of one egg is Rs 9 1/2. (6) A lady bought 5 2/5 metres of cloth. She used 1/3 of the cloth in making a TV cover. How much cloth did she use?

Answer: Step 1: (5) 13 dozen = 13 × 12 = 156 eggs. Each egg = 19/2 rupees.

Step 2: (5) 156 × 19/2 = 78 × 19 = Rs 1,482.

Step 3: (6) Used = 1/3 of 5 2/5 = 1/3 × 27/5 = 27/15 = 9/5 = 1 4/5 metres.

Final answer: (5) Rs 1,482 (6) 1 4/5 metres.

Question: (7) A cooler holds 8 bottles of water. If one bottle is holding 2 1/2 litre water, find the capacity of the full cooler. (8) What is the total length of the cloth, if it consists of 12 pieces of cloth and each piece is of length 1 1/2 metres?

Answer: Step 1: (7) Capacity = 8 × 2 1/2.

Step 2: (7) 8 × 5/2 = 40/2 = 20 litres.

Step 3: (8) Total length = 12 × 1 1/2 = 12 × 3/2 = 36/2 = 18 metres.

Final answer: (7) 20 litres (8) 18 metres.

Question: (9) The cost of 1 kg potatoes is Rs 45. Find the cost of 15 2/4 kg of those potatoes. (10) A piece of wire is 8 1/3 metre long. Find the total length of wire, if there are 13 1/5 such small pieces.

Answer: Step 1: (9) 15 2/4 = 31/2 kg. Cost = 45 × 31/2.

Step 2: (9) 45 × 31 = 1,395; 1,395 ÷ 2 = Rs 697.50.

Step 3: (10) 8 1/3 = 25/3 and 13 1/5 = 66/5. Total = 25/3 × 66/5 = 1,650/15 = 110 metres.

Final answer: (9) Rs 697.50 (10) 110 metres.

Exercise 3.6 — Reciprocals and Division

Question: A. Write the reciprocals of each of the following: (1) 2 (2) 4 (3) 1/3 (4) 1/5 (5) 1/10 (6) 3/4 (7) 2 6/7 (8) 2 3/5 (9) 2 1/3 (10) 4 2/3

Answer: Step 1: In the reciprocal, the numerator and denominator change places. First change a mixed fraction into an improper fraction.

Step 2: (1) 2 = 2/1, reciprocal 1/2. (2) 4 → 1/4. (3) 1/3 → 3. (4) 1/5 → 5. (5) 1/10 → 10.

Step 3: (6) 3/4 → 4/3. (7) 2 6/7 = 20/7 → 7/20. (8) 2 3/5 = 13/5 → 5/13. (9) 2 1/3 = 7/3 → 3/7. (10) 4 2/3 = 14/3 → 3/14.

Final answer: (1) 1/2 (2) 1/4 (3) 3 (4) 5 (5) 10 (6) 4/3 (7) 7/20 (8) 5/13 (9) 3/7 (10) 3/14.

Question: B. Solve: (1) 1/2 ÷ 5/6 (2) 2/3 ÷ 4 (3) 3/4 ÷ 6 (4) 4/5 ÷ 8 (5) 5/6 ÷ 10 (6) 4/5 ÷ 12 (7) 7/8 ÷ 14 (8) 8/9 ÷ 24 (9) 2 2/5 ÷ 36 (10) 1 1/9 ÷ 20 (11) 2 1/5 ÷ 55 (12) 3 1/3 ÷ 5

Answer: Step 1: Change the division into multiplication by the reciprocal of the second number.

Step 2: (1) 1/2 × 6/5 = 6/10 = 3/5. (2) 2/3 × 1/4 = 2/12 = 1/6. (3) 3/4 ×

1/6 = 3/24 = 1/8. (4) 4/5 × 1/8 = 4/40 = 1/10. (5) 5/6 × 1/10 = 5/60 = 1/12. (6) 4/5 × 1/12 = 4/60 = 1/15.

Step 3: (7) 7/8 × 1/14 = 7/112 = 1/16. (8) 8/9 × 1/24 = 8/216 = 1/27. (9) 12/5 × 1/36 = 12/180 = 1/15. (10) 10/9 × 1/20 = 10/180 = 1/18. (11) 11/5 × 1/55 = 11/275 = 1/25. (12) 10/3 × 1/5 = 10/15 = 2/3.

Final answer: (1) 3/5 (2) 1/6 (3) 1/8 (4) 1/10 (5) 1/12 (6) 1/15 (7) 1/16 (8) 1/27 (9) 1/15 (10) 1/18 (11) 1/25 (12) 2/3.

Exercise 3.7 — Dividing a Fraction by a Fraction

Question: Solve the following: (1) 1/2 ÷ 3/4 (2) 7/6 ÷ 4/9 (3) 3/4 ÷ 15/16 (4) 9/7 ÷ 81/14 (5) 1/6 ÷ 1/12 (6) 11/22 ÷ 11/22 (7) 16/9 ÷ 4/3 (8) 20/30 ÷ 40/30

Answer: Step 1: Take the reciprocal of the second fraction and multiply.

Step 2: (1) 1/2 × 4/3 = 4/6 = 2/3. (2) 7/6 × 9/4 = 63/24 = 2 5/8. (3) 3/4 × 16/15 = 48/60 = 4/5. (4) 9/7 × 14/81 = 126/567 = 2/9.

Step 3: (5) 1/6 × 12/1 = 2. (6) 11/22 × 22/11 = 1. (7) 16/9 × 3/4 = 48/36 = 1 1/3. (8) 20/30 × 30/40 = 600/1,200 = 1/2.

Final answer: (1) 2/3 (2) 2 5/8 (3) 4/5 (4) 2/9 (5) 2 (6) 1 (7) 1 1/3 (8) 1/2.

Question: Solve: (9) 1 1/4 ÷ 1/4 (10) 2/3 ÷ 4 3/4 (11) 8 1/2 ÷ 3 1/2 (12) 2 1/5 ÷ 1 1/6 (13) 9 3/5 ÷ 2 1/4 (14) 3 8/9 ÷ 1 1/9 (15) 10 1/9 ÷ 4 3/4

Answer: Step 1: Change mixed fractions into improper fractions, then multiply by the reciprocal.

Step 2: (9) 5/4 × 4/1 = 5. (10) 2/3 × 4/19 = 8/57. (11) 17/2 × 2/7 = 17/7 = 2 3/7. (12) 11/5 × 6/7 = 66/35 = 1 31/35.

Step 3: (13) 48/5 × 4/9 = 192/45 = 4 4/15. (14) 35/9 × 9/10 = 35/10 = 3 1/2. (15) 91/9 × 4/19 = 364/171 = 2 22/171.

Final answer: (9) 5 (10) 8/57 (11) 2 3/7 (12) 1 31/35 (13) 4 4/15 (14) 3 1/2

(15) 2 22/171.

Exercise 3.8 — Word Problems on Division

Question: (1) Talib buys 10 1/2 kg tomatoes for Rs 210. What is the price of 1 kg of tomatoes? (2) One-fifth of the 35 cars in the parking area are blue. How many blue cars are there in the parking area?

Answer: Step 1: (1) Price of 1 kg = 210 ÷ 10 1/2 = 210 ÷ 21/2.

Step 2: (1) 210 × 2/21 = 420/21 = Rs 20.

Step 3: (2) Blue cars = 1/5 × 35 = 7.

Final answer: (1) Rs 20 per kg (2) 7 blue cars.

Question: (3) Naila buys 7/3 m piece of a lace. She wants to make its small pieces at length 1/12 m. How many pieces can be made? (4) The camp cook made 1 3/4 kg of baked beans. Each serving of beans is 1/4 kg. How many servings of beans did the cook make?

Answer: Step 1: (3) Number of pieces = 7/3 ÷ 1/12.

Step 2: (3) 7/3 × 12/1 = 84/3 = 28 pieces.

Step 3: (4) Servings = 1 3/4 ÷ 1/4 = 7/4 × 4/1 = 7.

Final answer: (3) 28 pieces (4) 7 servings.

Question: (5) A road is 50 1/2 metre long. Half of it was damaged in rain. How much road is safe? (6) Rehan bought 82 1/2 metre cloth. He used this cloth for making uniform of 15 children of equal size. How much cloth is used in each dress?

Answer: Step 1: (5) Damaged = half = 50 1/2 ÷ 2; the safe part is also half.

Step 2: (5) 101/2 × 1/2 = 101/4 = 25 1/4 metres safe.

Step 3: (6) Each dress = 82 1/2 ÷ 15 = 165/2 × 1/15 = 165/30 = 11/2 = 5 1/2 metres.

Final answer: (5) 25 1/4 metres (6) 5 1/2 metres per dress.

Question: (7) A plastic drum holds 49 1/2 litres drinking water. How many bottles holding 1 1/2 litres of drinking water can be filled from it? (8) A boy completes 8 3/4 km in 2 1/2 hours on bicycle in a race. How much distance does he complete in 1 hour?

Answer: Step 1: (7) Bottles = 49 1/2 ÷ 1 1/2 = 99/2 ÷ 3/2.

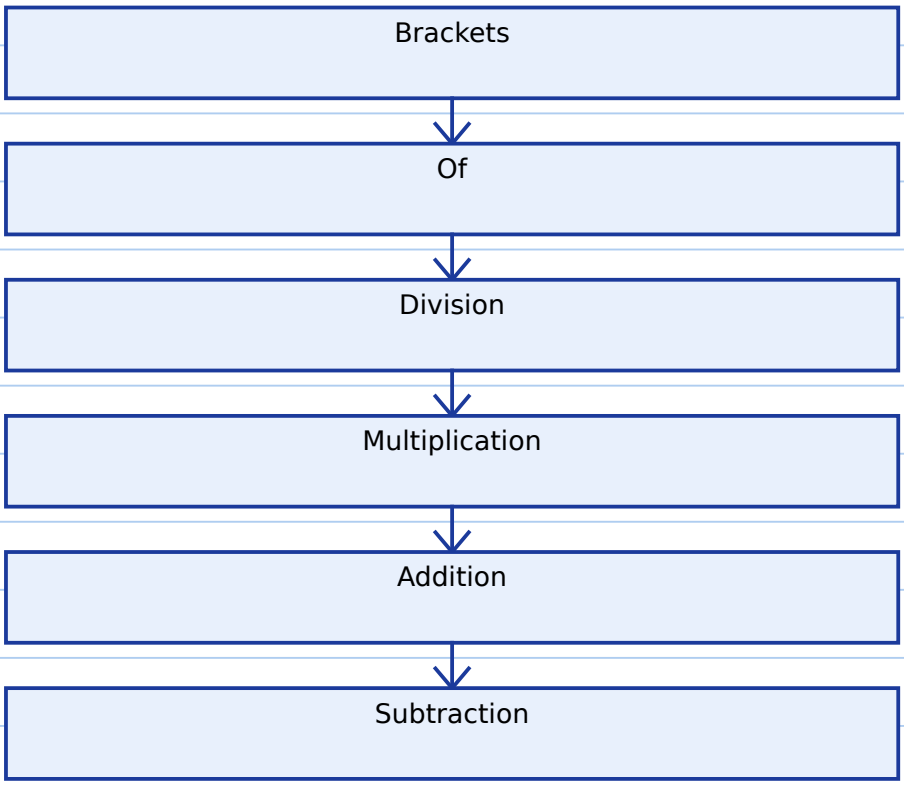
Step 2: (7) 99/2 × 2/3 = 198/6 = 33 bottles.

Step 3: (8) Distance in 1 hour = 8 3/4 ÷ 2 1/2 = 35/4 ÷ 5/2 = 35/4 × 2/5 = 70/20 = 3 1/2 km.

Final answer: (7) 33 bottles (8) 3 1/2 km per hour.

Exercise 3.9 — Simplify Fractions Using BODMAS Rule

BODMAS order



Question: Simplify by using BODMAS rule: (1) 3/4 + 2/9 × 4 1/3 ÷ 3 1/4 (2) 1/4 × (8/3 + 2/7) (3) 1/2 + (3/4 × 1 7/33 − 1/3) (4) (3 1/6 − 1 1/4) × 2

Answer: Step 1: BODMAS order: brackets first, then Of, Division, Multiplication, Addition, Subtraction.

Step 2: (1) Multiplication and division first: 2/9 × 13/3 ÷ 13/4 = 2/9 × 13/3 × 4/13 = 8/27; then 3/4 + 8/27 = 81/108 + 32/108 = 113/108 = 1 5/108. (2) Bracket: 8/3 + 2/7 = 62/21; 1/4 × 62/21 = 62/84 = 31/42.

Step 3: (3) Inside the bracket, multiplication first: 3/4 × 40/33 = 10/11; 10/11 − 1/3 = 19/33; 1/2 + 19/33 = 33/66 + 38/66 = 71/66 = 1 5/66. (4) Bracket: 19/6 − 5/4 = 38/12 − 15/12 = 23/12; 23/12 × 2 = 23/6 = 3 5/6.

Final answer: (1) 1 5/108 (2) 31/42 (3) 1 5/66 (4) 3 5/6.

Question: Simplify: (5) (4/5 − 3/10) × (1/2 + 3/4) (6) (2 1/2 ÷ 3/4) × 3/7 − (1/4 + 1/8) (7) 1 3/5 × (4/3 − 3/4 + 2 1/3) ÷ 1 2/7 (8) 1 1/6 + (2 3/4 × 3 1/3 ÷ 2 1/4) − 7 1/2

Answer: Step 1: Solve both brackets first.

Step 2: (5) First bracket: 4/5 − 3/10 = 1/2; second: 1/2 + 3/4 = 5/4; multiply: 1/2 × 5/4 = 5/8. (6) 2 1/2 ÷ 3/4 = 5/2 × 4/3 = 10/3; 10/3 × 3/7 = 10/7; bracket: 1/4 + 1/8 = 3/8; 10/7 − 3/8 = 80/56 − 21/56 = 59/56 = 1 3/56.

Step 3: (7) Bracket: 4/3 − 3/4 + 7/3 = 16/12 − 9/12 + 28/12 = 35/12; 8/5 × 35/12 = 14/3; 14/3 ÷ 9/7 = 14/3 × 7/9 = 98/27 = 3 17/27. (8) Bracket: 11/4 × 10/3 = 55/6; 55/6 ÷ 9/4 = 55/6 × 4/9 = 110/27; 7/6 + 110/27 − 15/2 = 63/54 + 220/54 − 405/54 = −122/54 = −61/27 = −2 7/27.

Final answer: (5) 5/8 (6) 1 3/56 (7) 3 17/27 (8) −2 7/27.

Question: Simplify: (9) (1 3/5 of 5/6) − (2 3/7 + 1 1/5) (10) (5/4 + 8/3) × (10/3 + 5/2) (11) 20 + 5 of 9 − (1 2/3 × 1/5) (12) 4 1/2 + (5 1/3 of 3) − 2 2/3

Answer: Step 1: Solve "of" before multiplication.

Step 2: (9) "Of": 8/5 × 5/6 = 4/3; bracket: 17/7 + 6/5 = 85/35 + 42/35 =

127/35; 4/3 − 127/35 = 140/105 − 381/105 = −241/105 = −2 31/105.

(10) Brackets: 5/4 + 8/3 = 47/12 and 10/3 + 5/2 = 35/6; multiply: 47/12 × 35/6 = 1,645/72 = 22 61/72.

Step 3: (11) "Of": 5 × 9 = 45; bracket: 5/3 × 1/5 = 1/3; 20 + 45 − 1/3 = 65 − 1/3 = 64 2/3. (12) "Of": 16/3 × 3 = 16; 9/2 + 16 − 8/3 = 27/6 + 96/6 − 16/6 = 107/6 = 17 5/6.

Final answer: (9) −2 31/105 (10) 22 61/72 (11) 64 2/3 (12) 17 5/6.

Review Exercise 3

Question: (1) Write 'T' for true and 'F' for false: (i) 4 2/3 is an example of an improper fraction. (ii) 1/3 ÷ 3/1 = 1/9 (iii) The lowest form of 12/4 = 1/3. (iv) 3/4 − 1/2 = 1/4 (v) 1/3, 2/6, 3/9 are equivalent fractions. (vi) The reciprocal of 2 1/2 is 2/5. (vii) 3/4 of a dozen is 9. (viii) 1 1/2 m is equal to 150 cm.

Answer: (i) ✗ False — 4 2/3 is a mixed fraction, not an improper fraction; its improper form is 14/3.

(ii) ✓ True — 1/3 ÷ 3 = 1/3 × 1/3 = 1/9.

(iii) ✗ False — the lowest form of 12/4 is 3, not 1/3.

(iv) ✓ True — 3/4 − 2/4 = 1/4.

(v) ✓ True — all three fractions are equal to 1/3.

(vi) ✓ True — 2 1/2 = 5/2, its reciprocal is 2/5.

(vii) ✓ True — 3/4 × 12 = 9.

(viii) ✓ True — 1 1/2 m = 150 cm.

Final answer: (i) F (ii) T (iii) F (iv) T (v) T (vi) T (vii) T (viii) T.

Question: (2) Solve the following: (i) Five ball points are bought for Rs 50 3/4. Find the price of one ball point. (ii) Afzal and Fazila bought a chocolate

bar. Afzal got 2/5 and Fazila got 3/10. Who got more and by how much?

Answer: Step 1: (i) Price of one = 50 3/4 ÷ 5 = 203/4 × 1/5.

Step 2: (i) 203/20 = Rs 10 3/20.

Step 3: (ii) Afzal: 2/5 = 4/10; Fazila: 3/10. 4/10 > 3/10, so Afzal got more. Difference = 4/10 − 3/10 = 1/10.

Final answer: (i) Rs 10 3/20 (ii) Afzal got more by 1/10.

Question: (iii) Find the price of 2 dozen packets of biscuit, if the price of one packet is Rs 12 1/2. (iv) Shumaila buys 2 1/4 m ribbon. She used 1 7/8 m of it. How much ribbon is left?

Answer: Step 1: (iii) 2 dozen = 24 packets. Price = 24 × 12 1/2 = 24 × 25/2.

Step 2: (iii) 24 × 25/2 = 12 × 25 = Rs 300.

Step 3: (iv) Left = 2 1/4 − 1 7/8 = 9/4 − 15/8 = 18/8 − 15/8 = 3/8 m.

Final answer: (iii) Rs 300 (iv) 3/8 m left.

Question: (v) 75 3/4 kg of tea is packed into packets. Each packet contains 3/8 kg of tea. How many packets are needed to fill the tea? (vi) At a charity show each student in a school purchases a ticket costing Rs 30. If 3/4 of the amount collected is Rs 24,600, find the number of tickets sold.

Answer: Step 1: (v) Packets = 75 3/4 ÷ 3/8 = 303/4 × 8/3.

Step 2: (v) 303/4 × 8/3 = 2,424/12 = 202 packets.

Step 3: (vi) Each ticket costs Rs 30; the given amount Rs 24,600 came from ticket sales, so tickets = 24,600 ÷ 30 = 820.

Final answer: (v) 202 packets (vi) 820 tickets.

Question: (vii) Simplify: (a) 2 1/3 − (16/5 ÷ 1 7/8 of 2 2/15) + 1 1/8 (b) 3/4 + 3/4 of 3/4 ÷ 3/4 − (3/4 × 3/4)

Answer: Step 1: BODMAS: brackets and "of" first.

Step 2: (a) "Of": 15/8 × 32/15 = 4; bracket: 16/5 ÷ 4 = 16/5 × 1/4 = 4/5; now 7/3 − 4/5 + 9/8; LCM 120: 280/120 − 96/120 + 135/120 = 319/120 = 2 79/120.

Step 3: (b) "Of": 3/4 × 3/4 = 9/16; division: 9/16 ÷ 3/4 = 9/16 × 4/3 = 3/4; bracket: 3/4 × 3/4 = 9/16; 3/4 + 3/4 − 9/16 = 24/16 − 9/16 = 15/16.

Final answer: (a) 2 79/120 (b) 15/16.

ڈھائي عدد سیکڑو4یونٹ :

Exercise 4.1 — Addition and Subtraction of Decimals

Question: A. Add the following: (1) 22.32 + 6.46 (2) 4.567 + 36.4 (3) 75.05 + 24.62 (4) 257.003 + 0.25 (5) 40.123 + 7.32 (6) 45.005 + 52.47

Answer: Step 1: Keep the decimal points exactly below each other; write 0 in empty places, then add column by column.

Step 2: (1) 22.32 + 6.46 = 28.78. (2) 4.567 + 36.400 = 40.967. (3) 75.05 + 24.62 = 99.67.

Step 3: (4) 257.003 + 0.250 = 257.253. (5) 40.123 + 7.320 = 47.443. (6) 45.005 + 52.470 = 97.475.

Final answer: (1) 28.78 (2) 40.967 (3) 99.67 (4) 257.253 (5) 47.443 (6) 97.475.

Question: Add: (7) 345.38 + 786.46 (8) 674.567 + 36.48 (9) 45.75 + 54.69 (10) 287.099 + 8.258 (11) 45.468 + 277.358 (12) 35.69 + 875.875

Answer: Step 1: Keep the points in one line and add column by column.

Step 2: (7) 345.38 + 786.46 = 1,131.84. (8) 674.567 + 36.480 = 711.047. (9) 45.75 + 54.69 = 100.44.

Step 3: (10) 287.099 + 8.258 = 295.357. (11) 45.468 + 277.358 = 322.826. (12) 35.690 + 875.875 = 911.565.

Final answer: (7) 1,131.84 (8) 711.047 (9) 100.44 (10) 295.357 (11) 322.826 (12) 911.565.

Question: B. Subtract the following: (1) 25.52 − 6.3 (2) 74.567 − 33.402 (3) 75.75 − 24.62 (4) 257.003 − 0.25 (5) 49.123 − 7.02 (6) 757.785 − 152.005

Answer: Step 1: Keep the points below each other, write 0 in empty places and subtract column by column.

Step 2: (1) 25.52 − 6.30 = 19.22. (2) 74.567 − 33.402 = 41.165. (3) 75.75 − 24.62 = 51.13.

Step 3: (4) 257.003 − 0.250 = 256.753. (5) 49.123 − 7.020 = 42.103. (6) 757.785 − 152.005 = 605.78.

Final answer: (1) 19.22 (2) 41.165 (3) 51.13 (4) 256.753 (5) 42.103 (6) 605.78.

Question: Subtract: (7) 786.46 − 345.38 (8) 674.567 − 36.48 (9) 85.75 − 54.65 (10) 287.099 − 174.055 (11) 845.468 − 234.358 (12) 935.69 − 805.365

Answer: Step 1: Same method: keep the points in line and subtract.

Step 2: (7) 786.46 − 345.38 = 441.08. (8) 674.567 − 36.480 = 638.087. (9) 85.75 − 54.65 = 31.10.

Step 3: (10) 287.099 − 174.055 = 113.044. (11) 845.468 − 234.358 = 611.11. (12) 935.690 − 805.365 = 130.325.

Final answer: (7) 441.08 (8) 638.087 (9) 31.10 (10) 113.044 (11) 611.11 (12) 130.325.

Exercise 4.2 — Decimal Places and Multiplying by 10, 100, 1000

Question: A. Separate the decimals of one, two and three decimal places from the following: 0.08, 2.123, 34.25, 0.6, 3.36, 52.302, 38.66 and 62.1.

B. Find the product of the following by using the rules of multiplication: (1) 0.175 x 10 (2) 0.175 x 100 (3) 0.175 x 1000 (4) 35.058 x 10 (5) 35.058 x 100 (6) 35.058 x 1000 (7) 8.15 x 10 (8) 8.15 x 100 (9) 8.15 x 1000 (10) 324.423 x 10 (11) 324.423 x 100 (12) 324.423 x 1000 (13) 0.0067 x 10 (14) 0.0067 x 100 (15) 0.0067 x 1000

Answer: Step 1: In A, count the digits after the point: 1 digit = one decimal place, 2 digits = two places, 3 digits = three places.

Step 2: A: One place: 0.6, 62.1. Two places: 0.08, 34.25, 3.36, 38.66. Three places: 2.123, 52.302.

Step 3: Rule for B: on x10 the point moves one place right, on x100 two places, on x1000 three places right. (1) 1.75 (2) 17.5 (3) 175 (4) 350.58 (5) 3,505.8 (6) 35,058.

Step 4: (7) 81.5 (8) 815 (9) 8,150 (10) 3,244.23 (11) 32,442.3 (12) 324,423 (13) 0.067 (14) 0.67 (15) 6.7.

Final answer: A: One place: 0.6, 62.1; two places: 0.08, 34.25, 3.36, 38.66; three places: 2.123, 52.302. B: (1) 1.75 (2) 17.5 (3) 175 (4) 350.58 (5) 3,505.8 (6) 35,058 (7) 81.5 (8) 815 (9) 8,150 (10) 3,244.23 (11) 32,442.3 (12) 324,423 (13) 0.067 (14) 0.67 (15) 6.7.

Exercise 4.3 — Divide by 10, 100 and 1000

Question: A. Solve the following by using the rules of division: (1) 6.675 ÷ 10 (2) 6.675 ÷ 100 (3) 6.675 ÷ 1000 (4) 35.89 ÷ 10 (5) 35.89 ÷ 100 (6) 35.89 ÷ 1000 (7) 815.4 ÷ 10 (8) 815.4 ÷ 100 (9) 815.4 ÷ 1000 (10) 0.085 ÷ 10 (11) 0.085 ÷ 100 (12) 0.085 ÷ 1000

Answer: Step 1: Rule: on ÷10 the point moves one place left, on ÷100 two places, on ÷1000 three places left; add zeros in front when needed.

Step 2: (1) 0.6675 (2) 0.06675 (3) 0.006675 (4) 3.589 (5) 0.3589 (6) 0.03589.

Step 3: (7) 81.54 (8) 8.154 (9) 0.8154 (10) 0.0085 (11) 0.00085 (12) 0.000085.

Final answer: (1) 0.6675 (2) 0.06675 (3) 0.006675 (4) 3.589 (5) 0.3589 (6) 0.03589 (7) 81.54 (8) 8.154 (9) 0.8154 (10) 0.0085 (11) 0.00085 (12)

0.000085.

Exercise 4.4 — Multiply a Decimal by a Whole Number

Question: Find the product: (1) 6.5 x 5 (2) 3.5 x 6 (3) 0.65 x 5 (4) 0.65 x 6 (5) 0.35 x 8 (6) 4.25 x 7 (7) 4.25 x 8 (8) 4.25 x 9 (9) 0.382 x 5

Answer: Step 1: First multiply without the point, then count as many decimal places as in the given number and place the point.

Step 2: (1) 65 x 5 = 325 → 32.5. (2) 35 x 6 = 210 → 21. (3) 65 x 5 = 325 → 3.25. (4) 65 x 6 = 390 → 3.9.

Step 3: (5) 35 x 8 = 280 → 2.8. (6) 425 x 7 = 2,975 → 29.75. (7) 425 x 8 = 3,400 → 34. (8) 425 x 9 = 3,825 → 38.25. (9) 382 x 5 = 1,910 → 1.91.

Final answer: (1) 32.5 (2) 21 (3) 3.25 (4) 3.9 (5) 2.8 (6) 29.75 (7) 34 (8) 38.25 (9) 1.91.

Question: Find the product: (10) 0.394 x 8 (11) 24.58 x 7 (12) 53.69 x 9 (13) 4.25 x 27 (14) 0.265 x 36 (15) 2.785 x 435 (16) 4.25 x 162 (17) 47.326 x 348 (18) 58.967 x 564

Answer: Step 1: Same method: multiply without the point, then count the decimal places and place the point.

Step 2: (10) 394 x 8 = 3,152 → 3.152. (11) 2,458 x 7 = 17,206 → 172.06. (12) 5,369 x 9 = 48,321 → 483.21. (13) 425 x 27 = 11,475 → 114.75.

Step 3: (14) 265 x 36 = 9,540 → 9.54. (15) 2,785 x 435 = 1,211,475 → 1,211.475. (16) 425 x 162 = 68,850 → 688.5. (17) 47,326 x 348 = 16,469,448 → 16,469.448. (18) 58,967 x 564 = 33,257,388 → 33,257.388.

Final answer: (10) 3.152 (11) 172.06 (12) 483.21 (13) 114.75 (14) 9.54 (15) 1,211.475 (16) 688.5 (17) 16,469.448 (18) 33,257.388.

Exercise 4.5 — Divide a Decimal by a Whole Number

Question: A. Solve: (1) 0.65 ÷ 5 (2) 6.5 ÷ 5 (3) 0.065 ÷ 5 (4) 3.6 ÷ 6 (5)

0.64 ÷ 8 (6) 4.27 ÷ 7 (7) 9.6 ÷ 8 (8) 44.1 ÷ 9 (9) 0.385 ÷ 5 (10) 8.394 ÷ 6 (11) 39.851 ÷ 7 (12) 87.03 ÷ 9

Answer: Step 1: Divide as in ordinary division; when you reach the decimal point, put the point in the answer at the same place.

Step 2: (1) 65 ÷ 5 = 13 → 0.13. (2) 6.5 ÷ 5 = 1.3. (3) 0.065 ÷ 5 = 0.013. (4) 3.6 ÷ 6 = 0.6. (5) 0.64 ÷ 8 = 0.08. (6) 4.27 ÷ 7 = 0.61.

Step 3: (7) 9.6 ÷ 8 = 1.2. (8) 44.1 ÷ 9 = 4.9. (9) 0.385 ÷ 5 = 0.077. (10) 8.394 ÷ 6 = 1.399. (11) 39.851 ÷ 7 = 5.693. (12) 87.03 ÷ 9 = 9.67.

Final answer: (1) 0.13 (2) 1.3 (3) 0.013 (4) 0.6 (5) 0.08 (6) 0.61 (7) 1.2 (8) 4.9 (9) 0.077 (10) 1.399 (11) 5.693 (12) 9.67.

Question: B. Solve up to 3-decimal places: (1) 40.25 ÷ 4 (2) 89.64 ÷ 6 (3) 98.58 ÷ 3 (4) 24.25 ÷ 10 (5) 0.265 ÷ 20 (6) 2.785 ÷ 25 (7) 3.16 ÷ 2.31 (8) 0.555 ÷ 5.5 (9) 14.04 ÷ 12.4

Answer: Step 1: Continue the division and stop the answer at 3 decimal places (the book's method: the extra digits are dropped).

Step 2: (1) 40.25 ÷ 4 = 10.0625 → 10.062. (2) 89.64 ÷ 6 = 14.94. (3) 98.58 ÷ 3 = 32.86. (4) 24.25 ÷ 10 = 2.425.

Step 3: (5) 0.265 ÷ 20 = 0.01325 → 0.013. (6) 2.785 ÷ 25 = 0.1114 → 0.111. (7) 3.16 ÷ 2.31 = 1.3679… → 1.367. (8) 0.555 ÷ 5.5 = 0.1009… → 0.100. (9) 14.04 ÷ 12.4 = 1.1322… → 1.132.

Final answer: (1) 10.062 (2) 14.94 (3) 32.86 (4) 2.425 (5) 0.013 (6) 0.111 (7) 1.367 (8) 0.100 (9) 1.132.

Exercise 4.6 — Multiply by Tenths and Hundredths

Question: Find the product of the following: (1) 2.6 x 0.1 (2) 3.87 x 0.2 (3) 42.5 x 0.3 (4) 59.95 x 0.4 (5) 0.3 x 0.5 (6) 0.01 x 0.6 (7) 1.1 x 0.4 (8) 1.05 x 0.2 (9) 2.6 x 0.03 (10) 0.3 x 0.01 (11) 0.01 x 0.01 (12) 1.1 x 0.08 (13)

3.87 x 0.06 (14) 42.5 x 0.05 (15) 59.95 x 0.07

Answer: Step 1: Multiply without the points; then count the total decimal places of both numbers and place the point.

Step 2: (1) 26 x 1 = 26, 2 places → 0.26. (2) 387 x 2 = 774, 3 places → 0.774. (3) 425 x 3 = 1,275, 2 places → 12.75. (4) 5,995 x 4 = 23,980, 3 places → 23.98. (5) 3 x 5 = 15, 2 places → 0.15.

Step 3: (6) 1 x 6 = 6, 3 places → 0.006. (7) 11 x 4 = 44, 2 places → 0.44. (8) 105 x 2 = 210, 3 places → 0.21. (9) 26 x 3 = 78, 3 places → 0.078. (10) 3 x 1 = 3, 3 places → 0.003.

Step 4: (11) 1 x 1 = 1, 4 places → 0.0001. (12) 11 x 8 = 88, 3 places → 0.088. (13) 387 x 6 = 2,322, 4 places → 0.2322. (14) 425 x 5 = 2,125, 3 places → 2.125. (15) 5,995 x 7 = 41,965, 4 places → 4.1965.

Final answer: (1) 0.26 (2) 0.774 (3) 12.75 (4) 23.98 (5) 0.15 (6) 0.006 (7) 0.44 (8) 0.21 (9) 0.078 (10) 0.003 (11) 0.0001 (12) 0.088 (13) 0.2322 (14) 2.125 (15) 4.1965.

Exercise 4.7 — Multiply a Decimal by a Decimal

Question: A. Put the decimal point at the appropriate place in each answer: (1) 46.7 x 1.2 = 5604 (2) 0.754 x 1.2 = 09048 (3) 3.57 x 1.2 = 4284 (4) 68.2 x 1.2 = 8184 (5) 0.402 x 1.2 = 04824 (6) 14.839 x 1.2 = 178068 (7) 361.4 x 3.54 = 1279356 (8) 219.241 x 2.5 = 5481025 (9) 402.58 x 2.3 = 925934

Answer: Step 1: Add the decimal places of both numbers; count that many places from the right in the answer and place the point.

Step 2: (1) 1+1 = 2 places → 56.04. (2) 3+1 = 4 places → 0.9048. (3) 2+1 = 3 places → 4.284. (4) 2 places → 81.84. (5) 4 places → 0.4824.

Step 3: (6) 3+1 = 4 places → 17.8068. (7) 1+2 = 3 places → 1,279.356. (8)

3+1 = 4 places → 548.1025. (9) 2+1 = 3 places → 925.934.

Final answer: (1) 56.04 (2) 0.9048 (3) 4.284 (4) 81.84 (5) 0.4824 (6) 17.8068 (7) 1,279.356 (8) 548.1025 (9) 925.934.

Question: B. Solve: (1) 0.28 x 0.4 (2) 0.45 x 0.5 (3) 0.2 x 0.023 (4) 0.28 x 0.05 (5) 0.065 x 0.25 (6) 0.005 x 0.12 (7) 0.002 x 0.08 (8) 0.105 x 0.09 (9) 0.31 x 0.052 (10) 0.22 x 0.057 (11) 0.25 x 0.05 (12) 0.755 x 0.14

Answer: Step 1: Multiply without the points and count the total decimal places.

Step 2: (1) 28 x 4 = 112, 3 places → 0.112. (2) 45 x 5 = 225, 3 places → 0.225. (3) 2 x 23 = 46, 4 places → 0.0046. (4) 28 x 5 = 140, 4 places → 0.014.

Step 3: (5) 65 x 25 = 1,625, 5 places → 0.01625. (6) 5 x 12 = 60, 5 places → 0.0006. (7) 2 x 8 = 16, 5 places → 0.00016. (8) 105 x 9 = 945, 5 places → 0.00945.

Step 4: (9) 31 x 52 = 1,612, 5 places → 0.01612. (10) 22 x 57 = 1,254, 5 places → 0.01254. (11) 25 x 5 = 125, 4 places → 0.0125. (12) 755 x 14 = 10,570, 5 places → 0.1057.

Final answer: (1) 0.112 (2) 0.225 (3) 0.0046 (4) 0.014 (5) 0.01625 (6) 0.0006 (7) 0.00016 (8) 0.00945 (9) 0.01612 (10) 0.01254 (11) 0.0125 (12) 0.1057.

Question: C. Find the product: (1) 1.4 x 2.5 (2) 1.05 x 2.6 (3) 3.8 x 0.7 (4) 0.3 x 2.01 (5) 4.45 x 1.8 (6) 8.84 x 1.5 (7) 6.24 x 7.1 (8) 5.25 x 4.4 (9) 7.05 x 2.6 (10) 9.06 x 5.5 (11) 25.08 x 8.5 (12) 9.2 x 7.85

Answer: Step 1: Same method.

Step 2: (1) 14 x 25 = 350, 2 places → 3.5. (2) 105 x 26 = 2,730, 3 places → 2.73. (3) 38 x 7 = 266, 2 places → 2.66. (4) 3 x 201 = 603, 3 places → 0.603.

Step 3: (5) 445 x 18 = 8,010, 3 places → 8.01. (6) 884 x 15 = 13,260, 3 places → 13.26. (7) 624 x 71 = 44,304, 3 places → 44.304. (8) 525 x 44 = 23,100, 3 places → 23.1.

Step 4: (9) 705 x 26 = 18,330, 3 places → 18.33. (10) 906 x 55 = 49,830, 3 places → 49.83. (11) 2,508 x 85 = 213,180, 3 places → 213.18. (12) 92 x 785 = 72,220, 3 places → 72.22.

Final answer: (1) 3.5 (2) 2.73 (3) 2.66 (4) 0.603 (5) 8.01 (6) 13.26 (7) 44.304 (8) 23.1 (9) 18.33 (10) 49.83 (11) 213.18 (12) 72.22.

Exercise 4.8 — Divide a Decimal by a Decimal

Question: A. Solve by converting decimals to fractions: (1) 2.16 ÷ 0.6 (2) 5.76 ÷ 0.24 (3) 4.41 ÷ 0.3 (4) 7.84 ÷ 0.07 (5) 0.6017 ÷ 1.1 (6) 4.905 ÷ 4.5 (7) 7.84 ÷ 0.14 (8) 78.4 ÷ 0.7 (9) 10.24 ÷ 0.08 (10) 3.5308 ÷ 0.13 (11) 97.578 ÷ 0.039 (12) 10.26 ÷ 0.18

Answer: Step 1: Change both into fractions (e.g. 2.16 = 216/100 and 0.6 = 6/10), then multiply by the reciprocal; or multiply both by the same number to make the divisor a whole number.

Step 2: (1) 216/60 = 3.6. (2) 576/24 = 24. (3) 441/30 = 14.7. (4) 784/7 = 112. (5) 6,017/11,000 = 0.547. (6) 4,905/4,500 = 1.09.

Step 3: (7) 784/14 = 56. (8) 784/7 = 112. (9) 1,024/8 = 128. (10) 35,308/1,300 = 27.16. (11) 97,578/39 = 2,502. (12) 1,026/18 = 57.

Final answer: (1) 3.6 (2) 24 (3) 14.7 (4) 112 (5) 0.547 (6) 1.09 (7) 56 (8) 112 (9) 128 (10) 27.16 (11) 2,502 (12) 57.

Question: B. Solve the following decimals using direct division method by moving decimal positions: (1) 0.016 ÷ 0.2 (2) 0.18 ÷ 0.3 (3) 0.36 ÷ 0.6 (4) 0.072 ÷ 0.8 (5) 0.121 ÷ 1.1 (6) 0.0169 ÷ 1.3 (7) 0.96 ÷ 0.4 (8) 9.018 ÷ 0.9 (9) 0.036 ÷ 1.2 (10) 0.0289 ÷ 1.7 (11) 0.0144 ÷ 1.2 (12) 0.072 ÷ 0.12

(13) 4.096 ÷ 0.16 (14) 0.325 ÷ 0.25 (15) 0.0196 ÷ 1.4

Answer: Step 1: Move the point in both numbers by the same places to make the divisor a whole number, then do ordinary division.

Step 2: (1) 0.16 ÷ 2 = 0.08. (2) 1.8 ÷ 3 = 0.6. (3) 3.6 ÷ 6 = 0.6. (4) 0.72 ÷ 8 = 0.09. (5) 1.21 ÷ 11 = 0.11.

Step 3: (6) 0.169 ÷ 13 = 0.013. (7) 9.6 ÷ 4 = 2.4. (8) 90.18 ÷ 9 = 10.02. (9) 0.36 ÷ 12 = 0.03. (10) 0.289 ÷ 17 = 0.017.

Step 4: (11) 0.144 ÷ 12 = 0.012. (12) 7.2 ÷ 12 = 0.6. (13) 409.6 ÷ 16 = 25.6. (14) 32.5 ÷ 25 = 1.3. (15) 0.196 ÷ 14 = 0.014.

Final answer: (1) 0.08 (2) 0.6 (3) 0.6 (4) 0.09 (5) 0.11 (6) 0.013 (7) 2.4 (8) 10.02 (9) 0.03 (10) 0.017 (11) 0.012 (12) 0.6 (13) 25.6 (14) 1.3 (15) 0.014.

Exercise 4.9 — Change Fractions into Decimals

Question: Change the following fractions into decimals: (1) 5/4 (2) 5/3 (3) 4/5 (4) 7/10 (5) 15/7 (6) 14/9 (7) 5/8 (8) 23/9 (9) 32/7 (10) 45/13

Answer: Step 1: Divide the numerator by the denominator; like the book, stop the answer at 3 decimal places.

Step 2: (1) 5 ÷ 4 = 1.25. (2) 5 ÷ 3 = 1.666… → 1.666. (3) 4 ÷ 5 = 0.8. (4) 7 ÷ 10 = 0.7. (5) 15 ÷ 7 = 2.142… → 2.142.

Step 3: (6) 14 ÷ 9 = 1.555… → 1.555. (7) 5 ÷ 8 = 0.625. (8) 23 ÷ 9 = 2.555… → 2.555. (9) 32 ÷ 7 = 4.571… → 4.571. (10) 45 ÷ 13 = 3.461… → 3.461.

Final answer: (1) 1.25 (2) 1.666 (3) 0.8 (4) 0.7 (5) 2.142 (6) 1.555 (7) 0.625 (8) 2.555 (9) 4.571 (10) 3.461.

Question: Change the fractions into decimals: (11) 50/15 (12) 9/14 (13) 11/16 (14) 5/12 (15) 17/20 (16) 125/60 (17) 245/26 (18) 250/8 (19) 300/250 (20) 23/25

Answer: Step 1: Same division method.

Step 2: (11) 50 ÷ 15 = 3.333... → 3.333. (12) 9 ÷ 14 = 0.642... → 0.642. (13) 11 ÷ 16 = 0.6875. (14) 5 ÷ 12 = 0.416... → 0.416. (15) 17 ÷ 20 = 0.85.

Step 3: (16) 125 ÷ 60 = 2.083... → 2.083. (17) 245 ÷ 26 = 9.423... → 9.423. (18) 250 ÷ 8 = 31.25. (19) 300 ÷ 250 = 1.2. (20) 23 ÷ 25 = 0.92.

Final answer: (11) 3.333 (12) 0.642 (13) 0.6875 (14) 0.416 (15) 0.85 (16) 2.083 (17) 9.423 (18) 31.25 (19) 1.2 (20) 0.92.

Exercise 4.10 — Simplify Decimal Expressions with Brackets

Question: Simplify: (1) 5.6 + 7.22 − (2.24 + 4.68) (2) (20.14 x 5.6) + 10.9335 (3) 4.6 + 6.07 + (23.35 − 8.30 + 8.34) (4) 14.3 − 2.8 + (1.84 + 3.29) (5) (5.24 + 2.02) − 0.96 − 7.45 + (9.405 − 2.24)

Answer: Step 1: BODMAS: brackets first, then multiplication/division, then addition/subtraction from left to right.

Step 2: (1) Bracket: 2.24 + 4.68 = 6.92; 5.6 + 7.22 = 12.82; 12.82 − 6.92 = 5.9. (2) Multiplication in the bracket: 20.14 x 5.6 = 112.784; 112.784 + 10.9335 = 123.7175. (3) Bracket: 23.35 − 8.30 + 8.34 = 23.39; 4.6 + 6.07 + 23.39 = 34.06.

Step 3: (4) Bracket: 1.84 + 3.29 = 5.13; 14.3 − 2.8 = 11.5; 11.5 + 5.13 = 16.63. (5) Brackets: 5.24 + 2.02 = 7.26 and 9.405 − 2.24 = 7.165; 7.26 − 0.96 − 7.45 + 7.165 = 6.015.

Final answer: (1) 5.9 (2) 123.7175 (3) 34.06 (4) 16.63 (5) 6.015.

Question: Simplify: (6) 5.6 x (25.5 − 12.2) + (2.3 + 2.6) (7) 45.234 + (18.024 − 6.66) − (0.457 + 9.945) (8) 3.45 x 8.56 − 4.23 + (2.2 − 1.12) (9) (6.6 x 3.59) − (1.12 + 0.1) − 1.02 (10) (230.24 + 23.028) − 72.72 ÷ (6.42 + (14.045 − 6.3))

Answer: Step 1: The innermost bracket first.

Step 2: (6) Brackets: 25.5 − 12.2 = 13.3 and 2.3 + 2.6 = 4.9; multiply: 5.6 x 13.3 = 74.48; 74.48 + 4.9 = 79.38. (7) Brackets: 18.024 − 6.66 = 11.364 and 0.457 + 9.945 = 10.402; 45.234 + 11.364 − 10.402 = 46.196.

Step 3: (8) Multiply: 3.45 x 8.56 = 29.532; bracket: 2.2 − 1.12 = 1.08; 29.532 − 4.23 + 1.08 = 26.382. (9) Brackets: 6.6 x 3.59 = 23.694 and 1.12 + 0.1 = 1.22; 23.694 − 1.22 − 1.02 = 21.454.

Step 4: (10) Innermost bracket: 14.045 − 6.3 = 7.745; next bracket: 6.42 + 7.745 = 14.165; first bracket: 230.24 + 23.028 = 253.268; division: 72.72 ÷ 14.165 ≈ 5.134; 253.268 − 5.134 = 248.134.

Final answer: (6) 79.38 (7) 46.196 (8) 26.382 (9) 21.454 (10) 248.134.

Exercise 4.11 — Round off Decimals

Question: 1. Round off the following decimals as whole numbers: (1) 2.3 (2) 5.6 (3) 7.7 (4) 6.6 (5) 9.9 (6) 8.3 (7) 7.8 (8) 50.2 (9) 58.6 (10) 78.2 (11) 81.7 (12) 99.9

Answer: Step 1: Rule: if the tenths digit is 5 or more, round up; if it is less than 5, keep it down.

Step 2: (1) 2.3 → 2 (2) 5.6 → 6 (3) 7.7 → 8 (4) 6.6 → 7 (5) 9.9 → 10 (6) 8.3 → 8.

Step 3: (7) 7.8 → 8 (8) 50.2 → 50 (9) 58.6 → 59 (10) 78.2 → 78 (11) 81.7 → 82 (12) 99.9 → 100.

Final answer: (1) 2 (2) 6 (3) 8 (4) 7 (5) 10 (6) 8 (7) 8 (8) 50 (9) 59 (10) 78 (11) 82 (12) 100.

Question: 2. Round off the following decimals up to one decimal place: (1) 32.38 (2) 25.156 (3) 6.17 (4) 6.42 (5) 76.798 (6) 95.24 (7) 12.86 (8) 5.95 (9) 3.432 (10) 11.7681 (11) 50.4752 (12) 60.1536

Answer: Step 1: Look at the second decimal digit (hundredths): 5 or more → increase the first digit, less → keep it the same.

Step 2: (1) 32.38 → 32.4 (2) 25.156 → 25.2 (3) 6.17 → 6.2 (4) 6.42 → 6.4 (5) 76.798 → 76.8 (6) 95.24 → 95.2.

Step 3: (7) 12.86 → 12.9 (8) 5.95 → 6.0 (9) 3.432 → 3.4 (10) 11.7681 → 11.8 (11) 50.4752 → 50.5 (12) 60.1536 → 60.2.

Final answer: (1) 32.4 (2) 25.2 (3) 6.2 (4) 6.4 (5) 76.8 (6) 95.2 (7) 12.9 (8) 6.0 (9) 3.4 (10) 11.8 (11) 50.5 (12) 60.2.

Question: 3. Round off the following decimals up to two decimal places: (1) 32.386 (2) 25.056 (3) 6.775 (4) 6.422 (5) 76.798 (6) 8.4832 (7) 0.9627 (8) 58.1905 (9) 4.0098 (10) 40.9807 (11) 70.4908 (12) 19.0185

Answer: Step 1: Look at the third decimal digit: 5 or more → increase the second digit, less → keep it the same.

Step 2: (1) 32.386 → 32.39 (2) 25.056 → 25.06 (3) 6.775 → 6.78 (4) 6.422 → 6.42 (5) 76.798 → 76.80 (6) 8.4832 → 8.48.

Step 3: (7) 0.9627 → 0.96 (8) 58.1905 → 58.19 (9) 4.0098 → 4.01 (10) 40.9807 → 40.98 (11) 70.4908 → 70.49 (12) 19.0185 → 19.02.

Final answer: (1) 32.39 (2) 25.06 (3) 6.78 (4) 6.42 (5) 76.80 (6) 8.48 (7) 0.96 (8) 58.19 (9) 4.01 (10) 40.98 (11) 70.49 (12) 19.02.

Exercise 4.12 — Convert Fractions and Decimals

Question: A. Convert the following fractions into decimals: (1) 7/8 (2) 9/10 (3) 2/5 (4) 17/25 (5) 61/50 (6) 19/20 (7) 49/40 (8) 71/80 (9) 451/500 (10) 79/250 (11) 83/100 (12) 111/400 (13) 777/800 (14) 551/1000 (15) 999/10000

Answer: Step 1: Divide the numerator by the denominator, or make the denominator 10, 100 or 1000 and place the point.

Step 2: (1) 7 ÷ 8 = 0.875. (2) 9 ÷ 10 = 0.9. (3) 2 ÷ 5 = 0.4. (4) 17 ÷ 25 = 0.68. (5) 61 ÷ 50 = 1.22. (6) 19 ÷ 20 = 0.95. (7) 49 ÷ 40 = 1.225. (8) 71 ÷ 80 = 0.8875.

Step 3: (9) 451 ÷ 500 = 0.902. (10) 79 ÷ 250 = 0.316. (11) 83 ÷ 100 = 0.83. (12) 111 ÷ 400 = 0.2775. (13) 777 ÷ 800 = 0.97125. (14) 551 ÷ 1,000 = 0.551. (15) 999 ÷ 10,000 = 0.0999.

Final answer: (1) 0.875 (2) 0.9 (3) 0.4 (4) 0.68 (5) 1.22 (6) 0.95 (7) 1.225 (8) 0.8875 (9) 0.902 (10) 0.316 (11) 0.83 (12) 0.2775 (13) 0.97125 (14) 0.551 (15) 0.0999.

Question: B. Convert the following decimals into fractions (in their lowest form): (1) 0.5 (2) 1.05 (3) 3.56 (4) 0.565 (5) 0.023 (6) 0.25 (7) 0.345 (8) 35.506 (9) 0.064 (10) 0.945 (11) 41.625 (12) 46.1024

Answer: Step 1: Write as many zeros below as there are decimal places to make the fraction, then cancel common factors to simplify.

Step 2: (1) 5/10 = 1/2. (2) 105/100 = 21/20 = 1 1/20. (3) 356/100 = 89/25 = 3 14/25. (4) 565/1,000 = 113/200. (5) 23/1,000. (6) 25/100 = 1/4.

Step 3: (7) 345/1,000 = 69/200. (8) 35,506/1,000 = 17,753/500 = 35 253/500. (9) 64/1,000 = 8/125. (10) 945/1,000 = 189/200. (11) 41,625/1,000 = 333/8 = 41 5/8. (12) 461,024/10,000 = 28,814/625 = 46 64/625.

Final answer: (1) 1/2 (2) 21/20 (3) 89/25 (4) 113/200 (5) 23/1000 (6) 1/4 (7) 69/200 (8) 17,753/500 (9) 8/125 (10) 189/200 (11) 333/8 (12) 28,814/625.

Exercise 4.13 — Word Problems on Decimals

Question: (1) Sahir bought two chickens of weight 1.450 kg and other of 1.685 kg respectively. Find the total weight of both chicken. (2) A general

store sold 200.750 kg of flour and 98.500 kg of sugar. How much more flour was sold than sugar? (3) Saira bought a necklace of gold of mass 2.565 g and bangles of mass 8.875 g. What is the total mass of these two items?

Answer: Step 1: (1) Total weight = 1.450 + 1.685.

Step 2: (1) 1.450 + 1.685 = 3.135 kg.

Step 3: (2) Difference = 200.750 − 98.500 = 102.25 kg more flour. (3) Total = 2.565 + 8.875 = 11.44 g.

Final answer: (1) 3.135 kg (2) 102.25 kg more (3) 11.44 g.

Question: (4) Aftab saved Rs 3,206.75 from monthly salary of Rs 35,200. Find his monthly expenditure. (5) Noor communications earns Rs 2,345.75 in a day by selling easy loading balance. How much amount will he earn in one month?

Answer: Step 1: (4) Expenditure = salary − savings = 35,200 − 3,206.75.

Step 2: (4) 35,200 − 3,206.75 = Rs 31,993.25.

Step 3: (5) One month = 30 days; earning = 2,345.75 x 30 = Rs 70,372.50.

Final answer: (4) Rs 31,993.25 (5) Rs 70,372.50.

Question: (6) A bundle contains 119.5 m of wire. If one piece of wire measuring 29.92 m is sold, find the length of remaining piece of wire. (7) The height of a pole is 18.75 m. We have ladder of 1/3 of pole. How much length is left out? (8) A school uniform is made from 4.5 m cloth. How many uniforms can be made from 31.5 m cloth?

Answer: Step 1: (6) Remaining = 119.5 − 29.92.

Step 2: (6) 119.5 − 29.92 = 89.58 m.

Step 3: (7) Ladder = 1/3 x 18.75 = 6.25 m; left above = 18.75 − 6.25 =

12.5 m. (8) Uniforms = 31.5 ÷ 4.5 = 315 ÷ 45 = 7.

Final answer: (6) 89.58 m (7) Ladder 6.25 m, 12.5 m of pole left above it (8) 7 uniforms.

Question: (9) A lady used 3.60 kg of salt in preparing different kinds of food for her family in the month of April. How much salt is used per day? (10) The price of 2.5 kg ghee is Rs 391.25. Find the price of 1 kg.

Answer: Step 1: (9) April has 30 days; per day = 3.60 ÷ 30.

Step 2: (9) 3.60 ÷ 30 = 0.12 kg per day.

Step 3: (10) 1 kg = 391.25 ÷ 2.5 = 3,912.5 ÷ 25 = Rs 156.50.

Final answer: (9) 0.12 kg per day (10) Rs 156.50 per kg.

Question: Activity (book page 80): Express the shaded parts as percentages in each 100-square grid: (i) 28 squares shaded (given in the book) (ii) the green squares (iii) the purple squares (iv) the pink shape (v) the blue colour (vi) the yellow-green colour

Answer: Step 1: Each grid has 100 squares, so the number of shaded squares is itself the percentage: percentage = shaded squares ÷ 100 x 100.

Step 2: Counting the squares: (ii) 17 squares → 17%. (iii) 8 squares → 8%. (iv) 49 squares → 49%.

Step 3: (v) 45 squares → 45%. (vi) 28 squares → 28%.

Final answer: (i) 28% (ii) 17% (iii) 8% (iv) 49% (v) 45% (vi) 28%.

Exercise 4.14 — Percentage, Fraction and Decimal

Question: A. Convert the given percentages into fraction and then into decimal: (1) 25% (2) 30% (3) 35% (4) 40% (5) 65% (6) 70% (7) 80% (8) 85% (9) 95% (10) 99% (11) 105% (12) 115%

Answer: Step 1: Divide the percentage by 100 to make the fraction and

simplify; dividing the fraction gives the decimal.

Step 2: (1) 25/100 = 1/4 = 0.25. (2) 30/100 = 3/10 = 0.3. (3) 35/100 = 7/20 = 0.35. (4) 40/100 = 2/5 = 0.4. (5) 65/100 = 13/20 = 0.65. (6) 70/100 = 7/10 = 0.7.

Step 3: (7) 80/100 = 4/5 = 0.8. (8) 85/100 = 17/20 = 0.85. (9) 95/100 = 19/20 = 0.95. (10) 99/100 = 0.99. (11) 105/100 = 21/20 = 1.05. (12) 115/100 = 23/20 = 1.15.

Final answer: (1) 1/4, 0.25 (2) 3/10, 0.3 (3) 7/20, 0.35 (4) 2/5, 0.4 (5) 13/20, 0.65 (6) 7/10, 0.7 (7) 4/5, 0.8 (8) 17/20, 0.85 (9) 19/20, 0.95 (10) 99/100, 0.99 (11) 21/20, 1.05 (12) 23/20, 1.15.

Question: B. Convert the given decimals into fractions and then convert into percentage: (1) 0.8 (2) 0.34 (3) 0.56 (4) 0.63 (5) 0.55 (6) 0.66 (7) 3.45 (8) 3.60 (9) 5.50 (10) 2.05 (11) 25.5 (12) 55.25 C. Express the following fractions as percentages, giving your answer correct to 1 decimal place where necessary: (1) 4/5 (2) 6/25 (3) 11/20 (4) 5/8 (5) 17/40 (6) 6/12 (7) 18/60 (8) 18/30 (9) 71/50 (10) 1 4/5

Answer: Step 1: B: Make the decimal a fraction over 100 and simplify; for the percentage, multiply the decimal by 100. C: Multiply the fraction by 100.

Step 2: B(1) 80/100 = 4/5, 80%. (2) 34/100 = 17/50, 34%. (3) 56/100 = 14/25, 56%. (4) 63/100, 63%. (5) 55/100 = 11/20, 55%. (6) 66/100 = 33/50, 66%.

Step 3: B(7) 345/100 = 69/20, 345%. (8) 360/100 = 18/5, 360%. (9) 550/100 = 11/2, 550%. (10) 205/100 = 41/20, 205%. (11) 2,550/100 = 51/2, 2,550%. (12) 5,525/100 = 221/4, 5,525%.

Step 4: C(1) 4/5 x 100 = 80%. (2) 6/25 x 100 = 24%. (3) 11/20 x 100 = 55%. (4) 5/8 x 100 = 62.5%. (5) 17/40 x 100 = 42.5%. (6) 6/12 x 100 =

50%. (7) 18/60 x 100 = 30%. (8) 18/30 x 100 = 60%. (9) 71/50 x 100 = 142%. (10) 9/5 x 100 = 180%.

Final answer: B: (1) 4/5, 80% (2) 17/50, 34% (3) 14/25, 56% (4) 63/100, 63% (5) 11/20, 55% (6) 33/50, 66% (7) 69/20, 345% (8) 18/5, 360% (9) 11/2, 550% (10) 41/20, 205% (11) 51/2, 2,550% (12) 221/4, 5,525%. C: (1) 80% (2) 24% (3) 55% (4) 62.5% (5) 42.5% (6) 50% (7) 30% (8) 60% (9) 142% (10) 180%.

Exercise 4.15 — Word Problems on Percentages

Question: (1) 480 students visited a book fair. Out of them 45 percent are boys. How many boys visited the book fair? (2) 825 students are studying in a school. 60 percent students were regular for whole of the session. How many students were regular in the school? (3) 900 employees are in a company. 70 percent employees were using English frequently. How many employees are using English frequently?

Answer: Step 1: Percentage means: number x percentage ÷ 100.

Step 2: (1) 480 x 45/100 = 21,600/100 = 216 boys. (2) 825 x 60/100 = 49,500/100 = 495 students.

Step 3: (3) 900 x 70/100 = 63,000/100 = 630 employees.

Final answer: (1) 216 boys (2) 495 students (3) 630 employees.

Question: (4) 65 percent of the houses in the colony have computers. There are 2,450 houses in all. How many houses have computers? (5) In a school total number of students are 1,200. In class V have 20% of the students. How many students in class V? (6) There are 450 cars in a car parking and 20% of these are white in colour. What is the number of white cars? (7) The original price of a dress is Rs 1,650. During a sale, it is sold at 9% discount. What is the price of a dress?

Answer: Step 1: Same percentage method.

Step 2: (4) 2,450 x 65/100 = 159,250/100 = 1,592.5 — the mathematical answer is 1,592.5 (a house cannot be half; the book's given number is like this). (5) 1,200 x 20/100 = 240 students.

Step 3: (6) 450 x 20/100 = 90 cars. (7) Discount = 1,650 x 9/100 = Rs 148.50; sale price = 1,650 − 148.50 = Rs 1,501.50.

Final answer: (4) 1,592.5 (mathematical) (5) 240 students (6) 90 white cars (7) Discount Rs 148.50, price Rs 1,501.50.

Review Exercise 4

Question: A. Fill in the blanks: (1) 42.0 ÷ ____ = 0.42 (2) 12.04 x ____ = 1204 (3) In decimals 65% = ____ (4) 4 + 0.4 + 0.04 + 0.004 = ____ (5) When we round off 4.956 to two decimal places, we get ____.

Answer: Step 1: (1) To make 0.42 from 42.0 the point moved two places left, i.e. ÷100. (2) To make 1204 from 12.04 the point moved two places right, i.e. x100.

Step 2: (3) 65% = 65/100 = 0.65. (4) 4 + 0.4 + 0.04 + 0.004 = 4.444. (5) The third digit is 6 (more than 5), so 4.956 → 4.96.

Final answer: (1) 100 (2) 100 (3) 0.65 (4) 4.444 (5) 4.96.

Question: B. Solve: (1) Arrange 0.016, 2.087, 1.995, 0.463 in ascending order. (2) Solve: 9.123 − 5.865 + 2.403 (3) Height of an electric pole is 14.95 m and height of mobile tower is 25.04 m. How much higher is the mobile tower then electric pole? (4) Verify: 4.5 x 3.2 = 45 x 0.32 (5) Divide 0.72 by 1.2

Answer: Step 1: (1) Compare whole parts first, then tenths and hundredths: 0.016 < 0.463 < 1.995 < 2.087.

Step 2: (2) 9.123 − 5.865 = 3.258; 3.258 + 2.403 = 5.661. (3) 25.04 − 14.95 = 10.09 m higher.

Step 3: (4) Left: 4.5 x 3.2 = 14.4; right: 45 x 0.32 = 14.4; both equal ✓ verified. (5) 0.72 ÷ 1.2 = 7.2 ÷ 12 = 0.6.

Final answer: (1) 0.016, 0.463, 1.995, 2.087 (2) 5.661 (3) 10.09 m (4) 14.4 = 14.4 ✓ (5) 0.6.

Question: C. Solve the problems: (1) 3.6 m cloth is required to make a dress. How many such dresses can be prepared from 14.4 m of cloth? (2) Aftab gets 84% marks in all subjects of computer science. If the total marks are 2000, find the marks he obtained. (3) Find the value of: (a) 3.5% of 450 (b) 0.45% of 760 (c) 110% of 220 (d) 1.1% of 1000

Answer: Step 1: (1) Dresses = 14.4 ÷ 3.6 = 144 ÷ 36.

Step 2: (1) 144 ÷ 36 = 4 dresses. (2) 84% of 2,000 = 2,000 x 84/100 = 168,000/100 = 1,680 marks.

Step 3: (3)(a) 450 x 3.5/100 = 15.75. (b) 760 x 0.45/100 = 3.42. (c) 220 x 110/100 = 242. (d) 1,000 x 1.1/100 = 11.

Final answer: (1) 4 dresses (2) 1,680 marks (3) (a) 15.75 (b) 3.42 (c) 242 (d) 11.

مفاصلو، وقت گرمي پد5يونٽ :

Question: Activity (book page 88): Convert the following: (1) 9 km = ____ m (2) 2,500 m = ____ km (3) 3,784 m = ____ km (4) 3,000 m = ____ km (5) 24 m = ____ cm (6) 350 cm = ____ m (7) 200 cm = ____ m (8) 4 m 58 cm = ____ m (9) 3 km 400 m = ____ m (10) 1,320 cm = ____ m (11) 425 cm = ____ m (12) 250 mm = ____ cm (13) 500 mm = ____ m (14) 10 mm = ____ cm (15) 28 cm 5 mm = ____ mm

Answer: Step 1: Rules: 1 km = 1,000 m; 1 m = 100 cm; 1 cm = 10 mm. From a bigger unit to a smaller one multiply, from a smaller to a bigger one divide.

Step 2: (1) 9 x 1,000 = 9,000 m (given in the book). (2) 2,500 ÷ 1,000 = 2.5 km (given). (3) 3,784 ÷ 1,000 = 3.784 km. (4) 3,000 ÷ 1,000 = 3 km. (5) 24 x 100 = 2,400 cm.

Step 3: (6) 350 ÷ 100 = 3.5 m. (7) 200 ÷ 100 = 2 m. (8) 4 m + 58 ÷ 100 = 4.58 m. (9) 3 x 1,000 + 400 = 3,400 m. (10) 1,320 ÷ 100 = 13.2 m.

Step 4: (11) 425 ÷ 100 = 4.25 m. (12) 250 ÷ 10 = 25 cm. (13) 500 mm = 50 cm = 0.5 m. (14) 10 ÷ 10 = 1 cm. (15) 28 x 10 + 5 = 285 mm.

Final answer: (1) 9,000 m (2) 2.5 km (3) 3.784 km (4) 3 km (5) 2,400 cm (6) 3.5 m (7) 2 m (8) 4.58 m (9) 3,400 m (10) 13.2 m (11) 4.25 m (12) 25 cm (13) 0.5 m (14) 1 cm (15) 285 mm.

Exercise 5.1 — Convert Units of Length

Question: A. How many kilometres and metres? (1) 1,600 m (2) 2,483 m (3) 1,386 m (4) 6,034 m (5) 8,324 m (6) 7,945 m B. Convert the lengths in metres and centimetres: (1) 400 cm (2) 750 cm (3) 385 cm (4) 810 cm (5)

205 cm (6) 567 cm (7) 684 cm (8) 998 cm C. Convert into centimetres and millimetres: (1) 35 mm (2) 634 mm (3) 593 mm (4) 400 mm (5) 295 mm (6) 447 mm (7) 609 mm (8) 899 mm

Answer: Step 1: Dividing metres by 1,000 gives kilometres and the remaining metres; dividing centimetres by 100 gives metres and the remaining centimetres; dividing millimetres by 10 gives centimetres and the remaining millimetres.

Step 2: A(1) 1,600 ÷ 1,000 = 1 km 600 m. (2) 2 km 483 m. (3) 1 km 386 m. (4) 6 km 34 m. (5) 8 km 324 m. (6) 7 km 945 m.

Step 3: B(1) 400 ÷ 100 = 4 m. (2) 7 m 50 cm. (3) 3 m 85 cm. (4) 8 m 10 cm. (5) 2 m 5 cm. (6) 5 m 67 cm. (7) 6 m 84 cm. (8) 9 m 98 cm.

Step 4: C(1) 35 ÷ 10 = 3 cm 5 mm. (2) 63 cm 4 mm. (3) 59 cm 3 mm. (4) 40 cm. (5) 29 cm 5 mm. (6) 44 cm 7 mm. (7) 60 cm 9 mm. (8) 89 cm 9 mm.

Final answer: A: (1) 1 km 600 m (2) 2 km 483 m (3) 1 km 386 m (4) 6 km 34 m (5) 8 km 324 m (6) 7 km 945 m. B: (1) 4 m (2) 7 m 50 cm (3) 3 m 85 cm (4) 8 m 10 cm (5) 2 m 5 cm (6) 5 m 67 cm (7) 6 m 84 cm (8) 9 m 98 cm. C: (1) 3 cm 5 mm (2) 63 cm 4 mm (3) 59 cm 3 mm (4) 40 cm (5) 29 cm 5 mm (6) 44 cm 7 mm (7) 60 cm 9 mm (8) 89 cm 9 mm.

Question: D. Convert the following: (1) 8 km to m (2) 20 km 340 m to m (3) 15 m to cm (4) 25 m 45 cm (5) 35 cm to mm (6) 1,200 m to km and m (7) 3,785 m to km and m (8) 1,520 cm to m and cm (9) 850 mm to cm (10) 4,725 mm to cm and mm E. How many metres? (1) 7,000 mm (2) 8,000 mm (3) 9,000 mm (4) 4,000 cm (5) 1,000 cm (6) 5,000 cm (7) 6,000 mm (8) 10,000 cm

Answer: Step 1: D: Multiply the bigger unit by its value and add the smaller part.

Step 2: D(1) 8 x 1,000 = 8,000 m. (2) 20 x 1,000 + 340 = 20,340 m. (3) 15 x 100 = 1,500 cm. (4) 25 x 100 + 45 = 2,545 cm. (5) 35 x 10 = 350 mm.

Step 3: D(6) 1,200 ÷ 1,000 = 1 km 200 m. (7) 3 km 785 m. (8) 1,520 ÷ 100 = 15 m 20 cm. (9) 850 ÷ 10 = 85 cm. (10) 4,725 ÷ 10 = 472 cm 5 mm.

Step 4: E: Divide millimetres by 1,000 and centimetres by 100: (1) 7 m (2) 8 m (3) 9 m (4) 40 m (5) 10 m (6) 50 m (7) 6 m (8) 100 m.

Final answer: D: (1) 8,000 m (2) 20,340 m (3) 1,500 cm (4) 2,545 cm (5) 350 mm (6) 1 km 200 m (7) 3 km 785 m (8) 15 m 20 cm (9) 85 cm (10) 472 cm 5 mm. E: (1) 7 m (2) 8 m (3) 9 m (4) 40 m (5) 10 m (6) 50 m (7) 6 m (8) 100 m.

Exercise 5.2 — Add and Subtract Distances

Question: A. Solve (add): 1. 43 km 15 m + 66 km 57 m 2. 428 m 15 cm + 257 m 29 cm 3. 5 km 860 m + 2 km 220 m 4. 8 m 70 cm + 5 m 60 cm 5. 3 km 918 m + 1 km 324 m 6. 45 cm 8 mm + 65 cm 7 mm 7. 8 km 750 m + 7 km 430 m 8. 9 m 58 cm + 4 m 64 cm

Answer: Step 1: Add each unit in its own column; if the smaller unit passes 1,000 m, 100 cm or 10 mm, carry one bigger unit forward.

Step 2: 1. 109 km 72 m. 2. 685 m 44 cm. 3. 7 km 1,080 m = 8 km 80 m. 4. 13 m 130 cm = 14 m 30 cm.

Step 3: 5. 4 km 1,242 m = 5 km 242 m. 6. 110 cm 15 mm = 111 cm 5 mm. 7. 15 km 1,180 m = 16 km 180 m. 8. 13 m 122 cm = 14 m 22 cm.

Final answer: 1. 109 km 72 m 2. 685 m 44 cm 3. 8 km 80 m 4. 14 m 30 cm 5. 5 km 242 m 6. 111 cm 5 mm 7. 16 km 180 m 8. 14 m 22 cm.

Question: B. Subtract: 1. 12 km 75 m − 3 km 84 m 2. 143 m 62 cm − 87

m 59 cm 3. 8 km 546 m − 6 km 804 m 4. 51 cm 3 mm − 27 cm 8 mm 5. 5 km 150 m − 2 km 730 m 6. 71 m 22 cm − 48 m 85 cm 7. 7 km 505 m − 4 km 700 m 8. 23 cm 2 mm − 17 cm 9 mm

Answer: Step 1: Subtract each unit separately; if the smaller unit is less, borrow one from the bigger unit (1 km = 1,000 m, 1 m = 100 cm, 1 cm = 10 mm).

Step 2: 1. 11 km 1,075 m − 3 km 84 m = 8 km 991 m. 2. 56 m 3 cm. 3. 7 km 1,546 m − 6 km 804 m = 1 km 742 m. 4. 50 cm 13 mm − 27 cm 8 mm = 23 cm 5 mm.

Step 3: 5. 4 km 1,150 m − 2 km 730 m = 2 km 420 m. 6. 70 m 122 cm − 48 m 85 cm = 22 m 37 cm. 7. 6 km 1,505 m − 4 km 700 m = 2 km 805 m. 8. 22 cm 12 mm − 17 cm 9 mm = 5 cm 3 mm.

Final answer: 1. 8 km 991 m 2. 56 m 3 cm 3. 1 km 742 m 4. 23 cm 5 mm 5. 2 km 420 m 6. 22 m 37 cm 7. 2 km 805 m 8. 5 cm 3 mm.

Exercise 5.3 — Word Problems on Distance

Question: Solve: (1) The length of a ribbon is 6 m 80 cm. How much ribbon is left, if 2m 88 cm has been cut off? (2) Jameel covered a distance of 589 m from his house to Jamia Masjid and then 868 m from Jamia Masjid to School. Find the total distance covered by him. (3) A car is 1 m 62 cm wide. A garage is 2m 41 cm wide. How much space is left when the car is in the garage?

Answer: Step 1: (1) Left = 6 m 80 cm − 2 m 88 cm; borrow: 5 m 180 cm.

Step 2: (1) 5 m 180 cm − 2 m 88 cm = 3 m 92 cm. (2) Total = 589 + 868 = 1,457 m.

Step 3: (3) Space left = 2 m 41 cm − 1 m 62 cm = 1 m 141 cm − 1 m 62 cm = 79 cm.

Final answer: (1) 3 m 92 cm (2) 1,457 m (3) 79 cm space.

Question: (4) Maria's home is distance as 375 m to the school. Railway station is 504 m far from her home. Which is farther from her home and how much? (5) The red part of a colour pencil is 65 mm long. The blue part is 57 mm long. What is the length of pencil? What is the answer in millimetres and centimetres?

Answer: Step 1: (4) Compare the two distances: 504 > 375.

Step 2: (4) Difference = 504 − 375 = 129 m; the railway station is 129 m farther.

Step 3: (5) Total = 65 + 57 = 122 mm = 12 cm 2 mm.

Final answer: (4) The railway station, 129 m farther (5) 122 mm = 12 cm 2 mm.

Question: (6) In a walking race, in an specified time Hamid walked 2 km 102 m, Hussain walked only 1 km 985 m. How far ahead of Hussain was Hamid? (7) In a 250 kilometre car race, a car get accident at a distance of 134 km from the winning point. What distance had the car drive before accident? (8) Naeem is 142 cm tall. His friend is 8 cm taller than Naeem. How tall is his friend? Give the answer in metres.

Answer: Step 1: (6) Difference = 2 km 102 m − 1 km 985 m; borrow: 1 km 1,102 m.

Step 2: (6) 1 km 1,102 m − 1 km 985 m = 117 m. (7) Distance driven = 250 − 134 = 116 km.

Step 3: (8) Friend = 142 + 8 = 150 cm = 1 m 50 cm.

Final answer: (6) 117 m ahead (7) 116 km (8) 150 cm = 1 m 50 cm.

Question: Activity (book page 93): Convert 2 days into hours, minutes and seconds.

Answer: Step 1: 1 day = 24 hours; 1 hour = 60 minutes; 1 minute = 60

seconds.

Step 2: 2 days = 2 x 24 = 48 hours.

Step 3: 48 x 60 = 2,880 minutes.

Step 4: 2,880 x 60 = 172,800 seconds.

Final answer: 2 days = 48 hours = 2,880 minutes = 172,800 seconds.

Exercise 5.4 — Convert Units of Time

Question: A. Convert into minutes: (1) 2 hours (2) 8 hours (3) 12 hours (4) 1 day 3 hours (5) 1 day 6 hours (6) 1 day 10 hours B. Convert into seconds: (1) 5 minutes (2) 10 minutes (3) 20 minutes (4) 45 minutes (5) 55 minutes (6) 1 hour (7) 1 hour 15 minutes (8) 1 hour 25 minutes

Answer: Step 1: Multiply hours by 60; first change the day into hours (1 day = 24 hours); multiply minutes by 60 for seconds.

Step 2: A(1) 2 x 60 = 120 minutes. (2) 8 x 60 = 480. (3) 12 x 60 = 720. (4) 27 hours x 60 = 1,620. (5) 30 x 60 = 1,800. (6) 34 x 60 = 2,040.

Step 3: B(1) 5 x 60 = 300 seconds. (2) 600. (3) 1,200. (4) 2,700. (5) 3,300. (6) 60 x 60 = 3,600. (7) 75 x 60 = 4,500. (8) 85 x 60 = 5,100.

Final answer: A: (1) 120 (2) 480 (3) 720 (4) 1,620 (5) 1,800 (6) 2,040 minutes. B: (1) 300 (2) 600 (3) 1,200 (4) 2,700 (5) 3,300 (6) 3,600 (7) 4,500 (8) 5,100 seconds.

Question: C. Convert into hours and minutes: (1) 1,180 minutes (2) 1,250 minutes (3) 1,490 minutes (4) 2,225 minutes (5) 1,815 minutes (6) 2,375 minutes D. Convert into minutes and seconds: (1) 2,185 seconds (2) 275 seconds (3) 350 seconds (4) 710 seconds (5) 990 seconds (6) 1,395 seconds E. Convert into hours, minutes and seconds: (1) 3,800 seconds (2) 4,100 seconds (3) 4,360 seconds (4) 4,595 seconds (5) 4,725 seconds (6) 4,915 seconds

Answer: Step 1: Dividing minutes by 60 gives hours and remaining minutes; dividing seconds by 60 gives minutes and remaining seconds; in E divide again by 60 to take out the hours.

Step 2: C(1) 1,180 ÷ 60 = 19 h 40 min. (2) 20 h 50 min. (3) 24 h 50 min. (4) 37 h 5 min. (5) 30 h 15 min. (6) 39 h 35 min.

Step 3: D(1) 2,185 ÷ 60 = 36 min 25 s. (2) 4 min 35 s. (3) 5 min 50 s. (4) 11 min 50 s. (5) 16 min 30 s. (6) 23 min 15 s.

Step 4: E(1) 3,800 ÷ 3,600 = 1 hour, remaining 200 s = 3 min 20 s → 1 h 3 min 20 s. (2) 1 h 8 min 20 s. (3) 1 h 12 min 40 s. (4) 1 h 16 min 35 s. (5) 1 h 18 min 45 s. (6) 1 h 21 min 55 s.

Final answer: C: (1) 19 h 40 min (2) 20 h 50 min (3) 24 h 50 min (4) 37 h 5 min (5) 30 h 15 min (6) 39 h 35 min. D: (1) 36 min 25 s (2) 4 min 35 s (3) 5 min 50 s (4) 11 min 50 s (5) 16 min 30 s (6) 23 min 15 s. E: (1) 1 h 3 min 20 s (2) 1 h 8 min 20 s (3) 1 h 12 min 40 s (4) 1 h 16 min 35 s (5) 1 h 18 min 45 s (6) 1 h 21 min 55 s.

Exercise 5.5 — Add and Subtract Time

Question: A. Add: (1) 30 minutes 38 seconds and 20 minutes 42 seconds (2) 47 minutes 25 seconds and 19 minutes 49 seconds (3) 4 hours 40 minutes and 3 hours 57 minutes (4) 3 hours 35 minutes 26 seconds and 2 hours 40 minutes 50 seconds (5) 2 hours 55 minutes 45 seconds and 3 hours 48 minutes 44 seconds

Answer: Step 1: Add each unit separately; if seconds or minutes pass 60, carry 1 forward.

Step 2: (1) 50 min 80 s = 51 min 20 s. (2) 66 min 74 s = 1 h 7 min 14 s. (3) 7 h 97 min = 8 h 37 min.

Step 3: (4) 5 h 75 min 76 s = 6 h 16 min 16 s. (5) 5 h 103 min 89 s = 6 h

44 min 29 s.

Final answer: (1) 51 min 20 s (2) 1 h 7 min 14 s (3) 8 h 37 min (4) 6 h 16 min 16 s (5) 6 h 44 min 29 s.

Question: B. Subtract: (1) 38 minutes 39 seconds from 62 minutes 20 seconds (2) 44 minutes 25 seconds from 50 minutes (3) 2 hours 58 minutes from 3 hours 5 minutes (4) 4 hours 32 minutes from 7 hours (5) 3 hours 45 minutes 50 seconds from 5 hours 30 minutes 40 seconds

Answer: Step 1: If the smaller unit is less, borrow 1 from the unit above (1 minute = 60 seconds, 1 hour = 60 minutes).

Step 2: (1) 61 min 80 s − 38 min 39 s = 23 min 41 s. (2) 49 min 60 s − 44 min 25 s = 5 min 35 s. (3) 2 h 65 min − 2 h 58 min = 7 min.

Step 3: (4) 6 h 60 min − 4 h 32 min = 2 h 28 min. (5) 4 h 89 min 100 s − 3 h 45 min 50 s = 1 h 44 min 50 s.

Final answer: (1) 23 min 41 s (2) 5 min 35 s (3) 7 min (4) 2 h 28 min (5) 1 h 44 min 50 s.

Exercise 5.6 — Convert Days, Weeks, Months and Years

Question: Convert: (1) 46 days into weeks and days (2) 80 days into weeks and days (3) 213 days into weeks and days (4) 2,343 days into weeks and days (5) 450 days into months and days (6) 800 days into months and days (7) 710 days into months and days (8) 650 days into months and days (9) 66 months into years and months (10) 82 months into years and months (11) 49 months into years and months (12) 244 months into years and months

Answer: Step 1: Rules: 1 week = 7 days; 1 month = 30 days (the book's rule); 1 year = 12 months. Divide days by 7 or 30, and months by 12; the remainder stays as the smaller unit.

Step 2: (1) 46 ÷ 7 = 6 weeks 4 days. (2) 80 ÷ 7 = 11 weeks 3 days. (3) 213 ÷ 7 = 30 weeks 3 days. (4) 2,343 ÷ 7 = 334 weeks 5 days.

Step 3: (5) 450 ÷ 30 = 15 months. (6) 800 ÷ 30 = 26 months 20 days. (7) 710 ÷ 30 = 23 months 20 days. (8) 650 ÷ 30 = 21 months 20 days.

Step 4: (9) 66 ÷ 12 = 5 years 6 months. (10) 82 ÷ 12 = 6 years 10 months. (11) 49 ÷ 12 = 4 years 1 month. (12) 244 ÷ 12 = 20 years 4 months.

Final answer: (1) 6 weeks 4 days (2) 11 weeks 3 days (3) 30 weeks 3 days (4) 334 weeks 5 days (5) 15 months (6) 26 months 20 days (7) 23 months 20 days (8) 21 months 20 days (9) 5 years 6 months (10) 6 years 10 months (11) 4 years 1 month (12) 20 years 4 months.

Exercise 5.7 — Word Problems on Time

Question: 1. Umair plays for 1 hour 15 minutes and Ali plays for 55 minutes. Who plays for shorter period and how much? 2. Fatima takes 1 hour 20 minutes to complete her Mathematics homework and 50 minutes to complete her English homework. How much total time she takes to complete the homework of both subjects?

Answer: Step 1: (1) Umair: 1 hour 15 minutes = 75 minutes; Ali: 55 minutes.

Step 2: (1) 55 < 75, so Ali plays for the shorter period; difference = 75 − 55 = 20 minutes.

Step 3: (2) Total = 1 hour 20 minutes + 50 minutes = 1 hour 70 minutes = 2 hours 10 minutes.

Final answer: 1. Ali, 20 minutes shorter 2. 2 hours 10 minutes.

Question: 3. School closes for summer vacations for a period of 2 months and 5 days, while the other holidays are for 27 days. How many days

school will remain closed? 4. At a library Hassan reads a newspaper for 36 minutes and then reads a magazine for 1 hour 50 minutes. How long does he spend in reading? 5. Adil takes 2 hours 9 minutes to complete a video game, while Ali takes 55 minutes to complete the same game. How much more time Adil takes as compared to Ali?

Answer: Step 1: (3) 1 month = 30 days (the book's rule); 2 months 5 days = 2 x 30 + 5 = 65 days.

Step 2: (3) Total closed = 65 + 27 = 92 days (= 3 months 2 days).

Step 3: (4) Total = 36 minutes + 1 hour 50 minutes = 1 hour 86 minutes = 2 hours 26 minutes. (5) Adil: 2 hours 9 minutes = 129 minutes; difference = 129 − 55 = 74 minutes = 1 hour 14 minutes.

Final answer: 3. 92 days 4. 2 hours 26 minutes 5. 1 hour 14 minutes more.

Exercise 5.8 — Temperature: Thermometer and Scale Conversion

Question: A. Write the readings from following thermometer: (1) the first thermometer (°C scale) (2) the second thermometer (°C scale) (3) the third thermometer (°F scale) (4) the fourth thermometer (°F scale)

Answer: Step 1: Read the scale up to the mark where the mercury stops in the thermometer.

Step 2: (1) The mercury is at the 20 mark → 20°C. (2) The mercury is at 35 → 35°C.

Step 3: (3) On the Fahrenheit scale the mercury is at 40 → 40°F. (4) The mercury is at 85 → 85°F.

Final answer: (1) 20°C (2) 35°C (3) 40°F (4) 85°F.

Question: B. Change to the Celsius scale: (1) 41°F (2) 77°F (3) 95°F (4) 68°F (5) 203°F (6) 230°F (7) 257°F (8) 275°F (9) 122°F (10) 167°F (11) 230°F (12) 248°F

Answer: Step 1: Formula: °C = (°F − 32) x 5/9 — first subtract 32, then multiply by 5/9.

Step 2: (1) (41 − 32) x 5/9 = 9 x 5/9 = 5°C. (2) 45 x 5/9 = 25°C. (3) 63 x 5/9 = 35°C. (4) 36 x 5/9 = 20°C.

Step 3: (5) 171 x 5/9 = 95°C. (6) 198 x 5/9 = 110°C. (7) 225 x 5/9 = 125°C. (8) 243 x 5/9 = 135°C.

Step 4: (9) 90 x 5/9 = 50°C. (10) 135 x 5/9 = 75°C. (11) 198 x 5/9 = 110°C. (12) 216 x 5/9 = 120°C.

Final answer: (1) 5°C (2) 25°C (3) 35°C (4) 20°C (5) 95°C (6) 110°C (7) 125°C (8) 135°C (9) 50°C (10) 75°C (11) 110°C (12) 120°C.

Question: C. Change to the Fahrenheit scale: (1) 30°C (2) 45°C (3) 85°C (4) 55°C (5) 90°C (6) 10°C (7) 20°C (8) 60°C (9) 80°C (10) 110°C

Answer: Step 1: Formula: °F = °C x 9/5 + 32 — first multiply by 9/5, then add 32.

Step 2: (1) 30 x 9/5 = 54; 54 + 32 = 86°F. (2) 45 x 9/5 = 81; +32 = 113°F. (3) 85 x 9/5 = 153; +32 = 185°F. (4) 55 x 9/5 = 99; +32 = 131°F.

Step 3: (5) 90 x 9/5 = 162; +32 = 194°F. (6) 10 x 9/5 = 18; +32 = 50°F. (7) 20 x 9/5 = 36; +32 = 68°F. (8) 60 x 9/5 = 108; +32 = 140°F.

Step 4: (9) 80 x 9/5 = 144; +32 = 176°F. (10) 110 x 9/5 = 198; +32 = 230°F.

Final answer: (1) 86°F (2) 113°F (3) 185°F (4) 131°F (5) 194°F (6) 50°F (7) 68°F (8) 140°F (9) 176°F (10) 230°F.

Exercise 5.9 — Word Problems on Temperature

Question: (1) Day the maximum temperature on a day in Hyderabad was 35°C. What was the temperature in degree Fahrenheit? (2) During summer on one day the temperature in Jacobabad was 113°F where as the

temperature in Hyderabad was 40°C. Which city had more temperature and how much?

Answer: Step 1: (1) °F = 35 x 9/5 + 32.

Step 2: (1) 35 x 9/5 = 63; 63 + 32 = 95°F.

Step 3: (2) Jacobabad: (113 − 32) x 5/9 = 81 x 5/9 = 45°C. Hyderabad is 40°C; 45 > 40, so Jacobabad was hotter; difference = 45 − 40 = 5°C (that is, 9°F more).

Final answer: (1) 95°F (2) Jacobabad, hotter by 5°C (9°F).

Question: (3) Ali was suffering from fever. His temperature at noon was 102°F. Convert the said temperature into degree Celsius. (4) In summer on one day the maximum temperature in Sukkur was 104°F, where as in Karachi it was 35°C. What was the difference in the temperatures of both the cities in degree Fahrenheit? (5) Day the maximum temperature on a day in Sehwan Sharif was 86°F. The minimum temperature on that day was 20°C. What was the difference in the both temperatures in degree Fahrenheit?

Answer: Step 1: (3) °C = (102 − 32) x 5/9 = 70 x 5/9.

Step 2: (3) 70 x 5/9 = 350/9 = 38.9°C (approximately).

Step 3: (4) Karachi: 35 x 9/5 + 32 = 95°F; difference = 104 − 95 = 9°F. (5) Minimum: 20 x 9/5 + 32 = 68°F; difference = 86 − 68 = 18°F.

Final answer: (3) 38.9°C (4) Difference of 9°F (5) Difference of 18°F.

Review Exercise 5

Question: 1. Convert: (i) 28km 648m to m (ii) 48m 97cm to cm (iii) 76cm 9mm to mm (iv) 6m 75cm 8mm to mm 2. Add: (i) 38 minutes 42 seconds and 25 minutes 55 seconds (ii) 20 hours 30 minutes and 16 hours 43 minutes (iii) 4m 70cm 9mm and 3m 80cm 5mm

Answer: Step 1: 1(i) 28 x 1,000 + 648. (ii) 48 x 100 + 97. (iii) 76 x 10 + 9. (iv) 6 x 1,000 + 75 x 10 + 8.

Step 2: 1(i) 28,648 m. (ii) 4,897 cm. (iii) 769 mm. (iv) 6,758 mm.

Step 3: 2(i) 63 min 97 s = 1 h 4 min 37 s. (ii) 36 h 73 min = 37 h 13 min (= 1 day 13 h 13 min). (iii) 7 m 150 cm 14 mm = 8 m 51 cm 4 mm.

Final answer: 1: (i) 28,648 m (ii) 4,897 cm (iii) 769 mm (iv) 6,758 mm. 2: (i) 1 h 4 min 37 s (ii) 37 h 13 min (iii) 8 m 51 cm 4 mm.

Question: 3. Subtract: (i) 22km 895m from 67km 472m (ii) 2m 85cm 8mm from 4m 13cm 2mm (iii) 39 hours 49 minutes from 76 hours 32 minutes 4. Choose the correct answer: (i) 5 km = (a) 500 m (b) 5000 m (c) 50 m (d) 555 m (ii) 3 metres = (a) 30 cm (b) 300 cm (c) 3000 cm (d) 30000 cm (iii) 6 weeks = (a) 40 days (b) 35 days (c) 30 days (d) 42 days (iv) 59°F = ____ °C (a) 15 (b) 25 (c) 30 (d) 35

Answer: Step 1: 3(i) Borrow and subtract: 66 km 1,472 m − 22 km 895 m. (ii) 3 m 112 cm 12 mm − 2 m 85 cm 8 mm. (iii) 75 h 92 min − 39 h 49 min.

Step 2: 3(i) 44 km 577 m. (ii) 1 m 27 cm 4 mm. (iii) 36 h 43 min.

Step 3: 4(i) (b) 5000 m — 5 x 1,000 = 5,000. (ii) (b) 300 cm — 3 x 100 = 300. (iii) (d) 42 days — 6 x 7 = 42. (iv) (a) 15 — (59 − 32) x 5/9 = 27 x 5/9 = 15°C.

Final answer: 3: (i) 44 km 577 m (ii) 1 m 27 cm 4 mm (iii) 36 h 43 min. 4: (i) (b) (ii) (b) (iii) (d) (iv) (a).

Question: 5. What will be the time? (i) 50 minutes earlier than 13:25 (ii) 45 minutes later than 9:45 6. How much time? (i) 5:30 to 6:15 (ii) 2:20 to 3:25

Answer: Step 1: 5(i) Go back 50 minutes from 13:25: first 25 minutes → 13:00, then 25 more minutes → 12:35. (ii) Add 45 minutes to 9:45: 9:45 +

15 minutes = 10:00, 30 more minutes = 10:30.

Step 2: 6(i) From 5:30 to 6:15: 5:30 → 6:00 = 30 minutes, 6:00 → 6:15 = 15 minutes; total 45 minutes. (ii) 2:20 → 3:20 = 1 hour, 3:20 → 3:25 = 5 minutes; total 1 hour 5 minutes.

Final answer: 5: (i) 12:35 (ii) 10:30. 6: (i) 45 minutes (ii) 1 hour 5 minutes.

ایکائي طریقو6یونٹ :

Exercise 6.1 — Unitary Method: Price of One or Many Things

Question: 1. The price of a pen is Rs 30. Find the price of 6 such pens. 2. One litre of petrol price is Rs 105.70. What is the price of 5 litres of petrol? 3. The price of 1 kg rice is Rs 110. What is the price of 7 kg rice? 4. The rent of a house for one month is Rs 5000. What is the rent for one year? 5. The price of a dozen bananas is Rs 66. What is the price of 1 banana?

Answer: Step 1: Unitary method: if the price of one thing is given, multiply by the number needed; if the price of many things is given, divide to get the price of one.

Step 2: 1. 30 x 6 = Rs 180. 2. 105.70 x 5 = Rs 528.50. 3. 110 x 7 = Rs 770.

Step 3: 4. 5,000 x 12 = Rs 60,000. 5. One dozen = 12 bananas; 66 ÷ 12 = Rs 5.50.

Final answer: 1. Rs 180 2. Rs 528.50 3. Rs 770 4. Rs 60,000 5. Rs 5.50.

Question: 6. If 12 books price Rs 480. What is the price of one book? 7. The price of 8 chocolates is Rs 36. What is the price of one chocolate? 8. The price of ten balls is Rs 205. What is the price of one ball? 9. The price of three mobile phones is Rs 38400. What is the price of one? 10. The price of 16 pairs of socks is Rs 1004.80. What is the price of one pair of socks?

Answer: Step 1: In each case divide the total price by the number of things.

Step 2: 6. 480 ÷ 12 = Rs 40. 7. 36 ÷ 8 = Rs 4.50. 8. 205 ÷ 10 = Rs 20.50.

Step 3: 9. 38,400 ÷ 3 = Rs 12,800. 10. 1,004.80 ÷ 16 = Rs 62.80.

Final answer: 6. Rs 40 7. Rs 4.50 8. Rs 20.50 9. Rs 12,800 10. Rs 62.80.

Exercise 6.2 — Unitary Method: Many to Many

Question: 1. The price of 6 balls is 240 rupees. Find the price of 10 such balls. 2. The cost of 10 books is Rs 240. What will be the cost of such 15 books? 3. The price of 2 dozen pencils is Rs 60. What is the price of 3 1/2 dozen pencils? 4. 6 farmers plough a field in 10 hours. How many hours will it take for 8 farmers to plough the same field? 5. Rent of a house for 3 months is Rs 18000. What is the rent for 8 months?

Answer: Step 1: First find the price/rate of one, then multiply by the number needed. Question 4 is inverse proportion: more farmers, less time.

Step 2: 1. One ball = 240 ÷ 6 = 40; 10 balls = 40 x 10 = Rs 400. 2. One book = 240 ÷ 10 = 24; 15 books = 24 x 15 = Rs 360.

Step 3: 3. One dozen = 60 ÷ 2 = 30; 3 1/2 dozen = 30 x 3.5 = Rs 105. 4. Total work = 6 x 10 = 60 farmer-hours; 60 ÷ 8 = 7.5 hours. 5. One month = 18,000 ÷ 3 = 6,000; 8 months = 6,000 x 8 = Rs 48,000.

Final answer: 1. Rs 400 2. Rs 360 3. Rs 105 4. 7.5 hours 5. Rs 48,000.

Question: 6. A car travels 45 km in 3 1/2 litres of petrol. How far would it travel with 364 litres of petrol? 7. 6 metres of cloth is required for 2 shirts. How many same shirts can be made form 42 metre? 8. 12 coaches of passengers carry 624 persons. How many passengers will be carried by 18 such coaches? 9. The weight of 16 bags of rice is 775.60 kg. What will be the weight of 24 such bags? 10. The bus fare of 10 persons from Karachi to Larkana is Rs 8300. What is the fare for 36 persons?

Answer: Step 1: In each case first find the rate per unit.

Step 2: 6. Distance per litre = 45 ÷ 3.5 = 12.857 km; 364 litres = 12.857 x

364 = 4,680 km. 7. Cloth per shirt = 6 ÷ 2 = 3 metres; 42 ÷ 3 = 14 shirts.

Step 3: 8. Per coach = 624 ÷ 12 = 52 passengers; 18 coaches = 52 x 18 = 936 passengers. 9. Per bag = 775.60 ÷ 16 = 48.475 kg; 24 bags = 48.475 x 24 = 1,163.40 kg. 10. Per person = 8,300 ÷ 10 = 830; 36 persons = 830 x 36 = Rs 29,880.

Final answer: 6. 4,680 km 7. 14 shirts 8. 936 passengers 9. 1,163.40 kg 10. Rs 29,880.

Question: Activity (book page 110): Identify and write direct or inverse proportion. (i) Larger crowd, more noise. (ii) More books, more money needed. (iii) Less labourers, more time to build a house. (iv) Less money, less toffees purchased.

Answer: Step 1: If both quantities increase together or decrease together, it is direct proportion; if one increases and the other decreases, it is inverse proportion.

Step 2: (i) Both increase → Direct proportion (given in the book). (ii) Both increase → Direct proportion.

Step 3: (iii) Labourers decrease, time increases → Inverse proportion. (iv) Both decrease → Direct proportion.

Final answer: (i) Direct (ii) Direct (iii) Inverse (iv) Direct proportion.

Exercise 6.3 — Ratio and Proportion

Question: 1. The ratio of pocket money of Bismah and Omaima is 3 : 5. Write true or False. (i) Omaima has less pocket money than Bismah. (ii) Bismah has less pocket money than Umaima. 2. Express the following as ratio. (i) 3 boys to 6 girls (ii) 25 minutes to 80 minutes (iii) 5 days to 3 weeks (iv) 250 kg to 1 1/2 kg 3. Identify and write direct or inverse proportion. (i) More laboures, more work. (ii) More laboures, less days

needed for work. (iii) Less speed of a car, more time taken. (iv) More farmers, more work done in the field.

Answer: Step 1: In 1, the ratio 3 : 5 means Bismah has 3 parts and Omaima 5 parts; more parts = more money. In 2, bring the ratio to the same units and simplify.

Step 2: 1(i) False — Omaima has more (5 parts). (ii) True — Bismah has less (3 parts).

Step 3: 2(i) 3 : 6 = 1 : 2. (ii) 25 : 80 = 5 : 16. (iii) 3 weeks = 21 days; 5 : 21. (iv) 250 : 1.5 = 500 : 3.

Step 4: 3(i) Direct proportion. (ii) Inverse proportion. (iii) Inverse proportion. (iv) Direct proportion.

Final answer: 1: (i) False (ii) True. 2: (i) 1 : 2 (ii) 5 : 16 (iii) 5 : 21 (iv) 500 : 3. 3: (i) Direct (ii) Inverse (iii) Inverse (iv) Direct.

Question: 4. If 2 packs of juice cost Rs 24. what is the cost of 4 packs of juice? 5. Aslam can walk 10.5 km in 2 hours. Find the distance he can travel in 5 hours. 6. If 9 workers can complete a work in 6 days. How many workers are required to complete the same work in 3 days? 7. A motorcycle covers 100 km in 2 1/2 litres of petorl. How many litres of petrol will it use to cover 300 km? 8. 180 soldiers have food stock for 6 days. How many soldiers can eat the same food for 9 days? 9. 44 farmers could reap a field in 15 days. How many farmers can reap the same field in 10 days?

Answer: Step 1: In direct proportion find the rate per unit; in inverse proportion the total quantity (workers x days etc.) stays the same.

Step 2: 4. One pack = 24 ÷ 2 = 12; 4 packs = 12 x 4 = Rs 48. 5. Per hour = 10.5 ÷ 2 = 5.25 km; 5 hours = 5.25 x 5 = 26.25 km.

Step 3: 6. Total work = 9 x 6 = 54 worker-days; 54 ÷ 3 = 18 workers. 7.

Per 100 km it uses 2.5 litres; 300 km = 2.5 x 3 = 7.5 litres.

Step 4: 8. Total food = 180 x 6 = 1,080 soldier-days; 1,080 ÷ 9 = 120 soldiers. 9. Total work = 44 x 15 = 660 farmer-days; 660 ÷ 10 = 66 farmers.

Final answer: 4. Rs 48 5. 26.25 km 6. 18 workers 7. 7.5 litres 8. 120 soldiers 9. 66 farmers.

Review Exercise 6

Question: 1. The cost of a sharpener is Rs. 4.50. Find the cost of a dozen sharpeners. 2. From Karachi to Nawabshah the fare of 10 persons in a bus is 6500 rupees. What is the fare of each person? 3. The cost of 2 dozen eggs is 220 rupees. What is the cost of 3 dozen eggs? 4. If 10 persons can do a work in 6 days. In how many days can 15 persons do the same work? 5. The price of 6 packets of chalk is Rs. 90. What is the price of 8 packets of chalk?

Answer: Step 1: Use the unitary method: first the price/rate of one, then the required quantity. Question 4 is inverse proportion.

Step 2: 1. 4.50 x 12 = Rs 54. 2. 6,500 ÷ 10 = Rs 650 per person. 3. One dozen = 220 ÷ 2 = 110; 3 dozen = 110 x 3 = Rs 330.

Step 3: 4. Total work = 10 x 6 = 60 person-days; 60 ÷ 15 = 4 days. 5. One packet = 90 ÷ 6 = 15; 8 packets = 15 x 8 = Rs 120.

Final answer: 1. Rs 54 2. Rs 650 3. Rs 330 4. 4 days 5. Rs 120.

Question: 6. A printer prints 7620 papers in 1 hour. Find how many papers will be printed by the same printer in 40 minutes? 7. A train travels 800 km in 6 hours. How much distance will be traveled in 15 hours with same speed?

Answer: Step 1: In 6, change the hour into minutes (1 hour = 60 minutes)

and find the rate per minute; in 7 find the speed per hour.

Step 2: 6. Per minute = 7,620 ÷ 60 = 127 papers; 40 minutes = 127 x 40 = 5,080 papers.

Step 3: 7. Speed = 800 ÷ 6 = 133.33 km per hour; 15 hours = 133.33 x 15 = 2,000 km.

Final answer: 6. 5,080 papers 7. 2,000 km.

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Exercise 7.1 — Names and Measures of Angles

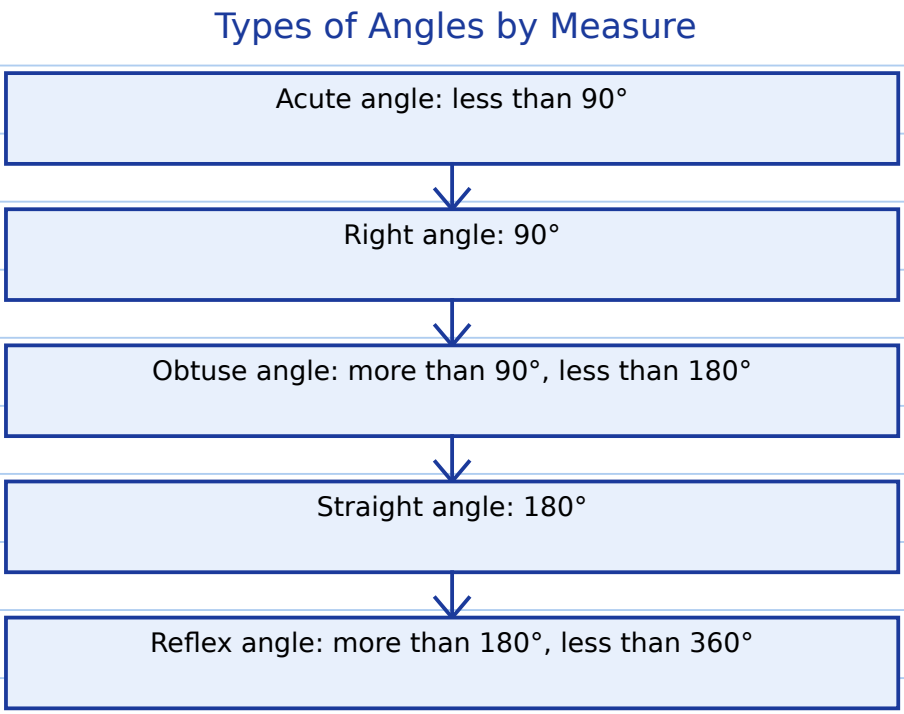
Question: 1. Name the angles then write their measurements (in the book's figures): (a) the first angle (b) the second angle (c) the third angle 2. Identify the angles as acute, right, obtuse, straight or reflex angle: (a) 50° (b) 120° (c) 90° (d) 280° (e) 180° (f) 85° (g) 200° (h) 150° (i) 90°

Answer: Step 1: Rules: an acute angle is less than 90°; a right angle is exactly 90°; an obtuse angle is between 90° and 180°; a straight angle is exactly 180°; a reflex angle is greater than 180° and less than 360°. Measure the angle with a protractor.

Step 2: 1(a) In the figure the two arms are perpendicular → right angle, 90°. (b) The angle is less than 90°; measured with a protractor it is about 30° → acute angle, 30°. (c) The angle is greater than 90°; it measures about 120° → obtuse angle, 120°.

Step 3: 2(a) 50° acute. (b) 120° obtuse. (c) 90° right. (d) 280° reflex. (e) 180° straight. (f) 85° acute. (g) 200° reflex. (h) 150° obtuse. (i) 90° right.

Final answer: 1: (a) Right angle 90° (b) Acute angle 30° (c) Obtuse angle 120°. 2: (a) acute (b) obtuse (c) right (d) reflex (e) straight (f) acute (g) reflex (h) obtuse (i) right.



Exercise 7.2 — Measuring and Drawing Reflex Angles

Question: 1. Measure the reflex angles by using protractor (in the book's figures): (i) the first angle (ii) the second angle (iii) the third angle (iv) the fourth angle 2. Draw and label the following angles: (i) ∠ABC (reflex angle) (ii) ∠DEF (straight angle) (iii) ∠GHI (reflex angle) (iv) ∠PQR (right angle) (v) ∠STU (straight angle) (vi) ∠XYZ (reflex angle) 3. Using protractor, draw the following: (i) Reflex ∠ABC of 310° (ii) Reflex ∠DEF of 280° (iii) Reflex ∠LMN of 340° (iv) Reflex ∠OPQ of 290°

Answer: Step 1: Reflex angle = 360° − the smaller angle. To measure, measure the smaller angle and subtract from 360°; to draw, first draw the smaller angle (360° − reflex) and mark the arc on the outside.

Step 2: 1(i) Smaller angle 25° → reflex = 360 − 25 = 335°. (ii) Smaller angle 30° → 360 − 30 = 330°. (iii) Smaller angle 90° → 360 − 90 = 270°. (iv) Smaller angle 80° → 360 − 80 = 280°.

Step 3: 2. For each angle: draw a ray, place the protractor at the vertex,

mark the required measure (180° for straight, 90° for right, and for reflex draw the smaller angle and arc outside) and draw the second arm and label it.

Step 4: 3(i) Smaller angle = 360 − 310 = 50°, draw it → reflex 310°. (ii) 360 − 280 = 80° → reflex 280°. (iii) 360 − 340 = 20° → reflex 340°. (iv) 360 − 290 = 70° → reflex 290°.

Final answer: 1: (i) 335° (ii) 330° (iii) 270° (iv) 280°. 3: reflex angles of (i) 310° (ii) 280° (iii) 340° (iv) 290° drawn.

Question: Activities (book page 119): (a) Find complement of m ∠DEF = 44° in the figure, where ∠DEF and ∠FEG are two complementary angles. (b) Find supplement of the given angle of 110°, where ∠CEF and ∠DEF are two supplementary angles and ∠DEF = 110°.

Answer: Step 1: Complementary angles add to 90°; supplementary angles add to 180°.

Step 2: (a) ∠FEG = 90° − 44° = 46°; so the complement of 44° is 46°.

Step 3: (b) ∠CEF = 180° − 110° = 70°; so the supplement of 110° is 70°.

Final answer: (a) 46° (b) 70°.

Exercise 7.3 — Adjacent, Complementary and Supplementary Angles

Question: 1. Look at the figure and answer the following: (i) Is ∠AOE adjacent to ∠DOE? (ii) Is ∠AOD adjacent to ∠COD? (iii) Is ∠AOE adjacent to ∠AOD? (iv) Is ∠DOE adjacent to ∠EOC? 2. Is ∠ABC adjacent to ∠BCD? Why or why not?

Answer: Step 1: Angles are adjacent when they have a common vertex, a common arm, and they do not overlap each other.

Step 2: 1(i) Yes — common vertex O, common arm OE, no overlap. (ii) Yes — common vertex O, common arm OD.

Step 3: 1(iii) No — ∠AOE lies inside ∠AOD; they overlap. (iv) No — ∠DOE overlaps inside ∠EOC.

Step 4: 2. No — ∠ABC has vertex B and ∠BCD has vertex C; adjacent angles need a common vertex and a common arm, here even the vertices are different.

Final answer: 1: (i) Yes (ii) Yes (iii) No (iv) No. 2: No, because the vertices are different (B and C).

Question: 3. Find the complement of each of the following angles: (i) 60° (ii) 76° (iii) 45° (iv) 38° (v) 15° 4. Find the supplement of each of the following angles: (i) 25° (ii) 45° (iii) 70° (iv) 98° (v) 143° 5. Identify the complementary and supplementary pair of angles: (i) 49°, 41° (ii) 154°, 26° (iii) 95°, 85° (iv) 32°, 58° (v) 111°, 69° (vi) 14°, 76°

Answer: Step 1: Complement = 90° − the angle; supplement = 180° − the angle. A pair adding to 90° is complementary, adding to 180° is supplementary.

Step 2: 3(i) 90 − 60 = 30°. (ii) 90 − 76 = 14°. (iii) 90 − 45 = 45°. (iv) 90 − 38 = 52°. (v) 90 − 15 = 75°.

Step 3: 4(i) 180 − 25 = 155°. (ii) 180 − 45 = 135°. (iii) 180 − 70 = 110°. (iv) 180 − 98 = 82°. (v) 180 − 143 = 37°.

Step 4: 5(i) 49 + 41 = 90 → complementary. (ii) 154 + 26 = 180 → supplementary. (iii) 95 + 85 = 180 → supplementary. (iv) 32 + 58 = 90 → complementary. (v) 111 + 69 = 180 → supplementary. (vi) 14 + 76 = 90 → complementary.

Final answer: 3: (i) 30° (ii) 14° (iii) 45° (iv) 52° (v) 75°. 4: (i) 155° (ii) 135° (iii) 110° (iv) 82° (v) 37°. 5: (i) complementary (ii) supplementary (iii) supplementary (iv) complementary (v) supplementary (vi) complementary.

Question: 6. (i) Find the angle which is equal to its complement. (ii) Find the angle which is equal to its supplement. 7. Can two angles be supplementary, if both of them are: (i) Obtuse (ii) Acute (iii) Right? 8. Are complementary angles also adjacent angles? Are supplementary angles also adjacent angles? Explain your answers.

Answer: Step 1: 6(i) angle + angle = 90° → angle = 90 ÷ 2 = 45°. (ii) angle + angle = 180° → angle = 180 ÷ 2 = 90°.

Step 2: 7(i) Two obtuse angles are each greater than 90°; their sum passes 180° → No. (ii) Two acute angles are each less than 90°; their sum stays below 180° → No. (iii) Two right angles: 90° + 90° = 180° → Yes, they can.

Step 3: 8. Not necessarily — complementary and supplementary describe the SUM of measures (90° or 180°), while adjacent describes position (common vertex and common arm). A pair of angles can be both, either one, or neither.

Final answer: 6: (i) 45° (ii) 90°. 7: (i) No (ii) No (iii) Yes. 8: No, not necessarily; complementary/supplementary is about the sum of measures, adjacent is about position.

Exercise 7.4 — Types of Triangles

Question: 1. Measure the sides of the following triangles and then write their names with respect to their sides (the book's figures (i) to (vi)). 2. Use protractor to measure the angles of the following triangles and write their names with respect to their angles (the book's figures (i) to (vi)).

Answer: Step 1: By sides: all three sides equal = equilateral; two sides equal = isosceles; all sides different = scalene. By angles: one angle greater than 90° = obtuse-angled; one angle exactly 90° = right-angled; all angles less than 90° = acute-angled.

Step 2: 1 Measuring the sides with a ruler: (i) all sides different → scalene.

(ii) all different → scalene. (iii) two sides equal → isosceles. (iv) all three sides equal → equilateral. (v) all different → scalene. (vi) two sides equal → isosceles.

Step 3: 2 Measuring the angles: (i) one angle greater than 90° → obtuse-angled. (ii) all angles acute → acute-angled. (iii) one angle 90° → right-angled. (iv) all acute → acute-angled. (v) one angle 90° → right-angled. (vi) all acute → acute-angled.

Final answer: 1: (i) scalene (ii) scalene (iii) isosceles (iv) equilateral (v) scalene (vi) isosceles. 2: (i) obtuse-angled (ii) acute-angled (iii) right-angled (iv) acute-angled (v) right-angled (vi) acute-angled.

Exercise 7.5 — Construction of Triangles

Question: 1. Using a ruler and a pair of compasses, construct the following equilateral triangles. (i) △ABC where mAB = mBC = mCA = 4 cm (ii) △DEF where mDE = mEF = mDF = 3.5 cm (iii) △PQR where mPQ = mQR = mPR = 5.2 cm 2. Using a ruler and a pair of compasses, construct the following isosceles triangles. (i) △ABC where mAB = mBC = 6 cm; mAC = 4 cm (ii) △DEF where mDE = 2.5 cm, mEF = mDF = 4 cm (iii) △PQR where mPQ = 4 cm, mQR = mPR = 3.5 cm 3. Using a ruler and a pair of compasses, construct the following scalene triangles. (i) △ABC where mAB = 4.8 cm, mBC = 3 cm and mAC = 5 cm (ii) △PQR where mPQ = 4.5 cm, mQR = 5 cm and mPR = 3.5 cm (iii) △EFG where mEF = 5.2 cm, mFG = 4.4 cm and mGE = 3 cm

Answer: Step 1: Construction steps (three-sides case, SSS): draw the base side with a ruler.

Step 2: Open the compasses to the other two side measures and draw arcs from both ends of the base so that they meet each other.

Step 3: Join the meeting point to both ends — this is the third vertex.

Step 4: Use the same steps for each triangle: 1(i) base 4 cm, arcs 4 and 4. 1(ii) base 3.5, arcs 3.5 and 3.5. 1(iii) base 5.2, arcs 5.2 and 5.2. 2(i) base AC = 4, arcs 6 and 6. 2(ii) base DE = 2.5, arcs 4 and 4. 2(iii) base PQ = 4, arcs 3.5 and 3.5. 3(i) base AB = 4.8, arcs 5 (from A) and 3 (from B). 3(ii) base PQ = 4.5, arcs 3.5 (from P) and 5 (from Q). 3(iii) base EF = 5.2, arcs 3 (from E) and 4.4 (from F).

Final answer: All the required triangles are constructed with the given sides.

Question: 4. Using ruler, protractor and a pencil to construct the following equilateral, isosceles and scalene triangles. Also measure the remaining two sides and one angle in each triangle. (i) An equilateral △ABC where mAB = 5.7 cm, m ∠ABC = m ∠BAC = 60°. (ii) An isosceles △LMN where mLM = 5 cm, m ∠LMN = m ∠MLN = 70°. (iii) A scalene △XYZ where mXY = 6 cm, m ∠XYZ = 60°, m ∠YXZ = 50°. (iv) An equilateral △RST where mRS = 4.8 cm and m ∠RST = m ∠TRS = 60°. (v) A scalene △EFG where mEF = 5.5 cm, m ∠EFG = 75° and m∠GEF = 65°. (vi) An isosceles △JKL where m ∠JLK = 45°, mJL = 4.6 cm and m ∠KJL = 45°.

Answer: Step 1: Construction steps (two angles and the included side, ASA): draw the side, make the given angles at both ends with a protractor, extend the arms until they meet; third angle = 180° − the sum of the two angles; measure the remaining sides with a ruler.

Step 2: (i) Third angle = 180 − 60 − 60 = 60°; remaining sides = 5.7 and 5.7 cm (equilateral). (ii) Third angle = 180 − 70 − 70 = 40°; the equal sides are about 7.3 and 7.3 cm. (iii) Third angle = 180 − 60 − 50 = 70°; YZ about 4.9 cm, XZ about 5.5 cm.

Step 3: (iv) Third angle = 60°; remaining sides = 4.8 and 4.8 cm. (v) Third angle = 180 − 75 − 65 = 40°; FG about 7.8 cm, EG about 8.3 cm. (vi)

Third angle = 180 − 45 − 45 = 90°; JK about 3.3 cm, KL about 3.3 cm.

Final answer: (i) 60°; 5.7, 5.7 cm (ii) 40°; 7.3, 7.3 cm (iii) 70°; 4.9, 5.5 cm (iv) 60°; 4.8, 4.8 cm (v) 40°; 7.8, 8.3 cm (vi) 90°; 3.3, 3.3 cm.

Question: 5. Look at the following right angled triangles. Name and measure the hypotenuse in each triangle: (i) △ABC (right angle at C) (ii) △PQR (right angle at Q) (iii) △XYZ (right angle at X) (iv) △LMN (right angle at N) (v) △DEF (right angle at E) (vi) △GHI (right angle at I) 6. Using a ruler and protractor to construct the following triangles. Measure the length of remaining two sides and one angle of the triangle. (i) △ABC where mAB = 5 cm, m∠BAC = 55°, m∠ABC = 35° (ii) △JKL where m∠KJL = 65°, m∠JKL = 25°, mJK = 4.8 cm (iii) △PQR where m∠QPR = 30°, m∠PQR = 60°, mQP = 4 cm (iv) △STU where mST = 5.3 cm, m∠STU = 75°, m∠TSU = 15°

Answer: Step 1: The hypotenuse is the side opposite the right angle and is the longest side of the triangle; measure its length with a ruler on the book's figure.

Step 2: 5(i) Hypotenuse AB. (ii) Hypotenuse PR. (iii) Hypotenuse YZ. (iv) Hypotenuse LM. (v) Hypotenuse DF. (vi) Hypotenuse GH.

Step 3: 6 Use the ASA steps (as in Question 4); third angle = 180° − the sum of the two given angles. (i) Third angle = 180 − 55 − 35 = 90°; BC about 4.1 cm, AC about 2.9 cm. (ii) Third angle = 180 − 65 − 25 = 90°; KL about 4.4 cm, JL about 2.0 cm.

Step 4: (iii) Third angle = 180 − 30 − 60 = 90°; QR = 2.0 cm, PR about 3.5 cm. (iv) Third angle = 180 − 75 − 15 = 90°; SU about 5.1 cm, TU about 1.4 cm.

Final answer: 5: (i) AB (ii) PR (iii) YZ (iv) LM (v) DF (vi) GH. 6: (i) 90°; 4.1, 2.9 cm (ii) 90°; 4.4, 2.0 cm (iii) 90°; 2.0, 3.5 cm (iv) 90°; 5.1, 1.4 cm.

Question: 7. Use protractor, a pair of compasses and ruler to construct the

following triangles: (i) Acute angled △ABC, when m∠BAC = 65°, mAB = 3.6 cm and mAC = 4.4 cm (ii) Right angled △DEF, when m∠DEF = 90°, sides 3 cm and mEF = 4 cm (iii) Obtuse angled △LMN, m∠NML = 110°, mMN = mML = 5 cm (iv) Acute angled △PQR, m∠PQR = 60°, mPQ = mQR = 4 cm (v) Right angled △XYZ, when m∠XYZ = 90°, mXY = 4.2 cm = mYZ (vi) Obtuse angled △STU, when m∠STU = 120°, mST = 3.6 cm and mTU = 4.2 cm

Answer: Step 1: Construction steps (one angle and its two adjacent sides, SAS): draw the first side.

Step 2: At one end of it, make the given angle with a protractor.

Step 3: Cut the second side's measure on the angle's arm with compasses/ruler and mark the point.

Step 4: Join the third vertex to the other end of the first side. Same steps: (i) Draw AB = 3.6, angle 65° at A, cut AC = 4.4 on the arm, join C to B. (ii) Draw one side 3 cm, angle 90° at E, cut EF = 4, join. (iii) Draw MN = 5, angle 110° at M, cut ML = 5, join L to N. (iv) Draw PQ = 4, angle 60° at Q, cut QR = 4, join R to P. (v) Draw XY = 4.2, angle 90° at Y, cut YZ = 4.2, join Z to X. (vi) Draw ST = 3.6, angle 120° at T, cut TU = 4.2, join U to S.

Final answer: All six required triangles (two acute, two right, two obtuse) are constructed.

Exercise 7.7 — Quadrilaterals

Question: 1. Recognize the names and give the answers of the following for each quadrilateral (the book's figures): (i) the first quadrilateral (ii) the second quadrilateral (iii) the third quadrilateral (iv) the fourth quadrilateral. How many? Sides ____, Angles ____, Vertex ____ and what is the sum of all the angles of each quadrilateral?

Answer: Step 1: Look at the properties of the figures: all sides equal and

opposite sides parallel → rhombus; two pairs of adjacent equal sides → kite; only one pair of parallel sides → trapezium.

Step 2: (i) Rhombus. (ii) Kite. (iii) Trapezium. (iv) Kite.

Step 3: Every quadrilateral has sides = 4, angles = 4, vertices = 4; the sum of the four angles = 360°.

Final answer: (i) Rhombus (ii) Kite (iii) Trapezium (iv) Kite; 4 sides, 4 angles, 4 vertices, angles sum to 360°.

Question: 2. Construct rectangle with sides given below with the help of a ruler and a protractor. (i) 4 cm, 3 cm (ii) 6 cm, 3.5 cm (iii) 5.5 cm, 2.8 cm 3. Construct a square with a side given below with the help of a ruler and a protractor. (i) 3 cm (ii) 4 cm (iii) 5.4 cm 4. Construct rectangles with sides given below with the help of a ruler and set squares. (i) 5 cm, 4 cm (ii) 6 cm, 3 cm (iii) 4.6 cm, 3.5 cm 5. Construct a square with a side given below with the help of a ruler and a set squares. (i) 4 cm (ii) 5 cm (iii) 4.4 cm

Answer: Step 1: Steps for a rectangle: draw the length side; make 90° angles at both ends (with a protractor or set squares); cut the breadth measure; join the top side. For a square use the same steps with all four sides equal.

Step 2: 2(i) Rectangle 4 x 3 cm constructed. (ii) Rectangle 6 x 3.5 constructed. (iii) Rectangle 5.5 x 2.8 constructed.

Step 3: 3(i) Square of side 3 cm constructed. (ii) Square of side 4 cm constructed. (iii) Square of side 5.4 cm constructed.

Step 4: 4(i) 5 x 4, (ii) 6 x 3, (iii) 4.6 x 3.5 cm rectangles constructed with set squares. 5(i) 4, (ii) 5, (iii) 4.4 cm squares constructed with set squares.

Final answer: All the required rectangles and squares are constructed with the given sides; every angle is 90°.

Review Exercise 7

Question: 1. Marium and Sakina start from a point A. Marium moves towards east up to E and Sakina moves towards south up to S. Draw their paths and name the kind of angle which will be formed between them. 2. State the kinds of angle that is formed between the following directions: (i) East and West (ii) East and North (iii) From North to West through East 3. Is it possible to have a triangle, in which: (i) Two of the angles are right angles? (ii) Two of the angles are acute? (iii) Two of the angles are obtuse? (iv) Each angle is less than 60°?

Answer: Step 1: Angles between directions: East to South = 90°; East and West are opposite directions = 180°; from North through East to West = 90° + 90° + 90° = 270°.

Step 2: 1. A right angle (90°) is formed between East and South. 2(i) Straight angle (180°). (ii) Right angle (90°). (iii) Reflex angle (270°).

Step 3: 3(i) No — two right angles alone make 90 + 90 = 180°, leaving nothing for the third angle. (ii) Yes — for example 50°, 60°, 70°; two angles can be acute. (iii) No — two obtuse angles alone pass 180°. (iv) No — if all three angles are less than 60°, the sum stays below 180°, but a triangle's angles must add to exactly 180°.

Final answer: 1. Right angle (90°). 2: (i) straight (180°) (ii) right (90°) (iii) reflex (270°). 3: (i) No (ii) Yes (iii) No (iv) No.

Question: 4. How many quadrilaterals are formed in the adjoining figure? Name them. (Figure: a big triangle ABC with F and E on the sides and D on the base; FE, FD and ED are joined.) 5. Write the names of the triangles and quadrilaterals formed in the figure. Also mention their kinds. (Figure 2: the figure of triangles and quadrilaterals with points A, B, C, D, E, F, G.) 6. Prove that a square is a rectangle but a rectangle is not a square. Give

reasons in your own words. 7. Two angles of a triangle are of measure 65° and 45°. Find the measure of the third angle.

Answer: Step 1: 4. Count the figures with four sides: AEDF, BEFD and DEFC — three quadrilaterals.

Step 2: 5. Equilateral triangles: ABC, FGE, GEC, ECD, AFE, BFG. Trapeziums: FEBC, FDGC, FACG, BAGE. Parallelogram: FDCB. Rhombuses: EDGC, FEBG, FAEG.

Step 3: 6. A square has all angles 90° and opposite sides equal, so it satisfies the definition of a rectangle — a square is a rectangle. But a rectangle's sides need not all be equal, so a rectangle is not necessarily a square.

Step 4: 7. Third angle = 180° − 65° − 45° = 180° − 110° = 70°.

Final answer: 4. 3 quadrilaterals: AEDF, BEFD, DEFC. 5. Triangles (equilateral): ABC, FGE, GEC, ECD, AFE, BFG; trapeziums: FEBC, FDGC, FACG, BAGE; parallelogram: FDCB; rhombuses: EDGC, FEBG, FAEG. 6. A square is a rectangle, but a rectangle is not always a square. 7. 70°.

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Exercise 8.1 — Closed and Open Figures, Units of Perimeter and Area

Question: A. Look at the following figures carefully. (✓) the shapes which are closed figures and (✗) the open figures (the book's figures (1) to (9)). B. Find the perimeter and write the unit of perimeter for each of the following: (1) a square of side 20 mm (2) a square of side 3 cm (3) a square of side 4 m C. Find the area by counting small unit squares and write the unit of area for each of the following: (1) rectangle 7 m x 2 m (divided in a grid) (2) square 40 mm x 40 mm (divided in a grid, each small square of side 10 mm) (3) rectangle 1 cm x 2 cm (divided in a grid) (4) rectangle 9 km x 1 km (divided in a grid)

Answer: Step 1: A closed figure has its boundary closed on every side; an open figure has a gap somewhere. Perimeter = sum of all sides; area is found by counting unit squares.

Step 2: A(1) ✗ open (one side incomplete). (2) ✓ closed. (3) ✓ closed. (4) ✓ closed. (5) ✗ open (two separate lines). (6) ✓ closed. (7) ✓ closed. (8) ✓ closed. (9) ✗ open (a spiral line, ends apart).

Step 3: B(1) Perimeter = 4 x 20 = 80 mm; unit: millimetre. (2) 4 x 3 = 12 cm; unit: centimetre. (3) 4 x 4 = 16 m; unit: metre.

Step 4: C(1) 7 x 2 = 14 unit squares; area = 14 sq m. (2) 4 x 4 = 16 unit squares; each square = 10 x 10 = 100 sq mm; area = 16 x 100 = 1,600 sq mm. (3) 2 unit squares; area = 2 sq cm. (4) 9 unit squares; area = 9 sq km.

Final answer: A: (1) ✗ (2) ✓ (3) ✓ (4) ✓ (5) ✗ (6) ✓ (7) ✓ (8) ✓ (9) ✗. B: (1)

80 mm (2) 12 cm (3) 16 m. C: (1) 14 sq m (2) 1,600 sq mm (3) 2 sq cm (4) 9 sq km.

Question: Activity 1 (book page 142): Look at the rectangle ABCD with the length of 5 cm and breadth 4 cm (in a grid of unit squares): (i) What is the length of rectangle ABCD? (ii) What is the breadth of rectangle ABCD? (iii) How many rows in the figure? (iv) How many unit squares in one row? (v) How many unit squares altogether?

Answer: Step 1: Look at the grid: 5 squares along the length and 4 rows along the breadth.

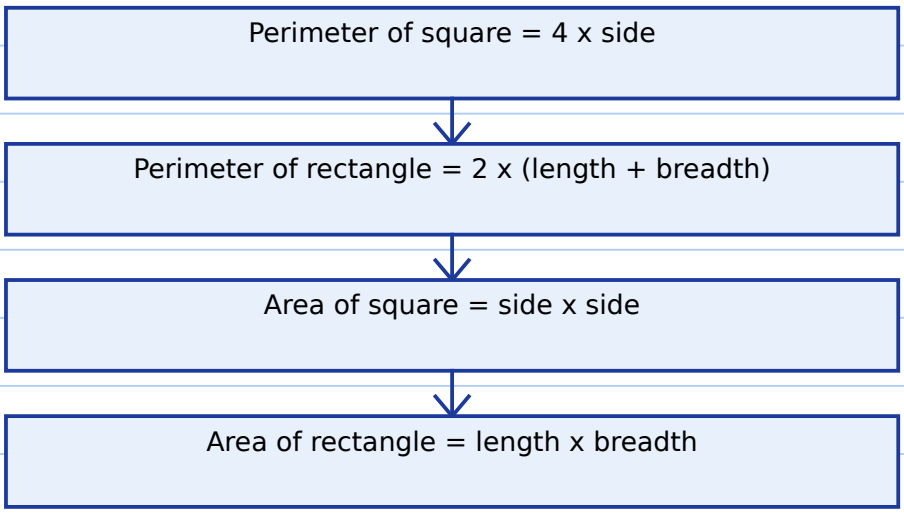
Step 2: (i) Length = 5 cm. (ii) Breadth = 4 cm. (iii) Rows = 4. (iv) 5 unit squares in one row.

Step 3: (v) Total = 5 x 4 = 20 unit squares; so the area = 20 sq cm.

Final answer: (i) 5 cm (ii) 4 cm (iii) 4 rows (iv) 5 (v) 20 unit squares, area 20 sq cm.

Exercise 8.2 — Perimeter and Area by Formulas

Perimeter and Area Formulas



Question: A. Find the perimeter of each of the following figures: (1) a triangle of sides 2 cm, 2 cm and 3 cm (2) a trapezium of sides 40 mm, 20

mm, 20 mm and 20 mm (3) a figure whose given sides are 2 cm, 3 cm, 20 mm and 10 mm B. Find the area of the following shapes by using formula: (1) 8 m x 800 cm square-like shape (2) 990 cm x 9.9 m shape (3) 7.7 m x 770 cm shape (4) rectangle 1650 cm x 9.2 m

Answer: Step 1: Perimeter = sum of all sides (first make the units the same); area = length x breadth (change units into metres).

Step 2: A(1) 2 + 2 + 3 = 7 cm. (2) 40 + 20 + 20 + 20 = 100 mm. (3) 20 mm = 2 cm and 10 mm = 1 cm; perimeter = 2 + 3 + 2 + 1 = 8 cm.

Step 3: B(1) 800 cm = 8 m; area = 8 x 8 = 64 sq m. (2) 990 cm = 9.9 m; 9.9 x 9.9 = 98.01 sq m. (3) 770 cm = 7.7 m; 7.7 x 7.7 = 59.29 sq m. (4) 1650 cm = 16.5 m; 16.5 x 9.2 = 151.8 sq m.

Final answer: A: (1) 7 cm (2) 100 mm (3) 8 cm. B: (1) 64 (2) 98.01 (3) 59.29 (4) 151.8 sq m.

Question: C. Find the area and perimeter of each of the following rectangles: (1) L = 3 cm, B = 2 cm (2) L = 5 cm, B = 1 cm (3) L = 4 cm, B = 3 cm (4) L = 8 cm, B = 2 cm (5) L = 9 cm, B = 5 cm (6) L = 7 cm, B = 4 cm (7) L = 4.5 cm, B = 2 cm (8) L = 8 cm, B = 3.5 cm D. Find area and perimeter of a square whose sides are given below: (1) 4 cm (2) 6 cm (3) 7.5 cm (4) 8.2 cm (5) 5 cm 6 mm (6) 9 cm 2 mm

Answer: Step 1: Rectangle: area = length x breadth; perimeter = 2 x (length + breadth). Square: area = side x side; perimeter = 4 x side.

Step 2: C(1) Area 3 x 2 = 6 sq cm, perimeter 2 x 5 = 10 cm. (2) 5, 12. (3) 12, 14. (4) 16, 20. (5) 45, 28. (6) 28, 22. (7) 9, 13. (8) 28, 23.

Step 3: D(1) 4 x 4 = 16 sq cm, perimeter 16 cm. (2) 36, 24. (3) 7.5 x 7.5 = 56.25, perimeter 30. (4) 8.2 x 8.2 = 67.24, perimeter 32.8.

Step 4: D(5) Side = 5.6 cm; area = 5.6 x 5.6 = 31.36 sq cm, perimeter = 22.4 cm. (6) Side = 9.2 cm; area = 84.64 sq cm, perimeter = 36.8 cm.

Final answer: C (area, perimeter): (1) 6, 10 (2) 5, 12 (3) 12, 14 (4) 16, 20 (5) 45, 28 (6) 28, 22 (7) 9, 13 (8) 28, 23. D (area, perimeter): (1) 16, 16 (2) 36, 24 (3) 56.25, 30 (4) 67.24, 32.8 (5) 31.36, 22.4 (6) 84.64, 36.8.

Question: E. Answer the questions (shapes A, B, C, D, E, F in the book's grid figure): the areas of the shapes in unit squares are: A = 6, B = 4, C = 3, D = 6, E = 4, F = 3. (1) Which shape has the same area as shape B? (2) Which shape has the same area as shape C? (3) Which rectangle shapes are equal in area as rectangle A? (4) Which rectangle shapes are equal in area as square E? (5) Which shapes have the same area?

Answer: Step 1: Count the unit squares of each shape in the grid: A = 6, B = 4, C = 3, D = 6, E = 4, F = 3.

Step 2: (1) B = 4; E also has 4 → E. (2) C = 3; F also has 3 → F.

Step 3: (3) A = 6; D also has 6 → D. (4) Square E = 4; B also has 4 → B.

Step 4: (5) Pairs with the same area: A and D (6), B and E (4), C and F (3).

Final answer: (1) E (2) F (3) D (4) B (5) A = D, B = E, C = F.

Exercise 8.3 — Word Problems on Perimeter and Area

Question: 1. A rectangular park is 84 m long and 56 m broad. Find the perimeter of the park. 2. A square room is 7 m wide. Find the area of the room. 3. A square picture is 60 cm wide. How long wooden frame will be required to reap it form all sides? 4. The length and breadth of a rectangular agriculture field are 190 m and 160 m respectively. Find the area and the perimeter of the field.

Answer: Step 1: Formulas: perimeter of rectangle = 2 x (length + breadth); area of square = side x side; frame = perimeter of square = 4 x side; area of rectangle = length x breadth.

Step 2: 1. Perimeter = 2 x (84 + 56) = 2 x 140 = 280 m. 2. Area = 7 x 7 =

49 sq m.

Step 3: 3. Frame = 4 x 60 = 240 cm. 4. Area = 190 x 160 = 30,400 sq m; perimeter = 2 x (190 + 160) = 2 x 350 = 700 m.

Final answer: 1. 280 m 2. 49 sq m 3. 240 cm 4. Area 30,400 sq m, perimeter 700 m.

Question: 5. How much lace is required to fix around a rectangular bed sheet, whose length is 2 m 80 cm and width is 1 m 50 cm? 6. Find the area of both fields; a square of 25 m side and a rectangle 30 m in length and 20 m in breadth. 7. A rectangular agriculture park is 75 m long and 40 m wide. Find the cost of laying green grass in it at the rate of Rs 25 per square metre. 8. Consider a room 15m long and 12m wide. A square carpet 10m x 10m is placed on it. (i) What is the area of the floor of the room? (ii) What is the area of the carpet? (iii) Which is the more in area; the floor or the carpet? And by how much?

Answer: Step 1: Lace = perimeter of the sheet; areas by formulas; cost = area x rate.

Step 2: 5. Length = 2.8 m, width = 1.5 m; lace = 2 x (2.8 + 1.5) = 2 x 4.3 = 8.6 m (= 860 cm). 6. Square = 25 x 25 = 625 sq m; rectangle = 30 x 20 = 600 sq m.

Step 3: 7. Area = 75 x 40 = 3,000 sq m; cost = 3,000 x 25 = Rs 75,000.

Step 4: 8(i) Floor = 15 x 12 = 180 sq m. (ii) Carpet = 10 x 10 = 100 sq m. (iii) The floor is more; difference = 180 − 100 = 80 sq m.

Final answer: 5. 8.6 m 6. Square 625 sq m; rectangle 600 sq m 7. Rs 75,000 8. (i) 180 sq m (ii) 100 sq m (iii) The floor, by 80 sq m.

Review Exercise 8

Question: A. Choose and tick (✓) the correct answer. 1. The space

occupied by the boundary of a shape is called: (a) Triangle (b) Square (c) Perimeter (d) Region 2. The distance along all the sides of a closed shape is called: (a) Triangle (b) Square (c) Perimeter (d) Region 3. The perimetre of a square of length of each side 4 cm is: (a) 16 m (b) 16 m² (c) 16 cm (d) 16 cm² 4. The area of a square having each side of 3 cm is: (a) 6 cm (b) 9 cm (c) 9 cm² (d) 12 cm 5. The area of a rectangle with length 4 cm and breadth 2 cm is: (a) 4 cm (b) 8 cm (c) 8 cm² (d) 12 cm 6. The perimeter of a rectangle with length 6 cm and breadth 3 cm is: (a) 6 cm (b) 18 cm (c) 9 cm (d) 15 cm B. Answer the following questions. 1. Write formula of area of a square. 2. Write formula of perimeter of a rectangle. 3. Find the perimeter and area of a square of side 7 cm. 4. Find the perimeter and area of a rectangle with length and breadth 8 cm and 5 cm respectively. Answer: Step 1: A 1. The space inside the boundary = Region → (d). 2. The distance along all sides = Perimeter → (c).

Step 2: 3. 4 x 4 = 16 cm → (c). 4. 3 x 3 = 9 sq cm → (c). 5. 4 x 2 = 8 sq cm → (c). 6. 2 x (6 + 3) = 18 cm → (b).

Step 3: B 1. Area of a square = side x side. 2. Perimeter of a rectangle = 2 x (length + breadth).

Step 4: B 3. Perimeter = 4 x 7 = 28 cm; area = 7 x 7 = 49 sq cm. 4. Perimeter = 2 x (8 + 5) = 26 cm; area = 8 x 5 = 40 sq cm.

Final answer: A: 1 (d) 2 (c) 3 (c) 4 (c) 5 (c) 6 (b). B: 1. side x side 2. 2 x (length + breadth) 3. Perimeter 28 cm, area 49 sq cm 4. Perimeter 26 cm, area 40 sq cm.

معلومات جو سنپال9یونٹ :

Question: Activities (book page 150): Activity 1: Find the average of first five even numbers. Activity 2: Find the average of first five odd numbers.

Answer: Step 1: Average = sum of numbers ÷ total number. First five even numbers: 2, 4, 6, 8, 10; first five odd numbers: 1, 3, 5, 7, 9.

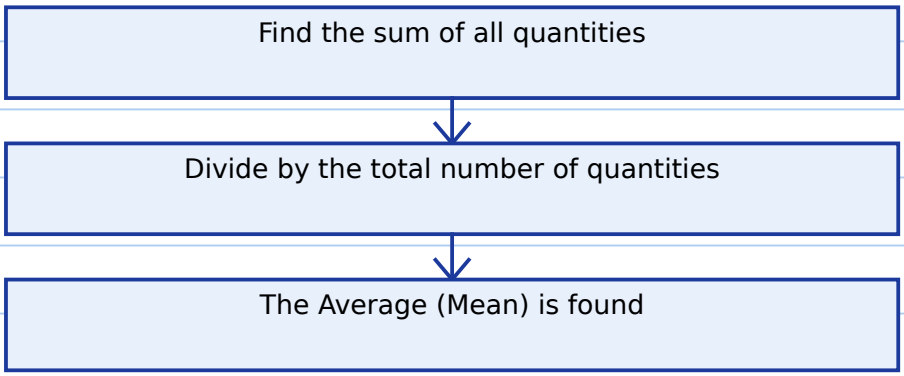
Step 2: Activity 1: Sum = 2 + 4 + 6 + 8 + 10 = 30; average = 30 ÷ 5 = 6.

Step 3: Activity 2: Sum = 1 + 3 + 5 + 7 + 9 = 25; average = 25 ÷ 5 = 5.

Final answer: Activity 1: 6; Activity 2: 5.

Exercise 9.1 — Find the Average (Mean)

Average Formula



Question: Find the average (mean) of the following numbers. (1) 12, 14, 16, 18 and 20 (2) 1, 2, 3, 4, 5, 6 and 7 (3) 6, 7, 8, 9, 7, 6, 5 and 15 (4) 2, 3, 5, 7, 11, 13, 17 and 19 (5) 1/2, 2/5, 3/4 and 4/5

Answer: Step 1: Average = sum of all numbers ÷ number of numbers.

Step 2: (1) Sum = 12 + 14 + 16 + 18 + 20 = 80; average = 80 ÷ 5 = 16. (2) Sum = 28; average = 28 ÷ 7 = 4.

Step 3: (3) Sum = 6 + 7 + 8 + 9 + 7 + 6 + 5 + 15 = 63; average = 63 ÷

8 = 7.875. (4) Sum = 77; average = 77 ÷ 8 = 9.625.

Step 4: (5) Change the fractions into decimals: 0.5 + 0.4 + 0.75 + 0.8 = 2.45; average = 2.45 ÷ 4 = 0.6125 = 49/80.

Final answer: (1) 16 (2) 4 (3) 7.875 (4) 9.625 (5) 49/80.

Question: Find the average of the following numbers. (6) 3/10, 7/20, 11/30, 13/40 and 17/50 (7) 1.1, 2.2, 3.3, 4.4, 5.5 and 6.6 (8) 13.5, 7.5, 6.5, 9.5, 5.5 and 7.5 (9) 1 2/3, 4 5/6, 7 8/9, 10 11/12 and 13 14/15 (10) 6 5/6, 7 1/2, 8 1/3 and 5

Answer: Step 1: For fractions first find the sum (by a common denominator or by changing into decimals), then divide by the count.

Step 2: (6) In decimals: 0.3 + 0.35 + 0.3667 + 0.325 + 0.34 = 1.6817; average = 1.6817 ÷ 5 = 0.33633 = 1009/3000. (7) Sum = 23.1; average = 23.1 ÷ 6 = 3.85.

Step 3: (8) Sum = 13.5 + 7.5 + 6.5 + 9.5 + 5.5 + 7.5 = 50; average = 50 ÷ 6 = 8.33 (25/3).

Step 4: (9) Sum in decimals ≈ 35.314; average ≈ 7.847 = 7063/900 = 7 763/900. (10) Sum ≈ 27.667; average = 27.667 ÷ 4 = 6.9167 = 83/12 = 6 11/12.

Final answer: (6) 1009/3000 (7) 3.85 (8) 8.33 (9) 7 763/900 (10) 6 11/12.

Exercise 9.2 — Real Life Problems on Average

Question: 1. Rabia obtained 65 marks in Maths, 72 marks in Urdu, 60 marks in Science, 75 marks in Sindhi and 70 marks in Islamiyat. Find her average marks. 2. Kalsoom saved rupees 13, 15, 12, 20, 25, 30 and 18 during a week. What is her average saving per day? 3. The maximum temperature of Sanghar city recorded during the week of June last year was 36.3C°, 42.7C°, 41.6C°, 38.5C°, 40.4C°, 41.9C° and 42.8C°. Find the

average temperature during that week. 4. A Qari wants to recite the Holy Qura'n in 15 days during the holy month of Ramzan. What will be the average number of Paras he has to recite daily? 5. Saleem scored 50, 70, 100, 60 runs in the four one day matches. What was his average score? 6. A labour earns Rs 577 on the first day, Rs 600 on the second day and Rs 725 on the third day. Find his average income per day.

Answer: Step 1: Average = total sum ÷ number of quantities (days/subjects).

Step 2: 1. Sum = 65 + 72 + 60 + 75 + 70 = 342; average = 342 ÷ 5 = 68.4 marks. 2. Sum = 13 + 15 + 12 + 20 + 25 + 30 + 18 = 133; average = 133 ÷ 7 = Rs 19 per day.

Step 3: 3. Sum = 36.3 + 42.7 + 41.6 + 38.5 + 40.4 + 41.9 + 42.8 = 284.2; average = 284.2 ÷ 7 = 40.6°C. 4. There are 30 Paras in the Holy Quran; average = 30 ÷ 15 = 2 Paras daily.

Step 4: 5. Sum = 50 + 70 + 100 + 60 = 280; average = 280 ÷ 4 = 70 runs. 6. Sum = 577 + 600 + 725 = 1,902; average = 1,902 ÷ 3 = Rs 634 per day.

Final answer: 1. 68.4 marks 2. Rs 19 per day 3. 40.6°C 4. 2 Paras daily 5. 70 runs 6. Rs 634 per day.

Question: 7. A train covers a distance of 560 km in 8 hours from Karachi to Rohri. What is its average speed? 8. A car covers a distance of 55 km in the first hour, 60 km in the second hour, 45 km in the third hour and 30 km in the fourth hour. Find the average speed of the car. 9. Akmal gets 8 wickets for 72 runs. Find the average runs he has given for each wicket. 10. Kulsoom's father told her that he would buy her a bicycle if she scored an average of 80 marks in her exams. The scores of all of her exams are below. Did she get the bicycle? Urdu: 85, English: 70, Math: 82, Science:

77, Social Studies: 75, Islamiyat: 89.

Answer: Step 1: Average speed = total distance ÷ total hours; average runs per wicket = runs ÷ wickets; average marks = total marks ÷ subjects.

Step 2: 7. 560 ÷ 8 = 70 km per hour. 8. Total distance = 55 + 60 + 45 + 30 = 190 km; average = 190 ÷ 4 = 47.5 km per hour.

Step 3: 9. 72 ÷ 8 = 9 runs per wicket.

Step 4: 10. Total = 85 + 70 + 82 + 77 + 75 + 89 = 478; average = 478 ÷ 6 = 79.67 marks. 79.67 is less than 80, so the condition is not met.

Final answer: 7. 70 km/h 8. 47.5 km/h 9. 9 runs per wicket 10. No, she does not get the bicycle (average 79.67, less than 80).

Question: Graph Activities 1 (book pages 154–155): (a) The block graph of favourite breakfast food of class V students: Fresh Fruit 10, Toast/Bread 5, Eggs 15, Paratha 10, Cereal 5. (i) How many students eat Paratha? (ii) Which breakfast is the most favourite? (iii) Which breakfast is least favourite? (b) The ages of Nazia's family members are: Majeed is 30 years old, Razia is 25 years old, Nazia is 10 years old, Hamza is 5 years old. Give the steps to draw their block graph and the height of each column.

Answer: Step 1: In a block graph read each column's height on the scale (1 block = 5 students); the tallest column is the most favourite, the shortest column is the least favourite.

Step 2: (a)(i) The Paratha column reaches 10 → 10 students. (ii) The Eggs column is the tallest (15) → Eggs. (iii) The Toast/Bread and Cereal columns are the shortest (5, 5) → Toast/Bread and Cereal.

Step 3: (b) Steps: write the ages on the vertical side (0 to 40, in steps of 5) and the family members on the horizontal side; then colour each member's column up to his/her age: Majeed 30, Razia 25, Nazia 10,

Hamza 5.

Final answer: (a) (i) 10 (ii) Eggs (iii) Toast/Bread and Cereal. (b) Columns: Majeed 30, Razia 25, Nazia 10, Hamza 5 years.

Question: Graph Activity 2 (book page 155): Read the vertical bar graph (students and what they ate at break time in school) and write the correct answer in the box. (1) How many children ate oranges? (2) How many children ate apples? (3) How many total children were in the school? (4) How many children ate samosa? (5) How many total ate chips? (6) How many children ate sandwich?

Answer: Step 1: Read each bar's height on the vertical scale (number of students): Samosa about 12, Apple 8, Sandwich about 9, Orange about 4, Chips about 1.

Step 2: (1) Oranges = 4. (2) Apples = 8. (4) Samosa = 12. (5) Chips = 1. (6) Sandwich = 9.

Step 3: (3) Total = 12 + 8 + 9 + 4 + 1 = 34 children.

Final answer: (1) 4 (2) 8 (3) 34 (4) 12 (5) 1 (6) 9.

Question: Graph Activity 3 (book page 156): Read the following horizontal bar graph. In this graph, the marks obtained by Yasir of Grade V in five subjects are given. (i) How many marks did Yasir get in English? (ii) How many marks did he get in Urdu? (iii) How many marks did he get in Mathematics? (iv) How many marks did he get in Science? (v) How many marks did he get in Social studies? (vi) In which subjects did he got the least number of marks? (vii) How many total marks did he get? (viii) In which subject did he get the highest marks? (ix) Which subjects he get equal marks?

Answer: Step 1: Read each bar's length on the horizontal scale (marks obtained): Mathematics 80, English 70, Urdu 60, Science 60, Social Studies

50.

Step 2: (i) English = 70 (given in the book). (ii) Urdu = 60. (iii) Mathematics = 80. (iv) Science = 60. (v) Social Studies = 50.

Step 3: (vi) The shortest bar is Social Studies (50) → Social Studies. (vii) Total = 70 + 60 + 80 + 60 + 50 = 320. (viii) The longest bar is Mathematics (80) → Mathematics. (ix) Urdu and Science both 60, 60 → equal.

Final answer: (i) 70 (ii) 60 (iii) 80 (iv) 60 (v) 50 (vi) Social Studies (vii) 320 (viii) Mathematics (ix) Urdu and Science.

Question: Graph Activity 4 (book page 157): The following bar graph shows the number of people who cannot read of a country in different years (in millions): 1970: 34 million, 1980: 25 million, 1990: 23 million, 2000: 21 million, 2010: 24 million. (i) How many people could not read in 1990? (ii) In which year was number of people who could not read highest? (iii) What was the number of people who could not read in 1980? (iv) How many people who could not read were in 2010? (v) In which year was 2nd highest people who could not read? (vi) How many fewer people illiterates in 2000 than in 1990?

Answer: Step 1: Read the value written on each year's bar (in millions): 1970 = 34, 1980 = 25, 1990 = 23, 2000 = 21, 2010 = 24.

Step 2: (i) 23 million (given in the book). (ii) The biggest value, 34 million, is in 1970 → 1970. (iii) 25 million. (iv) 24 million.

Step 3: (v) The second biggest value is 25 million → 1980. (vi) 23 − 21 = 2 million fewer.

Final answer: (i) 23 million (ii) 1970 (iii) 25 million (iv) 24 million (v) 1980 (vi) 2 million fewer.

Exercise 9.3 — Block, Column and Bar Graphs

Question: 1. Draw a block graph or column graph of the following (give the axes, scale and column heights of each graph): (i) Attendance of Faraz's class during a week: Monday 32, Tuesday 35, Wednesday 30, Thursday 30, Friday 25, Saturday 20 (ii) Result of examination of class V: Grade A1: 20 children, A: 25, B: 15, C: 20, D: 5 (iii) The marks obtained by Amjad during annual examination: Islamiyat 80, English 50, Urdu 65, Maths 90, S. Studies 40, Science 55

Answer: Step 1: Steps to draw a graph: write the names/days on the horizontal axis; write the numbers with a suitable scale on the vertical axis; draw a column of each item's value height.

Step 2: (i) Take the scale 1 block = 5 children; columns: Monday 32, Tuesday 35, Wednesday 30, Thursday 30, Friday 25, Saturday 20 — the tallest column is Tuesday's (35), the shortest Saturday's (20).

Step 3: (ii) Columns: A1 = 20, A = 25, B = 15, C = 20, D = 5 — tallest A (25), shortest D (5). (iii) Columns: Islamiyat 80, English 50, Urdu 65, Maths 90, S. Studies 40, Science 55 — tallest Maths (90), shortest S. Studies (40).

Final answer: All three graphs are drawn with columns of the given values: (i) 32, 35, 30, 30, 25, 20 (ii) 20, 25, 15, 20, 5 (iii) 80, 50, 65, 90, 40, 55.

Question: 2. The population of Dadu city is represented in the following bar graph (population in millions: 1990: 10, 1995: 15, 2000: 30, 2005: 35, 2010: 40). Read the graph and answer the following questions: (i) What was the population of the city in 1995? (ii) What was the population of the city in 2000? (iii) What was the population of the city in 2005? (iv) In which year the population was least? (v) In which year the population was highest? (vi) How many more people were there in 2010 than in 2005? 3.

The expenditure of a family for five months is reported by the following bar graph (rupees: January 7,000, February 5,500, March 8,000, April 7,000, May 10,000): (i) In which month expenditure was the least? (ii) In which month the expenditure was the greatest? (iii) What was the expenditure in the month of February? (iv) What was the expenditure in the month of April? (v) In which months the expenditure is equal? (vi) In which month the expenditure is Rs 8000? (vii) What is the highest amount of expenditure? (viii) What is the difference of expenditure between months of April and May?

Answer: Step 1: Read each bar's height on the vertical scale.

Step 2: 2(i) 15 million. (ii) 30 million. (iii) 35 million. (iv) The shortest bar is of 1990 (10 million) → 1990. (v) The tallest bar is of 2010 (40 million) → 2010. (vi) 40 − 35 = 5 million more.

Step 3: 3(i) February (the shortest bar). (ii) May (the tallest bar). (iii) Rs 5,500. (iv) Rs 7,000.

Step 4: 3(v) January and April (both 7,000). (vi) March (8,000). (vii) Rs 10,000 (May). (viii) 10,000 − 7,000 = Rs 3,000.

Final answer: 2: (i) 15 million (ii) 30 million (iii) 35 million (iv) 1990 (v) 2010 (vi) 5 million more. 3: (i) February (ii) May (iii) Rs 5,500 (iv) Rs 7,000 (v) January and April (vi) March (vii) Rs 10,000 (viii) Rs 3,000.

Question: Activity (book page 162): Organize the following information through tally method. The ages of students of class V in a school are: 11, 12, 11, 13, 10, 11, 13, 12, 13, 14, 12, 13, 11, 11, 14, 10, 12, 12, 11, 13.

Answer: Step 1: Write the ages in ascending order: 10, 10, 11, 11, 11, 11, 11, 11, 12, 12, 12, 12, 12, 13, 13, 13, 13, 13, 14, 14.

Step 2: Put a tally mark for each age (a group of five is shown by a crossed bundle).

Step 3: Counts: age 10 → tally 2 (two students); age 11 → tally 6; age 12 → tally 5; age 13 → tally 5; age 14 → tally 2.

Step 4: Check: 2 + 6 + 5 + 5 + 2 = 20 students — all counted.

Final answer: Tally table: age 10 → 2, age 11 → 6, age 12 → 5, age 13 → 5, age 14 → 2 students.

Exercise 9.4 — Organize Data Through Tally Method

Question: Organize the following survey information through tally method. 1. The pocket money of students of Class V is found as: Rs 30, Rs 40, Rs 50, Rs 40, Rs 50, Rs 60, Rs 50, Rs 50, Rs 40, Rs 70, Rs 60, Rs 50, Rs 40, Rs 30, Rs 40 and Rs 50. 2. The heights of students of Class V in a school are noted as: 110 cm, 115 cm, 100 cm, 105 cm, 104 cm, 110 cm, 108 cm, 100 cm, 100 cm, 105 cm, 110 cm, 104 cm, 105 cm, 100 cm, 115 cm, 105 cm, 110 cm, 105 cm, 100 cm and 105 cm. 3. The attendance of 20 days of class V students are: 15, 18, 16, 17, 18, 17, 18, 17, 16, 15, 16, 17, 18, 18, 17, 16, 18, 17, 16 and 17.

Answer: Step 1: Count a tally mark for each different value (one mark for each occurrence; make a crossed bundle for a group of five).

Step 2: 1. Rs 30 → 2, Rs 40 → 5, Rs 50 → 6, Rs 60 → 2, Rs 70 → 1. Check: 2 + 5 + 6 + 2 + 1 = 16 students.

Step 3: 2. 100 cm → 5, 104 → 2, 105 → 6, 108 → 1, 110 → 4, 115 → 2. Check: 5 + 2 + 6 + 1 + 4 + 2 = 20 students.

Step 4: 3. Attendance 15 → 2, 16 → 5, 17 → 7, 18 → 6. Check: 2 + 5 + 7 + 6 = 20 days.

Final answer: 1. 30→2, 40→5, 50→6, 60→2, 70→1. 2. 100→5, 104→2, 105→6, 108→1, 110→4, 115→2. 3. 15→2, 16→5, 17→7, 18→6.

Question: Organize the following information through tally method and

solve the last two questions. 4. A gardener planted different numbers of plants in 18 days as under: 2, 1, 3, 5, 6, 3, 2, 3, 1, 2, 3, 4, 3, 2, 5, 5, 2 and 3. 5. The number of patients daily checked by doctor in 15 days are: 35, 40, 36, 35, 35, 40, 37, 35, 40, 38, 35, 40, 39, 40 and 35. 6. A team played 6 matches. The score was 210, 189, 180, 205, 175 and 165 runs respectively. What is the average score? 7. The enrolment of a school is 460 students. If there are 8 class rooms, what is the average of students in each class room?

Answer: Step 1: Count the tallies as before; for average divide the total by the count.

Step 2: 4. Plants: 1 → 2, 2 → 5, 3 → 6, 4 → 1, 5 → 3, 6 → 1. Check: 2 + 5 + 6 + 1 + 3 + 1 = 18 days.

Step 3: 5. Patients: 35 → 6, 36 → 1, 37 → 1, 38 → 1, 39 → 1, 40 → 5. Check: 6 + 1 + 1 + 1 + 1 + 5 = 15 days.

Step 4: 6. Total score = 210 + 189 + 180 + 205 + 175 + 165 = 1,124; average = 1,124 ÷ 6 = 187.33 runs. 7. Average = 460 ÷ 8 = 57.5 students per class room.

Final answer: 4. 1→2, 2→5, 3→6, 4→1, 5→3, 6→1. 5. 35→6, 36→1, 37→1, 38→1, 39→1, 40→5. 6. 187.33 runs 7. 57.5 students.

Review Exercise 9

Question: 1. Find the average of following numbers: (i) 4, 3, 5, 8, 6, 10. (ii) 10, 20, 30, 40, 50 2. The six hours journey of a person on car is given below (Hours: Distance in kilometres — 1: 20, 2: 18, 3: 20, 4: 22, 5: 16, 6: 18). What is the average distance covered per hour? 3. A batsman made 160 runs in two innings. Find the average runs he made in each innings. 4. Read the graph and answer the following questions (sports items: Ball 3, Bats 5, Hockeys 2, Rackets 4). (i) How many balls are there? (ii) How many

bats are there? (iii) How many hockeys and bats are there? (iv) How many more rackets are there than hockeys? (v) How many fewer balls are there than bats? (vi) How many bats and rackets are there?

Answer: Step 1: Average = total sum ÷ number of quantities; read the graph bars on the scale.

Step 2: 1(i) Sum = 4 + 3 + 5 + 8 + 6 + 10 = 36; average = 36 ÷ 6 = 6. (ii) Sum = 150; average = 150 ÷ 5 = 30.

Step 3: 2. Total distance = 20 + 18 + 20 + 22 + 16 + 18 = 114 km; average = 114 ÷ 6 = 19 km per hour. 3. Average = 160 ÷ 2 = 80 runs per innings.

Step 4: 4(i) Balls = 3. (ii) Bats = 5. (iii) Hockeys + bats = 2 + 5 = 7. (iv) Rackets − hockeys = 4 − 2 = 2 more. (v) Bats − balls = 5 − 3 = 2 fewer. (vi) Bats + rackets = 5 + 4 = 9.

Final answer: 1: (i) 6 (ii) 30. 2. 19 km per hour 3. 80 runs 4: (i) 3 (ii) 5 (iii) 7 (iv) 2 more (v) 2 fewer (vi) 9.

Question: 5. Interpret the following vertical bar graph (number of apples: Asghar 30, Akbar 40, Huma 20, Hira 60). (i) How many apples does Asghar have? (ii) How many apples does Akbar have? (iii) How many apples does Huma have? (iv) How many apples does Hira have? (v) How many apples do Asghar and Akbar have? (vi) How many apples do Huma and Hira have? 6. Fill in the blanks. (i) The average of 5, 15, 30, 10 and 20 is ____ (ii) The average of 20, 40, 30 and 35 is ____ (iii) When we choose a suitable symbol to represent each part of information, we will use ____ graph.

Answer: Step 1: Read the bars' lengths on the scale; for the blanks use the average formula; a graph shown with symbols (pictures) is called a picture graph.

Step 2: 5(i) Asghar = 30. (ii) Akbar = 40. (iii) Huma = 20. (iv) Hira = 60. (v) 30 + 40 = 70. (vi) 20 + 60 = 80.

Step 3: 6(i) Sum = 5 + 15 + 30 + 10 + 20 = 80; average = 80 ÷ 5 = 16. (ii) Sum = 20 + 40 + 30 + 35 = 125; average = 125 ÷ 4 = 31.25. (iii) Picture graph.

Final answer: 5: (i) 30 (ii) 40 (iii) 20 (iv) 60 (v) 70 (vi) 80. 6: (i) 16 (ii) 31.25 (iii) Picture graph.